The Ins and Outs of Optimal Transport

Phillip Jiang Zhou

May 8, 2024

Submitted to the
Department of Mathematics and Statistics
of Amherst College
in partial fulfillment of the requirements
for the degree of
Bachelor of Arts with honors

Faculty Advisor: Professor Ryan Alvarado

Abstract

The theory of optimal transport describes how mass can be moved from one probability distribution to another in the least costly way. In this thesis, we discuss foundational results in the rapidly evolving field of optimal transport. In particular, we define a set of conditions under which optimal transport plans exist, show how the Kantorovich problem gives rise to the Wasserstein metric, and how optimal transport plans can be computationally approximated in practice. Optimal transport is an interdisciplinary field, and as such we draw on Probability Theory, Information Theory, Measure Theory, Functional Analysis, Convex Analysis, and Differential Geometry to develop the concepts discussed in this thesis.

Acknowledgements

First, I would like to thank Professor Alvarado for all his help on this thesis. Even though I have never embarked on a project of this magnitude before, Professor Alvarado provided critical feedback and moral support throughout the process of tackling this intellectually and emotionally challenging project. I could not be more grateful for his insights and vision for the thesis that he kindly provided throughout the academic year. Additionally, Professors Kraisler, Contreras, Moore, and Gong were all critical sources of feedback for the development of this thesis and I am extremely grateful for their time and expertise. Thank you to Ethan Meyers for providing invaluable support for the writing side of this thesis. I would like to thank Professor Benedetto for teaching me measure theory and for being such a responsive instructor. Even though that class kicked my ass, measure theory became the cornerstone of my thesis and one of my favorite math topics. I also want to thank Professor Contreras for his years of support and conversations with me through research on spectral graph theory, translating a Putnam textbook, and my independent study, Professor Daniels for welcoming me into the math department as a freshman, teaching my first math classes at Amherst, and being an amazing major advisor since my sophomore year, Professor Rasheed for being an absolute G in my complex analysis class, and Professor Leise for teaching a brilliant course on stochastic processes and her willingness to help me grow as a mathematical biologist. I also thank Jonah Botvinick-Greenhouse for looking through my thesis and helping me understand topics in optimal transport.

I want to acknowledge Dr. David McCandlish, Dr. Daniel Dwyer, Dr. Katerina Ragkousi, and Dr. Ian Mellis for guiding me towards a math thesis topic related to biology.

Finally, I give thanks to my parents, sister, family, and friends who have helped me grow over these past four years. I am tremendously grateful for you all.

Contents

List of Figures

List of Notation

(X, Σ, μ)	A measure space comprised of the set X, the σ-algebra Σ, and measure μ
(X, d)	A metric space comprised of the set X and metric d
$C(X)$	The space of continuous real-valued functions defined on the set X
$C_0(X)$	The space of continuous real-valued functions defined on the set X vanishing at infinity
$C_b(X)$	The space of bounded continuous real-valued functions defined on the set X
$T_{\#}\mu$	The pushforward measure of μ by T
$U(a, b)$	The space of admissible couplings of the vectors a, b
$\mu \ll \nu$	μ is absolutely continuous with respect to ν
$\mu \otimes \nu$	The product measure of μ and ν
μ, ν	Initial and final probability measures
φ, ψ	Dual functions (in the Kantorovich duality)
$L^p(X, \mu)$	The space of $p-$integrable $\mu-$measurable functions from X to $\mathbb{R}$
$\Pi(\mu, \nu)$	The set of admissible couplings of the probability measures μ, ν
$L\lfloor_{[0,1]}$	The Lebesgue measure on $\mathbb{R}$ restricted to the interval $[0, 1]$
δ_x	The Dirac measure on a point x
$W_p(\mu, \nu)$	The Wasserstein distance between μ and ν

$\mathcal{P}_p(X)$	The set of probability measures on X with bounded pth moment
$\mathcal{M}(X)$	The space of signed measures defined on a set X
$\mathbb{1}_d$	A column vector of d ones
$\mu_n \rightharpoonup \mu$	Weak convergence of measures
P	Coupling matrix
$H(P)$	Entropy of a coupling matrix
$\|\cdot\|_\infty$	The supremum norm on $C(X)$
$\|\cdot\|_{L^p(X,\mu)}$	The $L^p(X, \mu)$ norm on $L^p(X, \Sigma, \mu)$
$\|\cdot\|_{TV}$	The total variation norm

Chapter 1

Introduction

The theory of optimal transport establishes results about optimal ways of transporting mass from one probability distribution to another. The key elements of optimal transport problems are:

- The starting and ending probability distributions.
- The cost function, which determines how much effort is spent moving mass from one point in the starting distribution to another point in the ending distribution. Cost can correspond to distance, money, human labor, energy or other metrics of effort, and is defined *a priori*.
- The "transport scheme", which specifies which masses from the starting distribution get moved where in the ending distribution.

The idea of optimal transport was first formalized in Gaspard Monge's memoir, *Mémoire sur la théorie des déblais et des remblais*, published in 1776, where he considers how to move a mathematically abstracted pile of soil into a hole in the most efficient way. Monge's mathematical insights from this article were used by Napoleon Bonaparte during the French campaign in Egypt to efficiently move piles of soil for building embankments [35].

In Monge's formulation of the optimal transport problem, or the Monge problem, one seeks to construct a function $T : X \to Y$ called a **transport map** that takes mass from a probability measure μ defined on a metric space X and transports it to the probability measure ν defined on another metric space Y. However, this method of mass transport comes with the restriction that the mass from a point x in X cannot be split and moved to more than one point in Y. This restriction was later addressed by Leonid Kantorovich in the context of economics [36].

To motivate Kantorovich's formulation of optimal transport, let us consider a hypothetical city that has many bakeries and cafes. In order to fuel the bread consumption needs of the city's residents, a chain of bakeries delivers croissants to all of the cafes in the city daily. Each bakery and cafe have their own production capabilities for bread and differently sized customer bases respectively. The optimal transport problem in this case is to determine how the bakery should design its routes such that the least amount of effort is spent transporting bread from all its bakeries to all the cafes. Kantorovich's formulation of the optimal transport problem weakens the assumptions in the Monge problem by finding a joint probability measure π called a **transport plan** that contains the information for mass assignment and of the starting and ending distributions μ and ν as marginals.

The mathematicians Leonid Kantorovich and Leonid Wasserstein realized that the optimal transport problem could be used to quantify differences between probability distributions, or in other words, to define a metric on a space of probability measures with bounded pth moment. This metric is called the **Wasserstein distance**, and finds use in fields such as statistics and computer science [37] [49]. Equipped with the Wasserstein distance, a space of probability measures with bounded pth moment becomes a geodesic space, and can be even viewed as a Riemannian manifold [16, Chapter 18]. Such conceptual breakthroughs have led to interest in the optimal transport problem from pure mathematicians such as Alessio Figalli, who won a Fields Medal for his research on the subject [48].

Finally, in 2013, Marco Cuturi devised a computationally efficient method to find approximate solutions to the optimal transport problem [21] that has spurred the use of optimal transport in machine learning [20] and cell biology [26] called **entropic regularization**. Some of the foundational questions in optimal transport that arise from our previous discussion include:

1. When do solutions to the Monge/Kantorovich problems exist?
2. How can we use optimal transport to quantify distances between probability distributions?

3. How can we add a time component to optimal transport? How can optimal transport be used to transform probability distributions into each other?

4. How can we efficiently calculate optimal transport plans in practice?

The main goal of this thesis is to answer these questions. The thesis is organized as follows:

In Chapter 2, we define the **Monge and Kantorovich problems** and the conditions for when solutions to these problems exist. We also examine the solution to the optimal transport problem between one-dimensional (1-D) histograms and characterize their optimal transport plan. In Chapter 3, we define the **Wasserstein distance**, prove it is a metric, compare it to the Kullback-Leibler divergence, and prove that the space of probability measures equipped with this distance is a geodesic space. We will briefly touch on the dynamic formulation of the optimal transport problem as a preview of the potential applications of optimal transport in the study of Wasserstein gradient flows. Finally, in Chapter 4, we introduce **entropic regularization** as a way to approximate optimal transport plans and discuss how time-dependent optimal transport and entropic regularization enable biologists to discover cellular differentiation trajectories.

We have come a long way from using optimal transport for moving dirt and selling pastries to training adversarial neural networks, studying the weather [50], and decoding the genetic changes driving cell differentiation. Why has optimal transport stayed an active field of research from its inception in 1781 to the present day, and why is it applicable in so many different contexts? The mathematician Cedric Villani remarks that the diverse range of optimal transport applications in both pure and applied mathematics "...reflects the fact that optimal transport is a simple, meaningful, natural and therefore universal concept" [3].

Chapter 2

Monge and Kantorovich problems

In this chapter, we introduce the Monge and Kantorovich problems, both versions of the optimal transport problem. To minimize transport costs between two measures, the Monge problem produces a **transport map,** a function that transports masses from one distribution to another. However, since splitting of mass from the starting distribution is not allowed in the Monge problem, situations arise in which transport maps do not exist. The Kantorovich problem resolves this issue by generating a joint probability measure called a **transport plan** between two probability measures. To determine the conditions under which a transport plan exists, we prove a statement on the existence of transport plans in the continuous setting, and consider transport plans between one dimensional (1-D) measures. We assume in Chapter 2 and 3 that the reader is familiar with basic definitions and results from measure theory, many of which we provide in Appendix A.1, and which can also be found in [12].

2.1 Monge problem

Solving the Monge problem, in intuitive terms, is like figuring out the most efficient way to take a pile of dirt and shovel it into a hole of equal volume to the dirt. More rigorously, solving the Monge problem produces a mapping between two probability distributions that minimizes an integral representing the total cost of moving masses between distributions. The cost of moving mass from one location to another is determined *a priori* by a cost function, which can be the Euclidean distance or a more general function, such as a strictly convex function. For the formulation of the Monge problem in the general setting, μ and ν

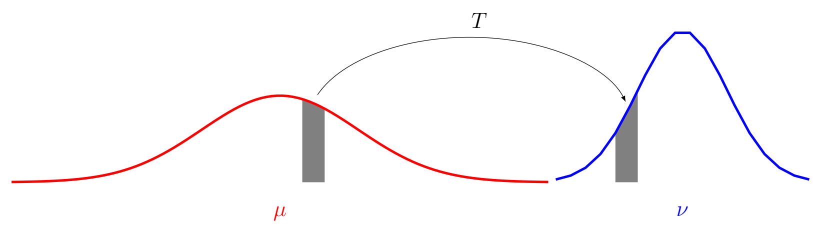


Figure 2.1: Suppose we have a starting distribution μ defined on a measurable space (X, Σ_1) and ending distribution ν defined on a measurable space (Y, Σ_2). Solving the Monge problem produces a surjective map $T : X \to Y$ that minimizes the cost incurred by transporting mass. The map T, commonly referred to as the **transport map**, dictates where the mass assigned to every point in X should be transported to in Y.

are Borel probability measures (Definition A.1.2).

A fundamental aspect of the Monge and Kantorovich problems is that they conserve mass. Transport plans are defined between probability measures μ and ν, which are themselves defined on the measurable spaces (X, Σ_1) and (Y, Σ_2) respectively. The total mass of each measure over the sets X and Y respectively is 1, i.e., $\mu(X) = \nu(Y) = 1$. The pushforward measure allows masses to be "pushed" from measure μ to ν by applying a measurable function $T : X \to Y$ (Definition A.1.5) to determine how much mass from the measure μ is assigned to a point in Y.

Definition 2.1.1 (Pushforward measure)**.** [14, Section 5.2] *Given measurable spaces (X, Σ_1) and (Y, Σ_2) where X and Y are separable metric spaces, $\mu \in \mathcal{P}(X)$, where $\mathcal{P}(\cdot)$ denotes the space of probability measures on a set, and a μ-measurable map $T : X \to Y$, the* **pushforward measure** *of μ by T denoted $T_{\#}\mu \in \mathcal{P}(Y)$ is defined by*

$$(T_{\#}\mu)(B) := \mu(T^{-1}(B)) \text{ for all Borel sets } B \subseteq Y.$$

By definition, for any bounded Borel function $f : Y \to \mathbb{R}$ (i.e., a measurable function defined on a Borel space) we also have

$$\int_X f(T(x))\, d\mu(x) = \int_Y f(y)\, d(T_{\#}\mu)(y). \tag{2.1}$$

Equality 2.1 is known as the "change of variables" formula.

Now that we have defined the pushforward measure, we present the full definition of the Monge problem.

Definition 2.1.2 (Monge problem)**.** [5, Problem 1.1] [18, Page 3] [8, Definition 1.3] *Let* (X, Σ_1) *and* (Y, Σ_2) *be two measurable spaces. Given probability measures* $\mu \in \mathcal{P}(X)$, $\nu \in \mathcal{P}(Y)$, *and a function* $c : X \times Y \to [0, \infty]$ *called the "cost function", find the transport map, a function* $T : X \to Y$ *that is the infimum of the quantity*

$$M(T) := \int_X c(x, T(x))\, d\mu(x),$$

where the infimum is taken over μ*-measurable maps* $T : X \to Y$ *satisfying* $T_{\#}\mu = \nu$.

The Monge problem was the first method of minimizing transport costs between probability distributions. However, there are some limitations to the Monge problem, namely, that mass at a single point in the set X cannot be split and moved to different locations.

2.1.1 Limitations of the Monge problem

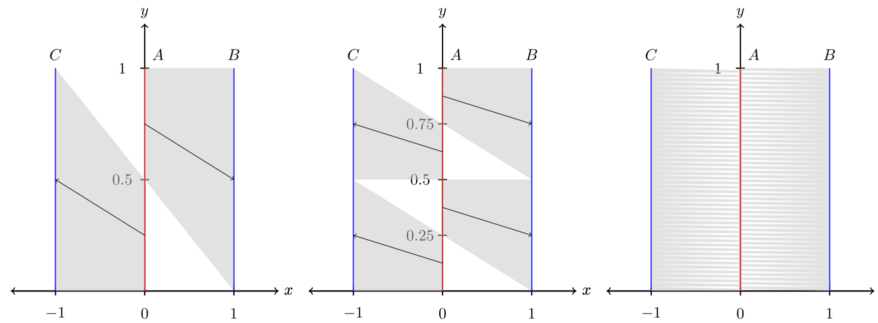
When considering the Monge problem, certain situations can arise where no transport map or optimal transport map exists. We present two here:

Example 1: No transport map exists

Consider the measure spaces $(\mathbb{R}, \mathcal{B}, \mu)$ and $(\mathbb{R}, \mathcal{B}, \nu)$, where $\mathcal{B}$ is the Borel σ-algebra (Definition A.1.2), and distinct points $x, y, z \in \mathbb{R}$. Construct the measures $\mu = \delta_x$, $\nu = \frac{1}{2}\delta_y + \frac{1}{2}\delta_z$, where the measures $\delta_x, \delta_y, \delta_z$ are Dirac measures (Definition B.1.1). There is no way to construct a surjective map mapping x to both y and z, so no transport map exists.

Example 2: No optimal transport map exists

There are also cases where transport maps exist between two measures, but no optimal one. Consider the "sawtooth" transport plan, defined in [4, Example 4.9] [5, Chapter 1.4], and depicted in Figure 2.2. Let $X = Y = \mathbb{R}^2$ where X and Y are equipped with the Borel σ-algebra, $c(x, y) = |x - y|^2$, $\mu = L^1\lfloor_{\{0\}\times[-1,1]}$, and $\nu = (1/2)L^1\lfloor_{\{-1,1\}\times[-1,1]}$, where L^1 is the 1-dimensional Lebesgue measure. In other words, we are taking mass uniformly distributed across the red line segment $\{0\} \times [-1, 1]$ denoted (A) and splitting it among two blue line segments $\{1\} \times [-1, 1]$ denoted (B) and $\{-1\} \times [-1, 1]$ denoted (C). There are many ways to create a transport map between these two measures. Since the source measure μ is a line



(a) T_1 (b) T_2 (c) T_N

Figure 2.2: The construction of the "sawtooth" scheme to transport mass from one line segment to two line segments. As the number of refinements of the central segment increases, the transportation cost decreases to 1, but can never attain that value.

segment that has twice as much mass as either of the two line segments it is transporting its mass to, one sensible way to move mass is to take the two halves of μ and move their mass onto the two line segments comprising ν. However, there is some inefficiency involved with doing so, because some masses will have to traverse diagonal paths and incur greater costs as a result. To reduce this inefficiency, we can subdivide the central interval even further and transport smaller units of mass to smaller increments of ν. We can repeat this process an arbitrary amount of times.

To formalize this construction, we define a sequence of maps $T_n : \{0\} \times [-1, 1] \to \{-1, 1\} \times [-1, 1]$ by dividing $\{0\} \times [-1, 1]$ into $2n$ equal segments and $\{-1\} \times [-1, 1]$ and $\{1\} \times [-1, 1]$ into n segments each. We can index the $2n$ segments of $\{0\} \times [-1, 1]$ and move the even ones to $\{-1\} \times [-1, 1]$ and odd to $\{1\} \times [-1, 1]$. As n grows larger, the cost of transporting mass approaches 1 but can never attain 1, as mass cannot be subdivided under the Monge problem. However, in the Kantorovich problem, which allows for mass splitting, this issue is circumvented by simply taking every point on $\{0\} \times [-1, 1]$, splitting its mass into two, and sending it horizontally to each interval B and C. The cost associated with the Kantorovich transport plan in this case is 1. Thus, we have an example of an optimal transport problem who has well-defined transport plans but no optimal transport plan.

2.2 Kantorovich problem

Both of the previous examples suggest that allowing for source masses to be split can remedy difficulties in finding optimal transport plans. The Kantorovich problem, which we now define, does exactly that.

Definition 2.2.1 (Kantorovich problem)**.** [5, Problem 1.2] *Let* (X, Σ_1) *and* (Y, Σ_2) *be two measurable spaces. Given probability measures* $\mu \in \mathcal{P}(X)$*,* $\nu \in \mathcal{P}(Y)$*, and a cost function* $c : X \times Y \to [0, \infty]$*, find the probability measure* $\pi \in \Pi(\mu, \nu)$ *for which the infimum in the following expression is attained:*

$$\inf_{\pi \in \Pi(\mu,\nu)} \left\{ \int_{X \times Y} c(x, y)\, d\pi(x, y) \right\},$$

where $\Pi(\mu, \nu)$ *is the space of probability measures* π *on* $X \times Y$ *such that for all measurable* $A \subset X$ *and* $B \subset Y$*, it is true that* $\pi(A \times Y) = \mu(A)$ *and* $\pi(X \times B) = \nu(B)$*.*

In other words, this definition says that π is a coupling measure on $X \times Y$ (Definition A.1.7). An equivalent condition for a measure γ to be an element of $\Pi(\mu, \nu)$ is if it satisfies $\pi_{x\#}\gamma = \mu$ and $\pi_{y\#}\gamma = \nu$, where $\pi_x(x, y) := x$ and $\pi_y(x, y) := y$ respectively [5, pg. 2]. In this case, μ and ν are referred to as the **marginals** of the measure γ. The Kantorovich problem is a relaxed version of the Monge problem. Historically, it was formulated as a response to the difficulties of utilizing the Monge problem, which is a reason why it is presented after the Monge problem in many optimal transport textbooks and in this thesis. From the definition alone, it is not immediately clear under what conditions an optimal transport plan for the Kantorovich problem exists. In the next section, we describe a setting in which the Kantorovich problem has a solution.

2.3 Existence of solution to the Kantorovich problem

A solution to the Kantorovich problem exists when X and Y are *Polish spaces*, $\mu \in \mathcal{P}(X), \nu \in \mathcal{P}(Y)$ and $c : X \times Y \to [0, +\infty]$ *lower semicontinuous* (Theorem 2.3.15). Let's break down this statement. What are Polish spaces, and what does it mean for a function to be lower semicontinuous?

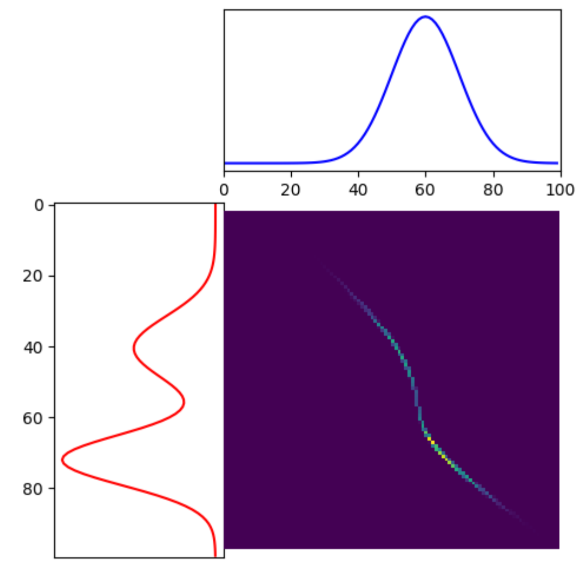


Figure 2.3: In this plot, the the center curve describes how mass from the red distribution is transported into the blue distribution. The gradient of colors from yellow to blue indicates how much mass is being transported from distribution to distribution. Each point corresponds to how much mass is transported from the red to the blue distribution.

Definition 2.3.1 (Polish space)**.** [3, pg. 10] *A* **Polish space** *is a complete separable metric space. In other words, every Cauchy sequence in a Polish space is convergent, and the Polish space contains a countable dense subset.*

Definition 2.3.2 (Lower semicontinuity)**.** [5, Box 1.1] *Let X be a metric space. A function $f : X \to \mathbb{R} \cup \{+\infty\}$ is* **lower semicontinuous** *if for any sequence $x_n \to x$, we have $f(x) \leq \liminf_n f(x_n)$.*

We will need some ideas from probability theory and functional analysis to prove this statement, namely, weak convergence, Weierstrass's theorem, Prokhorov's theorem, and the Banach-Alaoglu theorem.

Definition 2.3.3 (The function spaces C, C_0, C_b)**.** [38, pgs. 10, 33] *Let X be a compact set in a metric space. $C(X)$ is defined as the vector space of continuous functions on X to $\mathbb{R}$ or $\mathbb{C}$ equipped with the supremum norm (sup-norm) $\|f\|_\infty = \sup_{x \in X} |f(x)|$. $C_b(X)$ is the set of continuous bounded functions with the sup-norm. C_0 denotes the functions on X that vanish at infinity.*

Definition 2.3.4 (Weak convergence)**.** [14, 5.1.1] *Let (X, Σ) be a measurable space and $(\mu_i)_{i=1}^{\infty}$ and μ be finite Borel measures on (X, Σ). The sequence of measures $(\mu_i)_{i=1}^{\infty}$* **weakly converges** *to μ (denoted $\mu_i \rightharpoonup \mu$) if*

$$\int_X f \, d\mu_i \to \int_X f \, d\mu \text{ as } i \to \infty \text{ for all } f \in C_b(X),$$

where $C_b(X)$ is the space of continuous and bounded real-valued functions on f.

Theorem 2.3.5 (Weierstrass's theorem)**.** [5, Box 1.1] *If a function $f : X \to \mathbb{R} \cup \{+\infty\}$ is lower semicontinuous and X is a compact metric space, there exists $\overline{x} \in X$ such that $f(\overline{x}) = \min\{f(x) : x \in X\}$.*

Proof. Define $\ell := \inf\{f(x) : x \in X\} \in \mathbb{R} \cup \{-\infty\}$. There exists a minimizing sequence $(x_n)_{n=1}^{\infty}$, which is a sequence of points in X such that $f(x_n) \to \ell$. By compactness, there exists a subsequence (x_{n_k}) of this minimizing sequence which converges to some $\overline{x} \in X$.

Lower semicontinuity implies that $f(\overline{x}) \leq \liminf_n f(x_n) = \ell$, and by the definition of ℓ, $f(\overline{x}) \geq \ell$. Therefore, $\ell = f(\overline{x})$ as desired. □

Definition 2.3.6 (Weak topology)**.** [55, pgs. 369, 370] *Let X be a Polish space. Given a Borel measure $\mu \in \mathcal{P}(X)$, define*

$$S(\mu, \delta; \varphi_1, \ldots, \varphi_n) := \left\{ \nu \in \mathcal{P}(X) : \max_{1 \leq k \leq n} \left| \int_X \varphi_k \, d\nu - \int_X \varphi_k \, d\mu \right| < \delta \right\}$$

*for $n \in \mathbb{N}$, $\varphi_1, \ldots, \varphi_n \in C_b(X)$, $\delta > 0$, and ν Borel. The collection of all such sets $S(\mu, \delta; \varphi_1, \ldots, \varphi_n)$ is the **weak topology** on $\mathcal{P}(X)$.*

In the functional analytic sense, the collection of all such sets $S(\mu, \delta; \varphi_1, \ldots, \varphi_n)$ is actually the weak-$*$ topology, but is called the weak topology by probabilists.

Definition 2.3.7 (Tight probability measures)**.** [6, Definition 5.1] *A set Γ of Borel probability measures on X is* **tight** *if for any $\varepsilon > 0$ there exists a compact subset K of X such that*

$$\mu(K) \geq 1 - \varepsilon \text{ for all } \mu \in \Gamma.$$

Theorem 2.3.8 (Prokhorov's theorem)**.** [6, Theorem 5.2] *If X is a Polish space, then a set $\Gamma \subset \mathcal{P}(X)$ is tight if and only if $\overline{\Gamma}$ is tight, where $\overline{\Gamma}$ is the closure of Γ in the weak topology.*

Theorem 2.3.9 (Banach-Alaoglu Theorem, separable version)**.** [7, Theorem 14.25] *Suppose that X is separable. Then the unit ball of X^* is sequentially compact in the weak-$*$ topology.*

This is equivalent to stating that if X is separable and $(\xi_n)_n$ is a bounded sequence in X^*, then there exists a subsequence (ξ_{n_k}) of $(\xi_n)_n$ weakly converging to some $\xi \in X^*$. We now prove a weaker result of Theorem 2.3.15.

Theorem 2.3.10 (Existence of solution to Kantorovich problem for compact metric spaces)**.** [5, Theorem 1.4] *Let X and Y be compact metric spaces, $\mu \in \mathcal{P}(X), \nu \in \mathcal{P}(Y)$, and $c : X \times Y \to \mathbb{R}$ be a continuous function. Then the Kantorovich problem has a solution.*

Proof of Theorem 2.3.10. To apply Weierstrass's theorem, we show that $\Pi(\mu, \nu)$ is compact and that the map

$$f : (\Pi(\mu,\nu), \|\cdot\|_{TV}) \to \left(\left\{ \int c \, d\gamma : \gamma \in \Pi(\mu,\nu) \right\}, \|\cdot\|_\infty \right)$$

is continuous, where the supremum norm is defined as in Definition 2.3.3 and the total variation norm is defined as in Definition A.1.11.

In particular, our definition of convergence is the weak convergence of probability measures on the space of measures $\mathcal{M}(X \times Y)$ (see Definition A.1.3), which is dual to $C_0(X \times Y)$ [1, Appendix C].

Continuity

The definition of weak convergence allows us to conclude that the map is continuous, as if we assume weak convergence on a set of probability measures (μ_n) converging to a separate but distinct probability measure μ, it follows that for any continuous and bounded function f on $X \times Y$, $\int f d\mu_n$ converges to $\int f d\mu$. Because $c \in C(X \times Y)$ and because it is a function defined on a compact set, it must be bounded, so it is in the space of functions $C_b(X \times Y)$. By the definition of weak convergence, $\mu_n \rightharpoonup \mu \Rightarrow \int c \, d\mu_n \to \int c \, d\mu$.

Compactness

Take a sequence $\gamma_n \in \Pi(\mu, \nu)$. For γ to be an element of $\Pi(\mu, \nu)$, it must be the case that

$$\gamma \in \mathcal{P}(X \times Y) : (\pi_x)_\# \gamma = \mu, (\pi_y)_\# \gamma = \nu,$$

for $\pi_x(x,y) := x, \pi_y(x,y) := y$. All of the terms of this sequence are probability measures, implying that they are bounded in the dual of $C(X \times Y)$, or in other words, the space of signed measures $\mathcal{M}(X \times Y)$. This is because the total variation metric on signed measures cannot exceed 1 for probability measures, so the sequence is bounded. The Banach-Alaoglu theorem (Theorem A.3.3) guarantees the existence of a subsequence $\gamma_{n_k} \rightharpoonup \gamma$. To check that $\gamma \in \Pi(\mu, \nu)$, we fix $\phi \in C(X)$. Because all $\gamma_{n_k} \in \Pi(\mu, \nu)$ it follows that for any choice of n_k,

$$\begin{aligned} &\int \phi(x)\, d\gamma_{n_k} = \int \phi(x)\, d\mu \\ \Rightarrow &\int \phi(x)\, d\gamma = \lim_{n_k \to \infty} \int \phi(x)\, d\gamma_{n_k} = \int \phi(x)\, d\mu, \end{aligned}$$

where the first equality follows from Theorem A.1.7. Let π_x and π_y be the two projections of $X \times Y$ onto X and Y respectively. This implies that $(\pi_x)_\# \gamma = \mu$, and mutatis mutandis, $(\pi_y)_\# \gamma = \nu$. Therefore, $\gamma \in \Pi(\mu, \nu)$ and $\Pi(\mu, \nu)$ is compact. $\square$

Now that we have proven a weaker version of our theorem, we strengthen it by weakening the assumption that the function has to be continuous, and first prove results about lower semicontinuous functions.

Lemma 2.3.11 (Supremum of lower semicontinuous functions is lower semicontinuous)**.** [5, Box 1.5] *If $\{f_\alpha\}_\alpha$ is a family of lower semicontinuous functions on a metric space X, then the supremum of $\{f_\alpha\}_\alpha$ defined as $f(x) := \sup_\alpha f_\alpha(x)$ for all $x \in X$ is lower semicontinuous.*

Proof. Let $(x_n)_{n\in\mathbb{N}}$ be a sequence in X converging to $x \in X$, where x is arbitrarily chosen. By lower semicontinuity,

$$f_\alpha(x) \leq \liminf_n f_\alpha(x_n) \leq \liminf_n f(x_n),$$

where the first inequality follows by lower semicontinuity and the second from the fact that $f_\alpha(x_n) \leq f(x_n))$ by construction. Since $\liminf_n f(x_n)$ is an upper bound for all $f_\alpha(x)$ where $\alpha \in I$, the supremum of $\{f_\alpha\}_\alpha$ must be bounded by $\liminf_n f(x_n)$, i.e.,

$$\sup_\alpha f_\alpha(x) \leq \liminf_n f(x_n) \Leftrightarrow f(x) \leq \liminf_n f(x_n).$$

Therefore, $f(x)$ is lower semicontinuous. $\square$

Lemma 2.3.12. [5, Lemma 1.6] *If $f : X \to \mathbb{R} \cup \{+\infty\}$ is a lower semicontinuous function bounded from below, then the functional $J : \mathcal{M}_+(X) \to \mathbb{R} \cup \{+\infty\}$ defined on the set of positive measures on X through $J(\mu) := \int f \, d\mu$ is lower semicontinuous with respect to the weak convergence of measures.*

Proof. Consider a sequence $(f_k)_k$ of continuous and bounded functions increasing to f. Write $J(\mu) = \sup_k J_k(\mu) := \sup \int f_k \, d\mu$. Every J_k is continuous for the weak convergence, and hence J is lower semicontinuous by Lemma 2.3.11. □

We now weaken the restriction on the function from being continuous to lower semincontinuous.

Theorem 2.3.13. [5, Theorem 1.5] *Let X and Y be compact metric spaces, $\mu \in \mathcal{P}(X), \nu \in \mathcal{P}(Y)$, and $c : X \times Y \to \mathbb{R}$ be lower semicontinuous and bounded from below. Then the Kantorovich problem has a solution.*

This theorem is proved in [5, Theorem 1.5]. With our results about lower semicontinuity established, we now prove our main theorem. As before, we need several lemmas to prove this statement.

Lemma 2.3.14 (Prokhorov for sequences)**.** [5, Box 1.4] *Let $(\mu_n)_{n\in\mathbb{N}}$ be a tight sequence of probability measures over a Polish space (X, d). Then there exists a $\mu \in \mathcal{P}(X)$ and a subsequence (μ_{n_k}) of $(\mu_n)_{n\in\mathbb{N}}$ such that $\mu_{n_k} \rightharpoonup \mu$. Also, if a sequence $(\mu_n)_{n\in\mathbb{N}}$ converges to μ, then it is tight.*

Theorem 2.3.15. [5, Theorem 1.7] *Let X and Y be Polish spaces, $\mu \in \mathcal{P}(X), \nu \in \mathcal{P}(Y)$ and $c : X \times Y \to [0, +\infty]$ lower semicontinuous. Then the Kantorovich problem admits a solution.*

Proof. We can use Prokhorov's theorem for sequences to show that any sequence in $\Pi(\mu, \nu)$ is tight. Fix $\epsilon > 0$ and find two compact sets $K_X \subset X$ and $K_Y \subset Y$ such that $\mu(X \backslash K_X)$ and $\nu(Y \backslash K_Y)$ are less than $\frac{1}{2}\epsilon$. This can always be done because of Prokhorov's theorem, since both of the singleton sets $\{\mu\}$ and $\{\nu\}$ are compact. Therefore, the sets K_X and K_Y

can be constructed with our desired properties. The set $K_X \times K_Y$ is compact in $X \times Y$ and for any $\gamma_n \in \Pi(\mu, \nu)$, we have

$$\begin{aligned}\gamma_n((X \times Y)\backslash(K_X \times K_Y)) &\leq \gamma_n(((X\backslash K_X) \times Y) \cup (X \times (Y\backslash K_Y))) \\ &\leq \gamma_n((X\backslash K_X) \times Y) + \gamma_n(X \times (Y\backslash K_Y)) \\ &= \mu(X\backslash K_X) + \nu(Y\backslash K_Y) < \epsilon\end{aligned}$$

where the first inequality follows from Lemma B.7.6 and the second from Lemma B.7.7. This inequality shows tightness and therefore, compactness, of all sequences in $\Pi(\mu, \nu)$. $\square$

Thus, we have demonstrated that under certain constraints, a solution to the Kantorovich problem exists. We next introduce another natural formulation of the Kantorovich problem, the dual formulation.

2.4 Kantorovich duality

Solving the Kantorovich problem equates to *minimizing the cost incurred* by a company when transporting pastries from bakeries to cafes. The dual Kantorovich problem gets to the same answer but in a different way. Instead of a bread company coordinating the pastry transportation itself, it outsources transportation to a freight company, who charges a fee for their services based on their operational costs. They sell bread at cafes y_j from bakeries x_i. Let $\psi(x)$ be the price at which a basket of bread is bought at bakery x, and $\phi(y)$ the price at which it is sold at cafe y. Defining these two functions allows us to discard the cost function used in the original Kantorovich problem. The price which the bakery and cafe pays for the transport from cafe y_j to bakery x_i is now $\phi(y_j) - \psi(x_i)$ instead of the cost $c(x_i, y_j)$. The bread company does not outsource its transport if their services are at least as expensive as the freight company itself. If the bread company does decide to outsource its transportation, the freight company seeks to *maximize profits* as much as possible. These competing forces of the bread company limiting the cost to $c(x_i, y_j)$ and the freight company trying to maximize its profit lead to the dual Kantorovich problem's solution coinciding with the original Kantorovich problem. The dual Kantorovich problem is as follows:

Definition 2.4.1 (Dual Kantorovich Problem)**.** [3, 5.3] *Find functions $\phi \in L^1(X, \mu)$, $\psi \in L^1(Y, \nu)$ such that the following is attained:*

$$\sup \left\{ \int_Y \phi(y)\, d\nu(y) - \int_X \psi(x)\, d\mu(x) : \phi(y) - \psi(x) \leq c(x,y), \forall (x,y) \in X \times Y \right\}.$$

The shipment of bread must cost less than or equal to the transport cost. Because $\phi(y) - \psi(x) \leq c(x, y)$ for all $(x, y) \in X \times Y$,

$$\sup_{\phi - \psi \leq c} \left\{ \int_Y \phi(y)\, d\nu(y) - \int_X \psi(x)\, d\mu(x) \right\} \leq \inf_{\pi \in \Pi(\mu,\nu)} \left\{ \int_{X \times Y} c(x,y)\, d\pi(x,y) \right\}.$$

Can we show the opposite inequality, implying that solving the dual problem is equivalent to solving the original Kantorovich problem? Does minimizing a transport cost provide us the same value as maximizing an outsourcing company's profit? The following theorem on Kantorovich duality proves that this is the case.

Theorem 2.4.2 (Kantorovich Duality)**.** [4, Theorem 1.3] *Let X and Y be Polish spaces, $\mu \in \mathcal{P}(X)$ and $\nu \in \mathcal{P}(Y)$, and let $c : X \times Y \to [0, \infty]$ be a lower semicontinuous cost function. When $\pi \in \mathcal{P}(X \times Y)$ and $(\varphi, \psi) \in L^1(X, \mu) \times L^1(Y, \nu)$, define*

$$I[\pi] = \int_{X \times Y} c(x,y)\, d\pi(x,y), \quad J(\varphi, \psi) = \int_X \varphi\, d\mu + \int_Y \psi\, d\nu.$$

Let Φ_c be the set of all measurable functions $(\varphi, \psi) \in L^1(X, \mu) \times L^1(Y, \nu)$ satisfying $\varphi(x) + \psi(y) \leq c(x, y)$ for $d\mu$ almost all $x \in X$, $d\nu$ almost all $y \in Y$, and $\Pi(\mu, \nu)$ being defined as in Definition 2.2.1. Then

$$\inf_{\pi \in \Pi(\mu,\nu)} I[\pi] = \sup_{(\varphi,\psi) \in \Phi_c} J(\varphi, \psi).$$

Proof. Let us construct an indicator function for the space $\Pi(\mu, \nu)$. This is a function that takes on the value 0 if its input is in $\Pi(\mu, \nu)$, and ∞ otherwise. In fact, it turns out that

$$F(\pi) = \begin{cases} 0 \text{ if } \pi \in \Pi(\mu,\nu) \\ \infty \text{ otherwise} \end{cases} = \sup_{(\varphi,\psi)} \left[\int_X \varphi\, d\mu + \int_Y \psi\, d\nu - \int_{X \times Y} [\varphi(x) + \psi(y)]\, d\pi(x,y) \right],$$

where the supremum is taken over $(\varphi, \psi) \in C_b(X) \times C_b(Y)$.

To explain why, note that if $\pi \in \Pi(\mu, \nu)$, π is a coupling of μ and ν (Theorem A.1.7). As a result, it follows that

$$\int_X \varphi\, d\mu + \int_Y \psi\, d\nu = \int_{X \times Y} [\varphi(x) + \psi(y)]\, d\pi(x,y)$$

$$\Leftrightarrow \int_X \varphi\, d\mu + \int_Y \psi\, d\nu - \int_{X\times Y} [\varphi(x) + \psi(y)]\, d\pi(x,y) = 0.$$

Now suppose that $\pi \notin \Pi(\mu,\nu)$. This means that for some measurable functions φ and ψ on X and Y respectively, it must be that

$$\int_X \varphi\, d\mu + \int_Y \psi\, d\nu - \int_{X\times Y} [\varphi(x) + \psi(y)]\, d\pi(x,y) = k$$

for some nonzero k. Choose some real nonzero n such that $nk > 0$. Thus,

$$\int_X n\varphi(x)\, d\mu + \int_Y n\psi(y)\, d\nu - \int_{X\times Y} [n\varphi(x) + n\psi(y)]\, d\pi(x,y) = nk.$$

Taking the limit as $n \to \infty$ or $-\infty$ (depending on the sign of k), it follows that if $\pi \notin \Pi(\mu,\nu)$, then

$$\sup_{\varphi,\psi} \left[\int_X \varphi\, d\mu + \int_Y \psi\, d\nu - \int_{X\times Y} [\varphi(x) + \psi(y)]\, d\pi(x,y) \right] = \infty.$$

Now note that

$$\inf_{\pi\in\Pi(\mu,\nu)} I[\pi] = \inf_{\pi\in M_+(X\times Y)} I[\pi] + F(\pi),$$

as the indicator function $F(\pi)$ gets set to 0 by the infimum, where M_+ denotes the set of nonnegative Borel measures. Therefore,

$$\inf_{\pi\in\Pi(\mu,\nu)} I[\pi] = \inf_{\pi\in M_+(X\times Y)} \sup_{(\varphi,\psi)} \Bigg\{ \int_{X\times Y} c(x,y)\, d\pi(x,y) + \int_X \varphi\, d\mu + \int_Y \psi\, d\nu \\ - \int_{X\times Y} [\varphi(x) + \psi(y)]\, d\pi(x,y) \Bigg\}.$$

By applying a "minimax" principle, we can switch the order of the supremum and infimum. A proof of this principle is supplied in [4, pgs. 26, 27]. Therefore,

$$\begin{aligned} \inf_{\pi\in\Pi(\mu,\nu)} I[\pi] &= \sup_{(\varphi,\psi)} \inf_{\pi\in M_+(X\times Y)} \Bigg\{ \int_{X\times Y} c(x,y)\, d\pi(x,y) + \int_X \varphi\, d\mu + \\ &\qquad \int_Y \psi\, d\nu - \int_{X\times Y} [\varphi(x) + \psi(y)]\, d\pi(x,y) \Bigg\} \\ &= \sup \Bigg\{ \int_X \varphi\, d\mu + \int_Y \psi\, d\nu - \sup_{\pi\in M_+(X\times Y)} \int_{X\times Y} [\varphi(x) + \psi(y) - c(x,y)]\, d\pi(x,y) \Bigg\} \end{aligned}$$

Define the function $\zeta(x,y) = \varphi(x) + \psi(y) - c(x,y)$. Let us compute the following:

$$\sup_{\pi\in M_+(X\times Y)} \int_{X\times Y} [\varphi(x) + \psi(y) - c(x,y)]\, d\pi(x,y).$$

If $\zeta(x, y)$ takes on a positive value at some arbitrary point (x_0, y_0), then by choosing π to be the measure $\lambda\delta_{(x_0,y_0)}$ and letting $\lambda \to \infty$, we see that the supremum is infinity. If ζ is nonpositive, i.e. $\zeta(x, y) \leq 0$ for all $(x, y) \in X \times Y$, then the supremum is obtained when $\pi = 0$. Note that ζ being nonpositive is equivalent to the condition $\varphi(x) + \psi(y) \leq c(x, y)$, for all $(x, y) \in X \times Y$. Thus,

$$\sup_{\pi \in M_+(X \times Y)} \int_{X \times Y} [\phi(x) + \psi(y) - c(x, y)]\, d\pi(x, y) = \begin{cases} 0 \text{ if } (\varphi, \psi) \in \Phi_c \\ \infty \text{ otherwise.} \end{cases}$$

Therefore,

$$\inf_{\pi \in \Pi(\mu,\nu)} I[\pi] = \sup_{\pi \in M_+(X \times Y)} \left\{ \int_X \varphi\, d\mu + \int_Y \psi\, d\nu - \begin{cases} 0 \text{ if } (\varphi, \psi) \in \Phi_c \\ \infty \text{ otherwise} \end{cases} \right\} = \sup_{(\varphi,\psi) \in \Phi_c} J(\varphi, \psi).$$

□

Now that we have presented the Monge and Kantorovich problems and the dual Kantorovich problem, we discuss optimal transport between distributions defined on $\mathbb{R}$.

2.5 Monge and Kantorovich for 1-D distributions

As our results on the existence of Kantorovich transport plans apply on abstract metric spaces, we characterize transport plans on 1-dimensional (1-D) histograms to provide a concrete example of optimal transport. We additionally show that discrete transport plans always exist, and in these settings, the Monge and Kantorovich problems are equivalent (when a solution to the Monge problem exists.)

To define an optimal transport plan for 1-D histograms, we need vocabulary to understand their distributions and related functions. In particular, we define the cumulative distribution function for a probability measure and its associated generalized inverse.

Definition 2.5.1 (Cumulative distribution function)**.** [4, pg. 74] [8, pg. 9]

For a measure $\mu \in \mathcal{P}(\mathbb{R})$, its **cumulative distribution function***, or CDF, is defined as*

$$F(x) := \int_{-\infty}^{x} d\mu = \mu((-\infty, x]), \forall x \in \mathbb{R}.$$

For a measure $\pi \in \mathcal{P}(\mathbb{R}^2)$, its **cumulative distribution function** *is defined as*

$$H(x_0, y_0) := \int_{R(x_0,y_0)} d\pi = \pi[R(x_0, y_0)],$$

where $R(x_0, y_0)$ is the rectangle consisting of all points $(x, y) \in \mathbb{R}^2$ such that $(x, y) \in (-\infty, x_0) \times (-\infty, y_0]$.

As the name might suggest, the cumulative distribution function is a non-decreasing function. You can think of the cumulative distribution function as "accumulating" probability as it moves along the x-axis. Intuitively, $F(-\infty) = 0$ as there has been no opportunity to accumulate any probability, and $F(\infty) = 1$ after the entire mass of the probability measure has been accumulated. CDFs directly correspond to probability measures, in that if you have a probability measure there exists a corresponding CDF, and similarly, given a CDF $F(x)$, one can define a probability measure that satisfies $\mu((-\infty, x]) = F(x)$ for all x [17, Theorem 4], [29, Proposition 6.0.2]. We now define the generalized inverse, which is essential to formulating a solution to the Kantorovich problem in the 1-D setting.

Definition 2.5.2 (Generalized inverse of a CDF)**.** [8, pg. 9] *Let F be the cumulative distribution function of some probability measure μ. We define the* **generalized inverse** *of F as a function $F^{[-1]} : [0, 1] \to \mathbb{R}$,*

$$F^{[-1]}(t) = \inf\{x \in \mathbb{R} : F(x) > t\}, \forall t \in [0, 1].$$

The generalized inverse is also called the "quantile function." It is not to be confused with the inverse of a function in the general sense. The function $F^{[-1]}(t)$ is always non-decreasing. In general, we have $F^{[-1]}(F(x)) \geq x$ and $F(F^{[-1]}(t)) \geq t$. If F is invertible then the aforementioned inequalities become equalities.

Proposition 2.5.3. *$F^{[-1]}(t) \leq x$ if and only if $t \leq F(x)$.*

Proof. $F^{[-1]}(t) \leq x$ implies $x \in \{z : F(z) \geq t\}$ and so $t \leq F(x)$. Conversely, $t \leq F(x)$ implies that $x \in \{z : F(z) \geq t\}$ and therefore, $F^{[-1]}(t) \leq x$ since $F^{[-1]}(t)$ is a minimum. □

For similar reasons, it follows that $F^{[-1]}(t) \geq x$ if and only if $t \geq F(x)$. Now that we have defined the generalized inverse, we can use it to explicitly find the joint probability measure that solves the Kantorovich problem for 1-D distributions.

Definition 2.5.4 (Co-monotone transport plan)**.** [5, Definition 2.3] *Let F and G be the cumulative distribution functions of some probability measures μ and ν. Let $\pi^\dagger := (F^{[-1]} \times$*

$G^{[-1]})_{\#}L\lfloor_{[0,1]}$. *This transport plan is called the* **co-monotone transport plan** *between* μ *and* ν.

In fact, the CDF of the co-monotone transport plan is $\min\{F(x), G(y)\}$, which we now prove.

Proposition 2.5.5 (The co-monotone transport plan has CDF $\min\{F(x), G(y)\}$)**.** [5, Proposition 2.2]. *Given* $\mu, \nu \in \mathcal{P}(\mathbb{R})$, *if we set* $\pi^\dagger = (F^{[-1]} \times G^{[-1]})_{\#}L\lfloor_{[0,1]}$, *then* $\pi^\dagger \in \Pi(\mu, \nu)$ *and* $\pi^\dagger((-\infty, x] \times (-\infty, y]) = \min\{F(x), G(y))\}$.

Proof. The fact that $\pi^\dagger \in \Pi(\mu, \nu)$ is a coupling follows from the Knott-Smith optimality criterion [4, Page 77].

$$\begin{aligned}
\pi^\dagger((-\infty, x] \times (-\infty, y]) &= (F^{[-1]} \times G^{[-1]})_{\#}L\lfloor_{[0,1]}((-\infty, x] \times (-\infty, y]) \\
&= L\lfloor_{[0,1]}\Big((F^{[-1]} \times G^{[-1]})^{-1}((-\infty, x) \times (-\infty, y])\Big) \\
&= L\lfloor_{[0,1]}\Big(\{t : F^{[-1]}(t) \leq x \text{ and } G^{[-1]}(t) \leq y\}\Big) \\
&= L\lfloor_{[0,1]}(\{t : F(x) \geq t \text{ and } G(y) \geq t\}) \\
&= \min\{F(x), G(y)\}.
\end{aligned}$$

□

Theorem 2.5.6 (1-D Kantorovich Problem)**.** [8, Theorem 2.1] *Let* $\mu, \nu \in \mathcal{P}(\mathbb{R})$ *with cumulative distribution functions* F *and* G, *respectively. Define a cost function* $c(x, y) = d(x-y)$ *where* d *is convex and continuous. Let* $\pi^\dagger$ *be the measure on* $\mathbb{R}^2$ *with cumulative distribution function* $H(x, y) = \min\{F(x), G(y)\}$, *or in other words, the co-monotone transport plan between* μ *and* ν. *Then* $\pi^\dagger \in \Pi(\mu, \nu)$ *by Proposition 2.5.5 and is optimal for the Kantorovich problem with the specified cost function* c. *Additionally, the optimal transportation cost is*

$$\min_{\pi \in \Pi(\mu,\nu)} \int_{X \times Y} d(x - y)\, d\pi(x, y) = \int_0^1 d(F^{[-1]}(t) - G^{[-1]}(t))\, dt.$$

In order to prove this theorem, we will need Lemma 2.5.8, which is proved in [8, pgs. 12, 13], and the definition of a monotone set:

Definition 2.5.7 (Monotone)**.** [8, pg. 11] *A set* $\Gamma \subseteq \mathbb{R}^2$ *is* **monotone** *with respect to* d *if for all* $(x_1, y_1), (x_2, y_2) \in \Gamma$ *it holds that*

$$d(x_1 - y_1) + d(x_2 - y_2) \leq d(x_1 - y_2) + d(x_2 - y_1).$$

Lemma 2.5.8. [8, Proposition 2.3] *Let $\mu, \nu \in \mathcal{P}(\mathbb{R})$. Assume our cost function is $c(x,y) = d(x-y)$ where d is convex and continuous, and that π^* is an optimal transport plan for the Kantorovich problem. Then for all $(x_1, y_1), (x_2, y_2) \in supp(\pi^*)$, we have*

$$d(x_1 - y_1) + d(x_2 - y_2) \leq d(x_1 - y_2) + d(x_2 - y_1),$$

i.e., the support of $\pi^\dagger$, or the complement of the union of all sets with zero measure, is monotone.

We now continue with the proof of Theorem 2.5.6.

Proof. We first suppose that d is strictly convex (Definition A.4.2) and continuous. By Theorem 2.3.15, there exists some optimal transport plan $\pi^* \in \Pi(\mu, \nu)$ in the Kantorovich sense. Furthermore, by Lemma 2.5.8, $\Gamma = \operatorname{supp}(\pi^*)$ is monotone. We claim that for any $(x_1, x_2), (y_1, y_2) \in \Gamma$ and $x_1 < x_2$ that $y_1 \leq y_2$. Assume that $y_2 < y_1$ and let $a := x_1 - y_1$, $b := x_2 - y_2$, and $\delta := x_2 - x_1$. We know that

$$d(a) + d(b) \leq d(b - \delta) + d(a + \delta).$$

Let $t = \frac{\delta}{b-a}$. It follows that $t \in (0,1)$ and $b - \delta = (1-t)b + ta, a + \delta = tb + (1-t)a$. By strict convexity of d,

$$d(b - \delta) + d(a + \delta) < (1-t)d(b) + td(a) + td(b) + (1-t)d(a) = d(b) + d(a).$$

This is a contradiction, so $y_2 \geq y_1$.

Now we show that $\pi^*((-\infty, x] \times (-\infty, y]) = \min\{F(x), G(y)\}$, or in other words, that $\pi^* = \pi^\dagger$ (due to the 1-1 correspondence between CDFs and measures). Fix $x, y \in \mathbb{R}$. Let $A := (-\infty, x) \times (y, \infty)$, $B := (x, \infty) \times (-\infty, y]$. We know that if $(x_1, y_1), (x_2, y_2) \in \Gamma$ and $x_1 < x_2$ then $y_1 \leq y_2$. This implies that if $(x_0, y_0) \in \Gamma$ then

$$\Gamma \subset \{(x,y) : x \leq x_0, y \leq y_0\} \cup \{(x,y) : x \geq x_0, y \geq y_0\}.$$

Note that one of $\pi(A)$ and $\pi(B)$ has to be zero. Also observe that

$$\begin{aligned}\pi^*((-\infty, x] \times (-\infty, y]) = \min\{&\pi^*(((-\infty, x) \times (-\infty, y]) \cup A), \\ &\pi^*(((-\infty, x) \times (-\infty, y]) \cup B)\}.\end{aligned}$$

Since
$$\pi^*(((-\infty,x)\times(-\infty,y])\cup A)=\pi^*((-\infty,x]\times\mathbb{R})=F(x)$$
$$\pi^*(((-\infty,x)\times(-\infty,y])\cup B)=\pi^*(\mathbb{R}\times(-\infty,y])=G(y),$$

it follows that

$$\pi^*((-\infty,x]\times(-\infty,y])=\min\{F(x),G(y)\}.$$

We now move to the case where d is convex, but not strictly convex. Convex functions can be bounded below by affine functions. Therefore, there exists some a,b such that $d(x)\geq ax+b$. We can check that $f(x)=\frac{1}{2}\sqrt{4+(ax+b)^2}+\frac{1}{2}(ax+b)$ is strictly convex and satisfies $0\leq f(x)\leq 1+d(x)$, which is proven in Lemma B.7.4 in the Appendix.

We define $d_\varepsilon(x)=d(x)+\varepsilon f(x)$. This function is strictly convex (as the sum of a strictly convex function and convex function) and satisfies $d\leq d_\varepsilon\leq(1+\varepsilon)d+\varepsilon$. Let $\pi\in\Pi(\mu,\nu)$. It follows that

$$\begin{aligned}\int_{\mathbb{R}^2} d(x-y)\,d\pi^\dagger(x,y)\leq\int_{\mathbb{R}^2} d_\varepsilon(x-y)\,d\pi^\dagger(x,y)&\leq\int_{\mathbb{R}^2} d_\varepsilon(x-y)\,d\pi(x,y)\\&\leq(1+\varepsilon)\int_{\mathbb{R}^2} d(x-y)\,d\pi(x,y)+\varepsilon.\end{aligned}$$

When we take the limit as ϵ goes to 0, this proves that $\pi^\dagger$ is an optimal transport plan in the Kantorovich sense. Now we show that

$$\int_{\mathbb{R}^2} d(x-y)\,d\pi^\dagger(x,y)=\int_0^1 d(F^{[-1]}(t)-G^{[-1]}(t))\,dt.$$

Using the fact that $\pi^\dagger=(F^{[-1]}\times G^{[-1]})_\# L\lfloor_{[0,1]}$ and the change of variables formula (Definition 2.1.1), we conclude that

$$\int_{\mathbb{R}^2} d(x-y)\,d((F^{[-1]}\times G^{[-1]})_\# L\lfloor_{[0,1]})(x,y)=\int_0^1 d(F^{[-1]}(t)-G^{[-1]}(t))\,dt.$$

□

Now we can examine a consequence of this theorem, Corollary 2.5.10, which allows us to simplify the expression for the optimal transport distance between two probability measures if the cost is simply $c(x,y)=|x-y|$. To address this, we first establish a lemma on the equivalence of two sets that are used in the proof of this corollary.

Lemma 2.5.9. *Define $A_1 \subset \mathbb{R}^2$ as*

$$A_1 = \{(x,t) : \min\{F(x), G(x)\} \le t \le \max\{F(x), G(x)\}, x \in \mathbb{R}\},$$

and $A_2 \subset \mathbb{R}^2$ as

$$A_2 = \{(x,t) : \min\{F^{[-1]}(t), G^{[-1]}(t)\} \le x \le \max\{F^{[-1]}(t), G^{[-1]}(t)\}, t \in [0,1]\}.$$

We prove that $A_1 = A_2$.

Proof. $A_1 \subseteq A_2$: Let $(x_1, t_1) \in A_1$. Note that $t_1 \in [\min\{F(x_1), G(x_1)\}, \max\{F(x_1), G(x_1)\}]$ implies that $t_1 \in [0,1]$, as all CDFs are bounded by 0 and 1. Now we prove that $x_1 \in [\min\{F^{[-1]}(t_1), G^{[-1]}(t_1)\}, \max\{F^{[-1]}(t_1), G^{[-1]}(t_1)\}]$. We consider two cases.

Case 1: $F(x_1) \le G(x_1)$. Therefore, $F(x_1) \le t_1 \le G(x_1)$, and from the properties of the generalized inverse, $F^{[-1]}(t_1) \ge x_1 \ge G^{[-1]}(t_1)$.

Case 2: $F(x_1) \ge G(x_1)$. Therefore, $F(x_1) \ge t_1 \ge G(x_1)$, and from the properties of the generalized inverse, $F^{[-1]}(t_1) \le x_1 \le G^{[-1]}(t_1)$.

Putting these two cases together, it follows that

$$x_1 \in [\min\{F^{[-1]}(t), G^{[-1]}(t)\}, \max\{F^{[-1]}(t), G^{[-1]}(t)\}].$$

$A_1 \supseteq A_2$: Let $(x_2, t_2) \in A_2$. We know that $x_2 \in \mathbb{R}$. We must prove that

$$\min\{F(x_2), G(x_2)\} \le t_2 \le \max\{F(x_2), G(x_2)\}.$$

Case 3: $F^{[-1]}(t_2) \le x_2 \le G^{[-1]}(t_2) \Rightarrow G(x_2) \le t_2 \le F(x_2)$.

Case 4: $G^{[-1]}(t_2) \le x_2 \le F^{[-1]}(t_2) \Rightarrow F(x_2) \le t_2 \le G(x_2)$.

Combining these results, we find that

$$\min\{F(x_2), G(x_2)\} \le t_2 \le \max\{F(x_2), G(x_2)\}. \qquad \square$$

Corollary 2.5.10. [8, Corollary 2.2] *If $c(x,y) = |x-y|$ then the optimal transport distance is the L^1 distance between cumulative distribution functions:*

$$\inf_{\pi \in \Pi(\mu,\nu)} K(\pi) = \int_{\mathbb{R}} |F(x) - G(x)|\, dx.$$

Proof. Define $A \subset \mathbb{R}^2$ as

$$A = \{(x,t) : \min\{F(x), G(x)\} \leq t \leq \max\{F(x), G(x)\}, x \in \mathbb{R}\}.$$

We can also write this set using the definition of the generalized inverse as

$$A = \{(x,t) : \min\{F^{[-1]}(t), G^{[-1]}(t)\} \leq x \leq \max\{F^{[-1]}(t), G^{[-1]}(t)\}, t \in [0,1]\}.$$

Therefore,

$$\lambda(A) = \int_{\mathbb{R}} \int_{\min\{F(x),G(x)\}}^{\max\{F(x),G(x)\}} dt\, dx = \int_0^1 \int_{\min\{F^{[-1]}(t),G^{[-1]}(t)\}}^{\max\{F^{[-1]}(t),G^{[-1]}(t)\}} dx\, dt$$

where λ is the Lebesgue measure on $\mathbb{R}^2$. Note that

$$\int_0^1 \int_{\min\{F^{[-1]}(t),G^{[-1]}(t)\}}^{\max\{F^{[-1]}(t),G^{[-1]}(t)\}} dx\, dt = \int_{\mathbb{R}} \left| F^{[-1]}(t) - G^{[-1]}(t) \right| dt.$$

As $\max\{a,b\} - \min\{a,b\} = |a-b|$,

$$\begin{aligned}
\int_{\mathbb{R}} \int_{\min\{F(x),G(x)\}}^{\max\{F(x),G(x)\}} dt\, dx &= \int_{\mathbb{R}} \min\{F(x), G(x)\} - \max\{F(x), G(x)\}\, dx \\
&= \int_{\mathbb{R}} |F(x) - G(x)|\, dx.
\end{aligned}$$

□

Finally we show that if μ does not give mass to atoms, then there exists a Monge map between 1-D probability distributions.

Corollary 2.5.11. [8, Corollary 2.2] *If μ does not give mass to atoms then $\min_{T:T_{\#}\mu=\nu} M(T)$, the map $T^{\dagger} = G^{-1} \circ F$ is a minimiser to Monge's optimal transport problem, and*

$$\inf_{T:T_{\#}\mu=\nu} M(T) = M(T^{\dagger}).$$

Proof. Recall that $T_{\#}\mu = G^{-1}_{\#}(F_{\#}\mu)$. To show this result, we prove that $G^{[-1]}_{\#} L\lfloor_{[0,1]} = \nu$ and $L\lfloor_{[0,1]} = F_{\#}\mu$. Combining these results shows that $T_{\#}\mu = \nu$.
$G^{[-1]}_{\#} L\lfloor_{[0,1]} = \nu$:

$$\begin{aligned}
G^{[-1]}_{\#} L\lfloor_{[0,1]}((-\infty, y]) &= L\lfloor_{[0,1]}({G^{[-1]}}^{-1}(-\infty, y]) = L\lfloor_{[0,1]}(\{t : G^{[-1]}(t) \leq y\}) \\
&= L\lfloor_{[0,1]}(\{t : G(y) \geq t\}) = G(y) = \nu((-\infty, y]).
\end{aligned}$$

$L\lfloor_{[0,1]} = F_{\#}\mu$:

Note that $F^{-1}([0,t]) = (-\infty, x_t]$.

$$\forall t \in (0,1), F_{\#}\mu([0,t]) = \mu(\{x : F(x) \leq t\}) = \mu(\{x : x \leq x_t\}) = F(x_t) = t.$$

Putting these results together, we note that by Theorem 2.5.6,

$$\begin{aligned}
\inf_{\pi \in \Pi(\mu,\nu)} K(\pi) &= \int_0^1 d(F^{[-1]}(t) - G^{[-1]}(t))\, dt \\
&= \int_{\mathbb{R}} d(x - G^{-1}(F(x)))\, d\mu(x) \\
&= \int_{\mathbb{R}} d(x - T^{\dagger}(x))\, d\mu(x) \\
&\geq \inf_{T : T_{\#}\mu = \nu} M(T).
\end{aligned}$$

We know that in general

$$\inf_{T : T_{\#}\mu = \nu} M(T) \geq \min_{\pi \in \Pi(\mu,\nu)} K(\pi).$$

Therefore, the minima of MP and KP are the same, and $T^{\dagger}$ is an optimal Monge map. $\square$

Chapter 3

Wasserstein distances and spaces

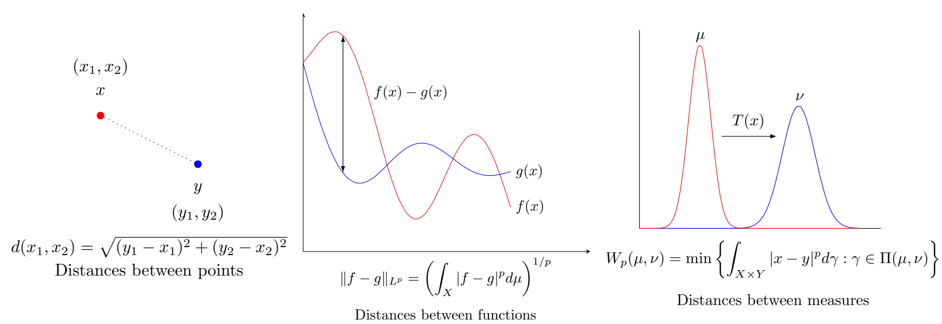


Figure 3.1: Just like how the Euclidean distance defines a distance between two points in $\mathbb{R}^n$, one can quantify distances between more complex objects. For instance, the L^p norm of the difference of two functions defines a notion of distance between functions, and the Wasserstein distance W_p defines a distance on probability measures. The p-Wasserstein distance W_p quantifies the extent to which probability measures differ.

Up to this point, we have discussed the Monge and Kantorovich problems and when there exist solutions for such problems. Solving the Monge or Kantorovich problem provides a cost-minimizing strategy for transporting mass. A natural next question to ask is "After we quantify the cost incurred by moving mass along the most efficient route, what can we do with it?" To answer this question, we introduce the **Wasserstein distance**, a metric on the space of probability distributions with bounded pth moment $\mathcal{P}_p(X)$ that is inspired by the Kantorovich problem. We will prove that the Wasserstein distance satisfies the properties of a metric and compare it to the **Kullback-Leibler divergence**, another way to quantify differences between distributions. While we will not fully develop such concepts, we will provide a sneak peek into the geometric and topological properties that make the

Wasserstein distance particularly useful for comparing distributions. Namely, we show that the space $(\mathcal{P}_p(X), W_p)$ is a geodesic space, where a length-minimizing path can be drawn from a starting measure μ to a target measure ν. This provides a "time-dependent" version of optimal transport known as McCann's **displacement interpolation**. To start, we first define the Wasserstein distance and the set it operates on.

Definition 3.0.1 (Wasserstein space)**.** [5, pg. 159] *Consider the space of probability measures on a Polish metric space (X, d) with bounded pth moment for $p \geq 1$, i.e.*

$$\mathcal{P}_p(X) := \left\{ \mu \in \mathcal{P}(x) : \int_X |x|^p \, d\mu(x) < \infty \right\}.$$

*The p-***Wasserstein distance** *is defined for $\mu, \nu \in \mathcal{P}_p(X)$ as the following:*

$$W_p(\mu, \nu) := \min_{\gamma \in \Pi(\mu,\nu)} \left\{ \int_{X \times X} |x - y|^p \, d\gamma \right\}^{1/p}.$$

In other words, $W_p(\mu, \nu)$ is the pth root of the minimum transport cost for the cost $c(x, y) = |x - y|^p$. The set and metric $(\mathcal{P}_p(X), W_p)$ constitute a **Wasserstein space**. Some results from previous chapters are immediately applicable here when studying Wasserstein distances between simple objects, such as one-dimensional probability distributions. In the simplest case, if we fix $x, y \in X$ and and consider the Dirac measures δ_x and δ_y, then $W_p(\delta_x, \delta_y) = d(x, y)$, which is a sensible way to compare distances between point masses. When we generalize to 1-D probability distributions, we can use the machinery established in previous chapters. Let $\mu, \nu \in \mathcal{P}(\mathbb{R})$ with cumulative distribution functions F and G respectively. Then the Wasserstein distance between the measures μ, ν is

$$W_p(\mu, \nu) := \left(\int_0^1 |F^{[-1]}(t) - G^{[-1]}(t)|^p \, dt \right)^{1/p}.$$

If $p = 1$, then

$$W_1(\mu, \nu) := \int_{\mathbb{R}} |F(x) - G(x)| \, dx.$$

These results directly follow from Theorem 2.5.6 and Corollary 2.5.10.

3.1 Wasserstein distance is a metric

In this section, we prove that the Wasserstein distance is a metric, taking special care to show the triangle inequality holds. The proof of the triangle inequality for Wasserstein distances requires the use of the gluing lemma and disintegration of measures.

Theorem 3.1.1 (Disintegration of measures)**.** [4, Lemma 7.6] *Let X, Y be Polish spaces, $\pi \in \Pi(\mu, \nu)$. There exists a family of probability measures $\{\pi_x\}_{x\in X}$ contained in $\mathcal{P}(Y)$ that "disintegrate" π, i.e., for all $u \in C_b(X \times Y)$,*

$$\pi = \int_X (\delta_x \otimes \pi_x)\, d\mu,$$

$$\int_{X\times Y} u(x,y)\, d\pi(x,y) = \int_X \left[\int_Y u(x,y)\, d\pi_x(y)\right] d\mu(x),$$

$$\pi(E) = \int_X \pi_x[A_x]\, d\mu(x) \text{ where } A_x = \{y \in Y : (x,y) \in A\},$$

where $\otimes$ denotes the product measure. The proof of this theorem can be found in [10]. Please see Appendix A.1.1 for an intuitive visualization of the disintegration of measure. The disintegration of measure is used to construct a probability measure on the product space $X_1 \times X_2 \times X_3$ with marginals defined on $X_1 \times X_2$ and $X_2 \times X_3$.

Lemma 3.1.2 (Gluing lemma)**.** [4, Lemma 7.6] *Let μ_1, μ_2, μ_3 be three probability measures on Polish spaces X_1, X_2, X_3. Let $\pi_{12} \in \Pi(\mu_1, \mu_2)$ and $\pi_{23} \in \Pi(\mu_2, \mu_3)$ be two transport plans. There exists a probability measure $\pi \in \mathcal{P}(X_1 \times X_2 \times X_3)$ with marginals π_{12} on $X_1 \times X_2$ and π_{23} on $X_2 \times X_3$.*

Proof. By disintegration of measures, we decompose π_{12} and π_{23} with respect to the measure μ_2, obtaining the families of measures $\{\pi_{12:x_2}\}_{x_2\in X_2}$ defined on X_1 and $\{\pi_{23:x_2}\}_{x_2\in X_2}$ defined on X_3 respectively. In other words,

$$\pi_{12} := \int_{X_2} \pi_{12:x_2} \otimes \delta_{x_2}\, d\mu_2(x_2) \text{ and } \pi_{23} := \int_{X_2} \delta_{x_2} \otimes \pi_{23:x_2}\, d\mu_2(x_2).$$

We define the measure $\pi \in \mathcal{P}(X_1 \times X_2 \times X_3)$ as

$$\pi := \int_{X_2} \pi_{12:x_2} \otimes \delta_{x_2} \otimes \pi_{23:x_2}\, d\mu_2(x_2).$$

To show that $\pi \in \mathcal{P}(X_1 \times X_2 \times X_3)$ has marginals π_{12} on $X_1 \times X_2$ and π_{23} on $X_2 \times X_3$, it suffices to show that for some measurable $A \subseteq X_1$, $B \subseteq X_2$, and $C \subseteq X_3$, $\pi(A \times B \times X_3) = \pi_{12}(A \times B)$ and $\pi(X_1 \times B \times C) = \pi_{23}(B \times C)$. Observe that

$$\pi(A \times B \times X_3) = \int_{X_2} (\pi_{12:x_2} \otimes \delta_{x_2} \otimes \pi_{23:x_2})(A \times B \times X_3)\, d\mu_2(x_2)$$

$$
\begin{aligned}
&= \int_{X_2} [\pi_{12:x_2}(A) \cdot \delta_{x_2}(B) \cdot \pi_{23:x_2}(X_3)]\, d\mu_2(x_2) \\
&= \int_{X_2} [\pi_{12:x_2}(A) \cdot \delta_{x_2}(B) \cdot 1]\, d\mu_2(x_2) \\
&= \int_{X_2} [\pi_{12:x_2}(A) \cdot \delta_{x_2}(B) \cdot 1]\, d\mu_2(x_2) \\
&= \int_{X_2} (\pi_{12:x_2} \otimes \delta_{x_2})(A \times B)\, d\mu_2(x_2) = \pi_{12}(A \times B).
\end{aligned}
$$

The proof that $\pi(X_1 \times B \times C) = \pi_{23}(B \times C)$ follows similarly. □

Now that we have constructed the measure π, we can apply it to the proof of the triangle inequality for the Wasserstein distance. Because all of the Wasserstein distances are integrals with respect to different measures, we cannot directly compare their integrands. Creating the measure π allows us to express $W_p(\mu_1, \mu_3), W_p(\mu_1, \mu_2)$, and $W_p(\mu_2, \mu_3)$ as integrals with respect to π, a single measure.

Lemma 3.1.3. *W_p satisfies the triangle inequality.* [4, pgs. 208, 209]

Proof. Armed with the gluing lemma, we can now construct a proof that the triangle inequality holds for the p-Wasserstein distance. In other words, for probability measures μ_1, μ_2, and μ_3 defined on X,

$$W_p(\mu_1, \mu_3) \leq W_p(\mu_1, \mu_2) + W_p(\mu_2, \mu_3).$$

We utilize the fact that the L^p norm (Appendix A.1.8) satisfies the triangle inequality (by Minkowski's inequality, Definition A.1.9), and that the fact that the $p-$Wasserstein distance minimizes the integral $\int_{X \times X} c(x, y) d\pi$. Let μ_1, μ_2, and μ_3 be defined as in the gluing lemma, where $X_1 = X_2 = X_3 = X$, as well as the coupling measure π. We choose $\pi_{12} \in \Pi(\mu_1, \mu_2)$ and $\pi_{23} \in \Pi(\mu_2, \mu_3)$ to be optimal transport plans, and also let π_{13} be the marginal of π on $X_1 \times X_3$, which is defined as

$$\pi_{13} := \int_{X_2} \pi_{12:x_2} \otimes \pi_{23:x_2}\, d\mu_2(x_2).$$

To prove π_{13} is a marginal of π, we show that $\pi(A \times X_2 \times C) = \pi_{13}(A \times C)$.

$$\pi_{13}(A \times C) = \int_{X_2} \pi_{12:x_2} \otimes \pi_{23:x_2}\, d\mu_2(x_2)$$

$$= \int_{X_2} [\pi_{12:x_2} \otimes \delta_{x_2} \otimes \pi_{23:x_2}](A \times X_2 \times C)\, d\mu_2(x_2) = \pi(A \times X_2 \times C).$$

Now that we have shown that π_{13} is a marginal, we prove that W_p satisfies the triangle inequality.

$$\begin{aligned}
W_p(\mu_1, \mu_3) &\leq \left(\int_{X_1 \times X_3} |x_1 - x_3|^p \, d\pi_{13} \right)^{1/p} = \left(\int_{X_1 \times X_2 \times X_3} |x_1 - x_3|^p \, d\pi(x_1, x_2, x_3) \right)^{1/p} \\
&= \|x_1 - x_3\|_{L^p(X_1 \times X_2 \times X_3, \pi)} \leq \|x_1 - x_2\|_{L^p(X_1 \times X_2 \times X_3, \pi)} + \|x_2 - x_3\|_{L^p(X_1 \times X_2 \times X_3, \pi)} \\
&= \left(\int_{X_1 \times X_2 \times X_3} |x_1 - x_2|^p \, d\pi(x_1, x_2, x_3) \right)^{1/p} + \left(\int_{X_1 \times X_2 \times X_3} |x_2 - x_3|^p \, d\pi(x_1, x_2, x_3) \right)^{1/p} \\
&= \left(\int_{X_1 \times X_2} |x_1 - x_2|^p \, d\pi_{12}(x_1, x_2) \right)^{1/p} + \left(\int_{X_2 \times X_3} |x_2 - x_3|^p \, d\pi_{23}(x_2, x_3) \right)^{1/p} \\
&= W_p(\mu_1, \mu_2) + W_p(\mu_2, \mu_3).
\end{aligned}$$

□

Theorem 3.1.4. [5, Proposition 5.1] *The Wasserstein distance is a metric.*

Proof. By construction, $W_p \geq 0$. Also, $W_p(\mu, \nu) = 0$ implies that there exists $\gamma \in \Pi(\mu, \nu)$ such that $\int |x - y|^p d\gamma = 0$. Such a $\gamma \in \Pi(\mu, \nu)$ is concentrated on $\{x = y\}$. This implies $\mu = \nu$, since for any test function ϕ (i.e. an infinitely differentiable function with compact support) we have

$$\int \phi \, d\mu = \int \phi(x) \, d\gamma = \int \phi(y) \, d\gamma = \int \phi \, d\nu.$$

The Wasserstein distance is clearly symmetric because the Kantorovich problem for cost function $c(x, y) = |x - y|^p$ does not change the infimum of the functional $I[\pi]$ as defined in Theorem 2.4.2. Finally, by Lemma 3.1.3, we have that the Wasserstein distance satisfies the triangle inequality. Thus, the $p-$Wasserstein distance is a metric. □

The Wasserstein distance's metric properties make it a useful tool for measuring distances between probability distributions. In the next section, we will discuss how the Wasserstein distance compares to the Kullback-Leibler divergence, another measure of similarity between probability distributions.

3.2 Wasserstein distance versus the Kullback-Leibler divergence

Besides the $1-$Wasserstein distance, also known as the "earth-mover's distance," a common method used to compare probability distributions is the Kullback-Leibler (KL) divergence.

The KL divergence is a member of a class of distance functions called φ-divergences, which are not true distances [33].

Definition 3.2.1 (Kullback-Leibler divergence)**.** [11, Definition 2.26, 8.46] *For discrete probability distributions P and Q defined on the same sample space X, the* **Kullback-Leibler divergence***, or relative entropy, is defined as*

$$D_{KL}(P\|Q) := \sum_{x \in X} P(x) \log\left(\frac{P(x)}{Q(x)}\right).$$

For continuous probability distributions P and Q defined on the same sample space X with probability density functions p and q respectively, the **Kullback-Leibler divergence** *is defined as*

$$D_{KL}(p\|q) := \int_{\mathbb{R}} p(x) \log\left(\frac{p(x)}{q(x)}\right) dx.$$

For probability measures μ and ν defined on X, where $\mu \ll \nu$, the **Kullback-Leibler divergence** *is defined as*

$$D_{KL}(\mu\|\nu) := \int_X \log\left(\frac{d\mu}{d\nu}\right) d\mu,$$

where $d\mu/d\nu$ is the **Radon-Nikodym derivative** *of μ with respect to ν. If μ is not absolutely continuous with respect to ν, then $D_{KL}(\mu\|\nu) = \infty$.*

In the context of statistics, the Kullback-Leibler divergence $D_{KL}(p\|q)$ measures the inefficiency of approximating the true distribution of data p with a theoretical distribution q. It is not, however, a metric, as the KL divergence does not satisfy symmetry. There is no *a priori* reason why we would expect the comparison of $D_{KL}(P\|Q)$ (the degree of "surprise" of witnessing Q given that the true distribution is P) to be equal to $D_{KL}(Q\|P)$ (the degree of "surprise" of witnessing P given that the true distribution is Q).

Further differences between the Kullback-Leibler divergence and the Wasserstein distance include the following:

- KL divergences can be defined on sets that are not metric spaces. For instance, if one defines a discrete probability distribution on the set $\{a, b, c, d, e\}$ with no obvious metric between any of the letters, the KL divergence can still be calculated, but the Wasserstein distance cannot.

- The *geometry* of KL and Wasserstein are different. Wasserstein distances change as you move masses farther from each other, whereas the Kullback-Leibler is agnostic to translation of mass. For instance, consider the sequence of Dirac measures $\left(\delta_{\frac{1}{n}}\right)_{n\in\mathbb{N}}$ defined on $(\mathbb{R}, \mathcal{B})$, as illustrated in Figure 3.2. One would expect that just as $\left(\delta_{\frac{1}{n}}\right)_{n\in\mathbb{N}}$ converges weakly to δ_0, a measure of distance between $\delta_{1/n}$ and δ_0 would similarly converge to zero. This illustrates how the Wasserstein distance **metrizes weak convergence,** i.e., that weak convergence to a measure and convergence to zero Wasserstein distance are equivalent conditions.

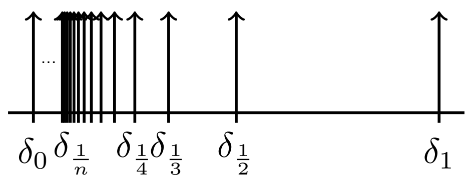


Figure 3.2: $W_p\left(\delta_0, \delta_{\frac{1}{n}}\right) \to 0$ as $n \to \infty$. However, $D_{KL}\left(\delta_0 \| \delta_{\frac{1}{n}}\right) = +\infty$ for all $n \in \mathbb{N}$ because δ_0 is not absolutely continuous with respect to $\delta_{1/n}$.

We may formalize metrization of weak convergence in the following theorem.

Theorem 3.2.2 (Wasserstein convergence metrizes weak convergence)**.** [3, Theorem 6.9] *Let* (X, d) *be a Polish space and* $p \in [1, \infty)$. *If* $(\mu_k)_{k\in\mathbb{N}}$ *is a sequence of measures in* $\mathcal{P}_p(X)$ *and* $\mu \in \mathcal{P}(X)$, *then* μ_k *converges weakly in* $\mathcal{P}_p(X)$ *to* μ *if and only if* $W_p(\mu_k, \mu) \to 0$.

This theorem in effect says that weak convergence, one notion in which measures can get arbitrarily "close" to each other, coincides with a geometric notion of closeness as determined by a metric (in this case, the Wasserstein metric).

While the Wasserstein distance metrizes weak convergence, there are other metrics that do so as well, namely the Lévy-Prokhorov distance, the bounded Lipschitz distance, weak-$*$ distance, and the Toscani distance [3, 6.5-6.8]. A final advantage that Wasserstein distances afford us over the previously mentioned metrics is their rich geometry. We will discuss this further in the subsequent sections and see how it allows us to gain a time-dependent and differential formulation of optimal transport.

3.3 Displacement interpolation

To get a richer picture of optimal transport, we now view the Wasserstein space $(\mathcal{P}_p(X), W_p)$ through the lens of differential geometry. Up to this point, we have discussed optimal transport in terms of creating a cost-minimizing transport plan between distributions. One way to think about the transport plan is that it "transforms" a probability distribution directly into another. Instead of transporting a distribution onto another in one step, can we define a way to more gradually transition from one distribution to another? Curves between probability measures in Wasserstein spaces represent ways to transition between probability measures, which we refer to as examples of **displacement interpolation**. We first establish the vocabulary needed to characterize such curves.

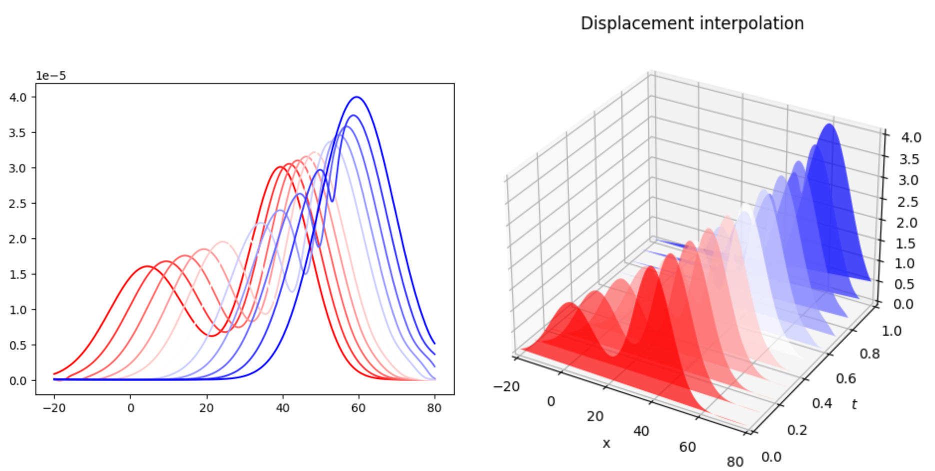


Figure 3.3: Displacement interpolation between two measures generates a curve of measures $(\mu_t)_{t\in[0,1]}$ that are "averages" between the two measures depending on the value of t. The red distribution is the initial distribution, and as the time parameter t goes from 0 to 1, the distribution transforms into the final blue distribution.

3.3.1 Differential geometry preliminaries

Some of the most elementary objects in differential geometry are curves. A continuous function, ω, defined on defined on an interval, commonly $[0, 1]$, and valued in a metric space (X, d) is called a **curve**. For a curve $\omega : [0, 1] \to (X, d)$, the **length** of ω is defined as

$$\ell(\omega) := \sup \left\{ \sum_{k=0}^{n-1} d(\omega(t_k), \omega(t_{k+1})) : n \geq 1, 0 = t_0 < t_1 \cdots < t_n = 1 \right\}.$$

This definition of length does not require the curve to be absolutely continuous [16, Definition 9.3]. A curve $\omega : [0,1] \to (X,d)$ is **absolutely continuous** if there exists $g \in L^1(0,1)$ such that

$$d(\omega(x), \omega(y)) \leq \int_x^y g(t)\, dt$$

for any $x, y \in X$. The set of all such curves is denoted by $AC([0,1];X)$ [16, Definition 9.1]. If $\omega : [0,1] \to (X,d)$ is a curve valued in the metric space (X,d), we define the **metric derivative** of ω at time t denoted by $|\omega'|(t)$ through

$$|\omega'|(t) := \lim_{h \to 0} \frac{d(\omega(t+h), \omega(t))}{|h|}.$$

For absolutely continuous curves, length can be defined as follows: $\ell(\omega) := \int_a^b |\omega'|(t) dt$.

The notion of metric derivative allows us to "differentiate" curves as we would other functions. If we are to think of a curve as parameterizing the position of a particle moving in time, the metric derivative corresponds to the speed of the particle [16, Theorem 9.2]. The endpoints of curves are not necessarily connected by the shortest possible path, i.e., the distance between the endpoints. Geodesics are curves that represent the shortest path between two points in a space, and generalize the notion of "straight lines" to more abstract contexts [16, Definition 9.7]. Let $\omega \in AC([a,b];X)$. The curve ω is a **geodesic** if $\ell(\omega) = d(\omega(a), \omega(b))$. If a geodesic satisfies $d(\omega(t), \omega(s)) = |t-s| d(\omega(0), \omega(1))$ for all $s, t \in [0,1]$, it is called a **constant speed geodesic.** We can also use curves and geodesics to characterize spaces and define metrics. A space (X,d) is a **length space** if

$$d(x,y) = \inf\{\ell(\omega) : \omega \in AC(X), \omega(0) = x, \omega(1) = y\}$$

or a **geodesic space** if

$$d(x,y) = \min\{\ell(\omega) : \omega \in AC(X), \omega(0) = x, \omega(1) = y\}.$$

In other words, a space is a length space if the distance between two points obtained by taking the infimum over all possible paths between them, and a geodesic space if the infimum is actually attained. We now show that Wasserstein spaces are geodesic spaces.

3.3.2 Geodesics in Wasserstein spaces connect probability measures

The following theorem characterizes geodesic curves in Wasserstein spaces. The curve of probability measures defined by this theorem is called **McCann's displacement interpolation**. The takeaway is that for every pair of measures μ, ν in the space $\mathcal{P}_p(\Omega)$, for convex Ω, there is a unique constant speed geodesic that connects these two measures of the form $\mu_t = ((1-t)\mu + t\nu)_{\#}\gamma$, where $t \in [0,1]$.

Theorem 3.3.1. [5, Theorem 5.27] *Let Ω be convex, $\mu, \nu \in \mathcal{P}_p(\Omega)$ and $\gamma \in \Pi(\mu, \nu)$ an optimal transport plan for the cost $c(x,y) = |x-y|^p$. Define $\pi_t : \Omega \times \Omega \to \Omega$ through $\pi_t(x,y) = (1-t)x + ty$. (By doing so, we ensure that $(\pi_0)_{\#}\gamma = \mu$ and $(\pi_1)_{\#}\gamma = \nu$.) Then the curve $\mu_t := (\pi_t)_{\#}\gamma$ is a constant speed geodesic in $\mathcal{W}_p$ connecting $\mu_0 = \mu$ to $\mu_1 = \nu$. As a result, $W_p(\Omega)$ is a geodesic space.*

Proof. Define $\gamma_t^s = (\pi_t, \pi_s)_{\#}\gamma \in \Pi(\mu_t, \mu_s)$. Note that $W_p(\mu_t, \mu_s) \leq \left(\int |x-y|^p \, d\gamma_t^s\right)^{1/p}$ because the Wasserstein distance is the optimal transport cost for any probability measure in $\Pi(\mu_t, \mu_s)$. Note that by using Definition 2.1.1,

$$
\begin{aligned}
&\left(\int |x-y|^p \, d\gamma_t^s\right)^{1/p} \\
&\quad = \left(\int |\pi_t(x,y) - \pi_s(x,y)|^p \, d\gamma\right)^{1/p} = \left(\int |(1-t)x + ty - (1-s)x - sy|^p \, d\gamma\right)^{1/p} \\
&\quad = \left(\int |(t-s)(x-y)|^p \, d\gamma\right)^{1/p} = |t-s| \left(\int |x-y|^p \, d\gamma\right)^{1/p} = |t-s| W_p(\mu, \nu).
\end{aligned}
$$

This implies that $W_p(\mu_t, \mu_s) \leq |t-s| W_p(\mu, \nu)$ for all $0 \leq s, t \leq 1$. Because this is the case, we also find that

$$W_p(\mu, \nu) \leq W_p(\mu, \mu_t) + W_p(\mu_t, \mu_s) + W_p(\mu_s, \nu) \leq W_p(\mu, \nu)(t + (s-t) + (1-s)) = W_p(\mu, \nu).$$

This implies that

$$W_p(\mu, \mu_t) + W_p(\mu_t, \mu_s) + W_p(\mu_s, \nu) = W_p(\mu, \nu).$$

We also know that

$$W_p(\mu, \mu_t) \leq |t| W_p(\mu, \nu)$$

$$W_p(\mu_t, \mu_s) \leq |s - t| W_p(\mu, \nu)$$
$$W_p(\mu_s, \nu) \leq |1 - s| W_p(\mu, \nu).$$

Without loss of generality, let $W_p(\mu, \mu_t) < |t| W_p(\mu, \nu)$. If this is true, then this implies that

$$W_p(\mu, \mu_t) + W_p(\mu_t, \mu_s) + W_p(\mu_s, \nu) < W_p(\mu, \nu),$$

which is a contradiction. Repeating this argument for the other two inequalities shows that $W_p(\mu_t, \mu_s) = |s - t| W_p(\mu, \nu)$ for $0 \leq s, t \leq 1$ as desired. □

A final note on this proof is that when an optimal transport map T exists between μ and ν, then the curve $(\mu_t)_{t \in [0,1]}$ is equal to $((1-t)\mathrm{Id} + tT)_{\#}\mu$, where Id corresponds to the identity transformation [56]. It follows that for every optimal coupling γ, there is a corresponding geodesic that can be defined as a curve of measures $(\mu_t)_{t \in [0,1]} = (\pi_t)\#\gamma$. Where in the original "static" Kantorovich problem as discussed in Chapter 2, a coupling measure defines the optimal transport between probability measure by minimizing the integral of a cost function, optimal transport in the time-dependent setting is now described by a geodesic, a curve minimizing the distance traversed between two measures μ_0 and μ_1 in $(\mathcal{P}_p(\Omega), W_p)$. Now that we have established that static optimal couplings correspond to geodesics in the space $(\mathcal{P}_p(\Omega), W_p)$, we can discuss **dynamic transport plans** in informal terms by considering the energy expenditure along curves in Wasserstein spaces. Here, the idea of optimal transport between two measures is captured as a curve of measures once again, except instead of minimizing distance, a dynamic transport plan minimizes kinetic energy. Drawing on the partial differential equations used in fluid dynamics, researchers such as Benamou and Brenier realized that such methods could also be used to reformulate the optimal transport problem. A more comprehensive discussion can be found in [4] or [5].

3.3.3 Dynamic optimal transport and Benamou-Brenier

The celebrated **Benamou-Brenier** reformulation of the optimal transport problem relates the the 2-Wasserstein distance to solutions of the continuity equation, which is a partial differential equation that models the flow or transport of mass. The continuity equation stands in for the traditional conservation of mass restriction in the Kantorovich problem.

To define the Benamou-Brenier formula, we need to define μ and ν as compactly supported probability measures on $\mathbb{R}^n$, which can be thought of as "density functions" of an ensemble of weighted particles at times 0 and 1. We will examine sequences of absolutely continuous measures $(\rho_t)_{t\in[0,1]}$ as in displacement interpolation that are defined such that $\rho_0 = \mu$ and $\rho_1 = \nu$. Let $X(t)$ represent the position of a particle. We also define the velocity field, a function $v : \mathbb{R}^n \times [0,1]$ as a function that assigns a velocity vector to each position at each point in time t as the time derivative of $X(t)$, i.e., $\frac{d}{dt}X(t) = v_t(X(t))$. When we refer to the velocity field, we set $v_t(x) := v(x,t))$. The density of particles at time t evolves as a weak solution of the continuity equation [3, Theorem 5.34]. At time t we can define the kinetic energy of the collection of particles: Disregarding the factor $\frac{1}{2}$, the kinetic energy is

$$E(t) = \int_{\mathbb{R}^n} \rho_t(x)|v_t(x)|^2\,dx.$$

We obtain the **action** of a pair $(\rho_t, v_t(x))_{t\in[0,1]}$ solving the continuity equation by taking the integral of the kinetic energy with respect to time, i.e.,

$$A[\rho, v] = \int_0^1 E(t)\,dt = \int_0^1 \left(\int_{\mathbb{R}^n} \rho_t(x)|v_t(x)|^2\,dx\right) dt$$

Finally, the continuity equation is $\frac{\partial \rho_t}{\partial t} + \nabla \cdot (v_t\rho_t) = 0$, which describes the time evolution of the measures $(\rho_t)_{t\in[0,1]}$ acted on by the velocity field $v_t(x)$. Intuitively, the term $\frac{\partial \rho_t}{\partial t}$ corresponds to the change in mass of the system, and the divergence term $\nabla \cdot (v_t\rho_t)$ corresponds to the difference in inward and outward mass flow.

Theorem 3.3.2 (Benamou-Brenier formula)**.** [54] *Given $\rho_0, \rho_1 \in \mathcal{P}(\mathbb{R}^n)$ having smooth densities with respect to L^n and bounded supports, then*

$$W_2(\rho_0, \rho_1)^2 = \min_{(\rho_t, v_t)} A[\rho_t, v_t],$$

for (ρ_t, v_t) weakly satisfying (defined in Appendix) the continuity equation and where $\rho_0 = \mu, \rho_1 = \nu$.

A proof for this theorem can be found in [4, Theorem 8.1]. The Benamou-Brenier theorem serves as inspiration for thinking about Wasserstein spaces from a Riemannian geometry standpoint. Felix Otto developed a system for viewing the space of probability measures

$\mathcal{P}_2(\mathbb{R}^n)$ equipped with the 2-Wasserstein distance $(\mathcal{P}_2(\mathbb{R}^n), W_2)$ as an infinite-dimensional Riemannian manifold, which is referred to as the **Otto calculus**. In particular, for $\mu \in \mathcal{P}_2(\mathbb{R}^n)$, the **tangent bundle** of μ denoted T_μ is defined as $\overline{\{\nabla\psi : \psi : \mathbb{R}^n \to \mathbb{R}\}}^{L^p(\mu, \mathcal{P}(\mathbb{R}^n))}$ when ψ are smooth, compactly supported functions. The corresponding norm on the tangent space T_μ is the $L^2(\mu)$ norm, which induces an inner product. Indeed, Otto shows that the induced metric d on $\mathcal{P}_2(\mathbb{R}^n)$ with the tangent bundles and inner product previously specified is the 2-Wasserstein distance, i.e., the Benamou-Brenier formula [57]. The Otto calculus has inspired other innovations in the field of optimal transport such as the **JKO scheme**, which relates the Fokker-Planck equation, a partial differential equation that models the probability density of a particle's velocity subject to random diffusion, to optimal transport. In particular, Jordan, Kinderlehrer, and Otto construct an iteratively defined sequence of probability measures that converge to the solution of the Fokker-Planck equation[58], which we explicitly define in Appendix section B.5. In other words, they show that the Fokker-Planck equation is a gradient flow in the space of probability measures with bounded 2nd moment. The JKO scheme is one example of the theoretical advances that have resulted from combining the theory of optimal transport and Riemannian geometry, and has led to theoretical advances in the study of cancer treatment, among other applications [61]. Another exciting application of optimal transport is its use in modern computer science, which has been accelerated by the advent of entropically regularized optimal transport.

Chapter 4

Entropically regularized optimal transport (EROT)

We have previously defined conditions for which solutions to optimal transport problems exist. In this chapter, we provide insight into how mathematicians and computer scientists compute optimal transport plans in practice. We will discuss the discrete formulation of the so called entropically regularized optimal transport (EROT) problem, which is a computationally efficient way to approximately solve the Kantorovich problem. We use the discrete formulation of optimal transport in this chapter for the purposes of exposition and forgo discussion of continuous entropically regularized optimal transport, which is formulated analogously to the discrete version. A discussion of continuous EROT can be found in [20].

4.1 Disadvantages of the Wasserstein distance

The Wasserstein distance has merits as a distance between probability measures, but calculating Wasserstein distances is a slow process, and the Wasserstein distance suffers from the curse of dimensionality. Before we can discuss the computational costs of the Wasserstein distance, we define "big O notation", which allows us to quantify the speed at which a function grows with increasing input data.

Definition 4.1.1 (Big O and $\tilde{O}$ notation). [23, pgs. 47, 63] *For a given real-valued function $g(n)$, $O(g(n))$ denotes the set of real-valued functions $f(n)$ such that there exists positive $c, n_0 \in \mathbb{R}$ such that $0 \leq f(n) \leq cg(n)$ for all $n \geq n_0$. If $h(n) \in O(g(n))$, we say that $h(n)$*

is **big** O *of* $g(n)$. $\tilde{O}$ *notation is big-O notation with logarithmic factors ignored, i.e., we say that* $h(n)$ *is* ***soft-****O of* $g(n)$ *if it belongs to the set of real-valued functions* $f(n)$ *such that there exists positive* c, k, n_0 *such that* $0 \leq f(n) \leq cg(n)\log^k(n)$ *for all* $n \geq n_0$. *We write* $f(n) = O(g(n))$ *to indicate that a function* $f(n)$ *is an element of the set* $O(g(n))$ *and* $f(n) = \tilde{O}(g(n))$ *to indicate that a function* $f(n)$ *is an element of the set* $\tilde{O}(g(n))$.

Network flow methods are generally used to solve discrete optimal transport problems [22]. Such algorithms are $O(n^3 \log(n))$, meaning that computational complexity rapidly grows as the number of points in discrete probability measures being compared increase [27, pg. 2]. In addition, Wasserstein distances suffer from the "curse of dimensionality", meaning that they treat high-dimensional data differently from low-dimensional data. In particular, it gets harder to estimate a probability measure $\mu \in \mathcal{M}(\mathbb{R}^d)$ from its an approximating distribution called an *empirical distribution* as the dimension d increases.

Definition 4.1.2 (Empirical distribution)**.** [28, pg. 10] *Let* $X_1, \ldots, X_n$ *be independent and identically distributed random variables which are draws from a probability measure* μ *on a sample space* X. *The* **empirical measure** *of* μ *is defined as* $\hat{\mu}_n := \frac{1}{n}\sum_{i=1}^{n} \delta_{X_i}$.

As we would expect, as the number of samples in the empirical distribution n increases, the Wasserstein distance between the empirical distribution and the measure μ gets closer to 0. In fact, as $n \to \infty$, the empirical measure $\hat{\mu}_n$ weakly converges to μ, and because the Wasserstein distance metrizes weak convergence, it follows that $W_p(\hat{\mu}_n, \mu) \to 0$ [25, Theorem 3]. As the dimension d of the space on which the measure μ is defined on increases, the rate of convergence of $\hat{\mu}_n$ decreases in turn. Specifically, the expected value of $W_p(\mu, \hat{\mu}_n)$ is $O(n^{-1/d})$. This complicates the use of Wasserstein distances to calculate distances between high dimensional data, such as single cell transcriptomics data [26]. For these reasons and more, there has been a recent shift to apply entropic regularization to calculating transport plans, spurred on by Marco Cuturi's seminal paper from 2013 [21].

4.2 An introduction to entropic regularization

A major advancement in the application of optimal transport to machine learning is the usage of entropic regularization for optimal transport. EROT provides approximate solutions to the optimal transport problem and “smears out” the precise optimal transport plan generated between histograms.

What is entropy, and why do we use it for regularization?

Entropy is a function that comes from information theory, and in informal terms is a measure of “surprise”. The application of entropic regularization to optimal transport is inspired by a problem in physics called the **Schrödinger bridge problem**. The Schrödinger bridge problem studies the random movement of particles, where the concepts of information entropy and entropy in statistical mechanics coincide [32].

In the context of EROT, entropy appears as a term to be maximized, according to the *principle of maximum entropy.* This principle states that, given data approximable by a probability distribution, the maximum entropy distribution makes the least assumptions about the data and is therefore the best candidate for approximation. The maximum entropy principle was first proposed in the context of statistical mechanics by E.T. Jaynes [41, Chapter 11]. *Regularization* is a common technique in machine learning to prevent a machine learning algorithm from overfitting or underfitting on training data [31, pg. 10]. Intuitively, adding a regularization term in EROT creates a “tug-of-war” between minimizing $\langle C, P\rangle$ and maximizing the entropy of the transport matrix $H(P)$. Not only does the addition of the entropy term have theoretical motivation, it allows an approximate optimal transport plan to be calculated using Sinkhorn's algorithm more quickly than solving the conventional Kantorovich problem.

To define the EROT problem, we present the discrete analogs of probability measures and coupling measures. In particular, $U(a, b)$ and C_{ij} are the discrete analogs of the space of coupling measures $\Pi(\mu, \nu)$ and the cost function $c(x, y)$ discussed in Chapter 2. For more details on discrete optimal transport, we encourage the reader to refer to Appendix B.

Definition 4.2.1 (Probability simplex, admissible coupling, discrete Kantorovich problem)**.** [2, pg. 7, 14, Section 2.3] *The probability simplex* Σ_n *is defined as*

$$\Sigma_n := \left\{ a \in \mathbb{R}^n_+ : \sum_{i=1}^{n} a_i = 1. \right\}, \tag{4.1}$$

where $\mathbb{R}^n_+$ is defined as the set of positive n-dimensional vectors.

Let $a = (a_1, \ldots, a_n)$ and $b = (b_1, \ldots, b_n) \in \Sigma_n$. The space of **admissible couplings** *or* **transportation polytope** *$U(a, b)$ is defined as*

$$U(a,b) := \left\{ P \in M_+^{n\times m}(\mathbb{R}) : \left(\sum_j P_{ij} \right)_i = a, \left(\sum_i P_{ij} \right)_j = b \right\}, \tag{4.2}$$

where $M_+^{n\times m}(\mathbb{R})$ represents the set of nonnegative real-valued $n \times m$ matrices.

The **discrete Kantorovich problem** *is the following: Define $\alpha = \sum_{i=1}^n a_i \delta_{x_i}$ and $\beta = \sum_{j=1}^m b_j \delta_{y_j}$, for $(x_1, \ldots, x_n) \in (X, \Sigma_2)$ and $(y_1, \ldots, y_m) \in (Y, \Sigma_2)$. For a specified transportation cost matrix $C \in M_+^{n\times m}(\mathbb{R})$ where $C_{ij} = c(x_i, y_j)$ for a function $c : X \times Y \to [0, \infty)$, find a coupling matrix $P \in U(a, b)$ satisfying*

$$L_C(a,b) := \min_{P\in U(a,b)} \langle C, P\rangle := \min_{P\in U(a,b)} \sum_{i,j} C_{ij} P_{ij}. \tag{4.3}$$

We now define entropy in the context of our discrete optimal transport problem.

Definition 4.2.2 (Entropy)**.** [2, pg. 57] *Let P be an admissible coupling. The* **discrete entropy** *of P is defined as*

$$H(P) = -\sum_{i,j} P_{ij}(\log(P_{ij}) - 1), \tag{4.4}$$

where $H(P) = -\infty$ if one of the entries P_{ij} is 0.

H is $1-$strongly concave (by Proposition B.7.8), which means that there exists a unique solution to the entropic regularization problem. A quirk of this definition of entropy is that -1 is added to $\log(P_{ij})$, which is not present in the standard definition of entropy in information theory textbooks or in [21]. This change is introduced to make computations for Proposition 4.3.2 easier, and does not otherwise change the properties of this definition relative to other definitions of entropy.

Definition 4.2.3 (Entropically regularized optimal transport)**.** [2, pg. 58] *Let $\varepsilon > 0$, $a \in \mathbb{R}^n, b \in \mathbb{R}^m$, C a cost matrix defined by a function $c(x, y)$ and $P \in U(a, b)$. Then the* **entropically regularized optimal transport problem***, or* **EROT problem***, is*

$$L_C^\varepsilon(a,b) = \min_{P\in U(a,b)} \langle P, C\rangle - \varepsilon H(P). \tag{4.5}$$

The entropically regularized optimal transport problem's solution looks like a smudged out version of the solution to the Kantorovich problem. By including the entropy term, the matrix that solves the EROT problem is less sparse than that of the one solving the Kantorovich problem.

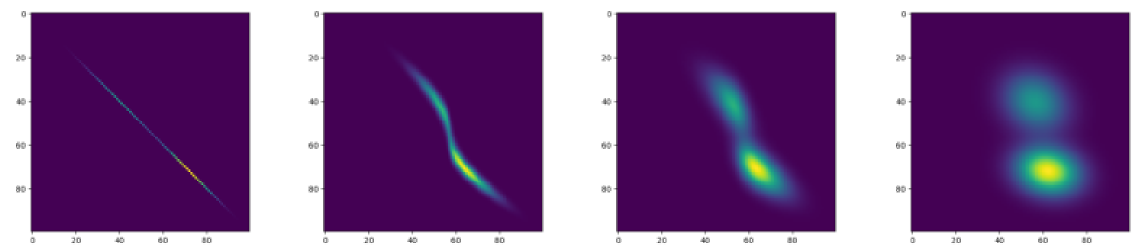


Figure 4.1: Entropic regularized optimal transport between the two probability distributions from Figure 2.3 with ε set to $10^{-9}, 10^{-3}, 10^{-2}, 10^{-1}$ respectively. As ε increases, the transport plan gets more and more spread out.

Analogous to the Wasserstein distance, we can define a distance inspired by entropically regularized optimal transport, called the **Sinkhorn distance.** Here we provide alternative definitions for the entropies of histograms and admissible couplings in Definition 4.2.4 and Lemma 4.2.5. For an element r of the probability simplex Σ_n, its **entropy** is defined as $H(r) = -\sum_{i=1}^{n} r_i \log(r_i)$. For P, an admissible coupling of vectors $a, b \in \Sigma_n$, its **entropy** is defined as $H(P) = -\sum_{i,j=1}^{n} P_{ij} \log(P_{ij})$.

Definition 4.2.4 (Sinkhorn distance)**.** [21, pg. 4] *Let $a, b \in \Sigma_n$, C be a cost matrix, and $\alpha \in \mathbb{R}^+$. The convex set $U_\alpha(a,b) \subset U(a,b)$ is defined as*

$$U_\alpha(a,b) := \{P \in U(a,b) : KL(P \| ab^T) \leq \alpha\} = \{P \in U(a,b) : h(P) \geq h(a) + h(b) - \alpha\},$$

where $KL(P\|ab^T)$ denotes the Kullback-Leibler divergence between P and ab^T. Then the **Sinkhorn distance** *is defined as $d_{M,\alpha}(a,b) := \min_{P \in U_\alpha(a,b)} \langle P, C \rangle$.*

The Sinkhorn distance satisfies the properties of a metric much like the Wasserstein distance [21, Theorem 1]. In addition, when α is made large enough, the Sinkhorn distance becomes the standard discrete Wasserstein distance [21, Property 1].

For EROT to be a useful weakening of the Kantorovich problem, as the regularization term diminishes in importance (i.e., as $\varepsilon \to 0$), entropically regularized optimal transport should produce an optimal transport plan in the Kantorovich sense. When the regularization term $\varepsilon H(P)$ is much greater in magnitude than $\langle P, C\rangle$, solving the EROT problem should provide the admissible coupling with the maximum entropy. This is exactly what we see in Proposition 4.2.6 below. We first prove that the coupling measure $(a \otimes b)$ between two probability vectors a and b has maximum entropy.

Lemma 4.2.5 ($(a \otimes b)$ has maximum entropy)**.** *Let $a, b \in \Sigma_n$ and $U(a, b)$ be the space of admissible couplings of a, b. Then*

$$\max_{P \in U(a,b)} H(P) = H(a \otimes b),$$

where $a \otimes b \in U(a, b)$ is defined by $(a \otimes b)_{i,j} = (a_i b_j)$ for all $i, j \in \{1, \ldots, d\}$.

Proof. In the context of this problem, we define entropies on probability vectors and couplings as in [21]. The proof is practically identical if Definition 4.2.2 is used.

We can interpret a, b, and P through a probabilistic lens. Let X and Y be random variables taking values in $\{1, \ldots, d\}$ distributed according to a and b respectively, i.e. $p(X = i) = a_i$ and $p(Y = j) = b_j$ for all $i, j \in \{1, \ldots, d\}$. The matrix $P \in U(a, b)$ corresponds to a joint probability mass function such that $p(X = i, Y = j) = p_{ij}$. In this way, our notion of entropies on probability vectors a, b and couplings P is equivalent to that of entropies on random variables as discussed in [11, 2.1, 2.8]. From [11, Theorem 2.6.6], we observe that for any $P \in U(a, b)$, it is true that $H(P) \leq H(a) + H(b)$. We now show that $H(a) + H(b) = H(a \otimes b)$.

$$\begin{aligned}
H(a \otimes b) = H((a_i b_j)_{i,j}) &= -\sum_{i=1}^{d}\sum_{j=1}^{d}(a_i b_j)\log(a_i b_j) \\
&= -\sum_{i=1}^{d}\sum_{j=1}^{d}(a_i b_j)\log(a_i) - \sum_{i=1}^{d}\sum_{j=1}^{d}(a_i b_j)\log(b_j) \\
&= -\sum_{j=1}^{d} b_j \left[\sum_{i=1}^{d} a_i \log(a_i)\right] - \sum_{i=1}^{d} a_i \left[\sum_{j=1}^{d} b_j \log(b_j)\right] \\
&= -\left[\sum_{i=1}^{d} a_i \log(a_i)\right] \cdot \sum_{j=1}^{d} b_j - \left[\sum_{j=1}^{d} b_j \log(b_j)\right] \cdot \sum_{i=1}^{d} a_i \\
&= -\sum_{i=1}^{d} a_i \log(a_i) - \sum_{j=1}^{d} b_j \log(b_j) = H(a) + H(b).
\end{aligned}$$

□

The measure $(a \otimes b)$ is perhaps the simplest coupling measure for the Kantorovich problem: it takes mass from any point in the initial distribution and distributes it over the entire target distribution proportionally [4, pg. 2], so a solution to the EROT problem with large ε will in turn be diffuse as in Figure 4.1.

Proposition 4.2.6. [2, Proposition 4.1] *The unique solution P_ϵ of Definition 4.2.3 converges to the optimal transport plan (within the set of all optimal solutions of the Kantorovich problem) with maximal entropy, i.e. as $\varepsilon \to 0$, $L_C^\varepsilon(a,b) \to L_c(a,b)$, where L_c is defined as in* (4.3)*, and*

$$P_\varepsilon \to \operatorname{argmin}_P\{-H(P) : P \in U(a,b), \langle P, C\rangle = L_c(a,b)\}. \tag{4.6}$$

As $\varepsilon \to \infty$, $P_\varepsilon \to a \otimes b = ab^T = (a_i b_j)_{i,j}$.

Proof. Let us define a sequence $(\varepsilon_\ell)_{\ell\in\mathbb{N}}$ such that $\varepsilon_\ell \to 0$, and $\varepsilon_\ell > 0$ for all $\ell \in \mathbb{N}$. Define $P_\ell = L_C^{\varepsilon_\ell}(a,b)$ for all $\ell \in \mathbb{N}$, which gives us another sequence $(P_\ell)_{\ell\in\mathbb{N}}$. Since $U(a,b)$ is a bounded set for fixed a, b, we can extract a subsequence of $(P_\ell)_{\ell\in\mathbb{N}}$ which we call (P_{ℓ_k}) that converges to a matrix P^* [2, Page 14]. Furthermore, since $U(a,b)$ is closed, $P^* \in U(a,b)$. Let P be an arbitrary coupling matrix that satisfies $\langle C, P\rangle = L_C(a,b)$. The matrices P and P_{ℓ_k} are optimal for the Kantorovich problem and entropically regularized optimal transport with $\varepsilon = \varepsilon_{\ell_k}$, or in other words, for all $Q \in U(a,b)$,

$$\langle C, P\rangle \le \langle C, Q\rangle \text{ and } \langle C, P_{\ell_k}\rangle - \varepsilon_{\ell_k} H(P_{\ell_k}) \le \langle C, Q\rangle - \varepsilon_{\ell_k} H(Q).$$

This implies that for all $\ell_k \in \mathbb{N}$,

$$0 \le \langle C, P_{\ell_k}\rangle - \langle C, P\rangle \le \varepsilon_{\ell_k}(H(P_{\ell_k}) - H(P)) \tag{4.7}$$

$$\Rightarrow \lim_{\ell_k\to\infty} 0 \le \lim_{\ell_k\to\infty} \langle C, P_{\ell_k}\rangle - \langle C, P\rangle \le \lim_{\ell_k\to\infty} \varepsilon_{\ell_k}(H(P_{\ell_k}) - H(P))$$
$$\Rightarrow 0 \le \langle C, P^*\rangle - \langle C, P\rangle \le 0 \Rightarrow \langle C, P^*\rangle = \langle C, P\rangle,$$

where $\lim_{\ell_k\to\infty} \varepsilon_{\ell_k}(H(P_\ell) - H(P))$ follows from the fact that $H(P)$ and $H(P_{\ell_k})$ are bounded by $H(a \otimes b)$ according to Lemma 4.2.5. This implies that P^* is an optimal coupling for the two measures a, b. In addition, when we divide the expression (4.7) by ε_{ℓ_k}, we see that

$$\lim_{\ell_k\to\infty} 0 \le \lim_{\ell_k\to\infty} (H(P_{\ell_k}) - H(P)) \Rightarrow -H(P^*) \le -H(P),$$

which implies that P^* minimizes negative entropy. Since the solution of 4.5 is unique, the whole sequence $(P_\ell)_{\ell\in\mathbb{N}}$ converges to P^*. In the limit $\varepsilon \to \infty$, a similar proof shows that one should consider the problem

$$\min_{P\in U(a,b)} -H(P),$$

the solution of which is $a \otimes b$ according to Lemma 4.2.5. $\square$

4.3 Sinkhorn's Algorithm and its convergence

In this section, we seek to show that for an admissible coupling solving 4.2.3, every element P_{ij} can be expressed as the matrix product of u_i, K_{ij}, and v_j, which we produce from formulating the entropically regularized optimal transport as an optimization problem and taking the derivative of its Lagrangian. Before doing this, we have to define the Karush-Kuhn-Tucker conditions.

Theorem 4.3.1 (Karush-Kuhn-Tucker (KKT) conditions)**.** [30, pgs. 215, 243] *Define $f, g_i, h_j : \mathbb{R}^n \to \mathbb{R}$ for $i \in \{1, \ldots, m\}$ and $j \in \{1, \ldots, p\}$. Consider an optimization problem in the standard form:*

$$\text{minimize } f(x) \text{ subject to } g_i(x) \leq 0, i \in \{1, \ldots, m\} \textit{ and } h_j(x) = 0, j \in \{1, \ldots, p\},$$

with variable $x \in \mathbb{R}^n$. Let the objective function f and the constraint functions g_i and h_j have subderivatives at a local optimum $x^* \in \mathbb{R}^n$. If the optimization problem satisfies regularity conditions, such as the linearity constraint qualification (g_i and h_j are affine for all $i \in \{1, \ldots, m\}$ and $j \in \{1, \ldots, p\}$), then there exist real constants λ_i and ν_i such that the following conditions hold, called the **Karush-Kuhn-Tucker (KKT) conditions:**

$$\nabla f(x^*) + \sum_{i=1}^{m} \lambda_i \nabla g_i(x^*) + \sum_{i=1}^{p} \nu_i \nabla h_i(x^*) = 0,$$

$$\lambda_i g_i(x^*) = 0, i \in \{1, \ldots, m\} \textit{ and } \lambda_i \geq 0, i \in \{1, \ldots, m\}$$

$$g_i(x^*) \leq 0, i \in \{1, \ldots, m\} \textit{ and } h_j(x^*) = 0, j \in \{1, \ldots, p\}.$$

Proposition 4.3.2. [2, Proposition 4.3] *The solution to Definition 4.2.3 is unique and has the form $P_{ij} = u_i K_{ij} v_j$ for all $(i,j) \in \{1, \ldots, n\} \times \{1, \ldots, m\}$, where $(u, v) \in \mathbb{R}^n_+ \times \mathbb{R}^m_+$ and $K \in M^{n\times m}_+(\mathbb{R})$.*

Proof. We introduce two variables $\alpha \in \mathbb{R}^n$ and $\beta \in \mathbb{R}^m$. The entropically regularized optimal transport problem can be written as an optimization problem in the standard form for some arbitrary $P \in U(a, b)$:

$$\begin{aligned}
\text{minimize } L_C^\varepsilon &:= \sum_{i,j} P_{ij}C_{ij} + \varepsilon \sum_{i,j} P_{ij}(\log P_{ij} - 1) \\
\text{subject to } &(P\mathbb{1}_m - a)_i = 0 \text{ for } i \in \{1, \ldots, n\}, \\
&(P^T\mathbb{1}_n - b)_j = 0 \text{ for } j \in \{1, \ldots, m\}, \\
&-P_{ij} \leq 0 \text{ for } (i, j) \in \{1, \ldots, n\} \times \{1, \ldots, m\}.
\end{aligned}$$

Let P be a solution of (4.2.3). Because all our constraints are affine and differentiable, we can produce real constants α_i, β_j, and ν_{ij} such that the KKT conditions are satisfied, i.e.,

$$\nabla\left(\sum_{i,j} P_{ij}C_{ij} + \varepsilon \sum_{i,j} P_{ij}(\log P_{ij} - 1) + \alpha^T(P\mathbb{1}_m - a) + \beta^T(P^T\mathbb{1}_n - b) + \sum_{i,j} \nu_{ij}P_{ij}\right) = 0. \tag{4.8}$$

Because the term $\log(P_{ij}) - 1$ forces $P_{ij} > 0$ for all $i, j \in \{1, \ldots, n\} \times \{1, \ldots, m\}$, it follows by the condition $\sum_{i,j} \nu_{ij}P_{ij} = 0$ that $\nu_{ij} = 0$ for all $i, j \in \{1, \ldots, n\} \times \{1, \ldots, m\}$. The term that the gradient is being applied to in (4.8) is called the **Lagrangian** of the function L_c^ε, denoted $L(L_c^\varepsilon, \alpha, \beta)$. By Theorem 4.3.1, the solution of EROT can be found by taking the derivative of the Lagrangian with respect to P_{ij} and finding the coupling for which the Lagrangian is zero. The partial deriative of the Lagrangian with respect to P_{ij} is

$$\frac{\partial L(L_c^\varepsilon, \alpha, \beta)}{\partial P_{ij}} = c_{ij} + \varepsilon \log(P_{ij}) + \alpha_i + \beta_j.$$

Setting the partial derivative equal to zero, we find that $P_{ij} = e^{\frac{\alpha_i}{\varepsilon}} e^{-\frac{c_{ij}}{\varepsilon}} e^{\frac{\beta_j}{\varepsilon}}$, where $u_i := e^{\frac{\alpha_i}{\varepsilon}}$, $K_{ij} := e^{-\frac{c_{ij}}{\varepsilon}}$, and $v_j := e^{\frac{\beta_j}{\varepsilon}}$. □

Let us convert the vectors u_i and v_j into matrices, where the values of u_i and v_j are on the diagonals and all other values are zero. We refer to these matrices as $\text{diag}(u_i)$ and $\text{diag}(v_j)$ respectively. Proposition 4.3.2 gives us a "factorization" for the optimal solution P as $\text{diag}(u)K\text{diag}(v)$. The variables (u, v) satisfy the following constraints, corresponding to the original mass conservation constraints of optimal transport:

$$P\mathbb{1}_m = \mathrm{diag}(u)K\mathrm{diag}(v)\mathbb{1}_m = a \text{ and } P^T\mathbb{1}_n = \mathrm{diag}(u)K^T\mathrm{diag}(v)\mathbb{1}_n = b.$$

We can simplify these equations. Note that $\mathrm{diag}(v)\mathbb{1}_m$ is just the vector v and $\mathrm{diag}(u)$ times Kv is the element-wise product of the two matrices (denoted $\odot$). Therefore,

$$u \odot (Kv) = a \text{ and } v \odot (K^T u) = b. \tag{4.9}$$

In practice, if we want to find u, v, we can construct an iterative algorithm known as Sinkhorn's algorithm that switches off between refining the approximate values of u and v. Sinkhorn's algorithm for EROT has a computational complexity of $\tilde{O}(n^2/\varepsilon^2)$, which is preferable to the network flow algorithms for standard optimal transport [39]. After initializing the algorithm with a vector $v^0 := \mathbb{1}_m$, the successive iterations are as follows:

$$u^{(\ell+1)} := \frac{a}{Kv^{(\ell)}} \text{ and } v^{(\ell+1)} := \frac{b}{K^T u^{(\ell+1)}}. \tag{4.10}$$

Sinkhorn's algorithm converges to the EROT solution in the sense of the Hilbert metric. The general framework for proving convergence was first presented in [40, Section 3], and we refer to it in the proof of Theorem 4.3.5. To do so, we first define the Hilbert metric between vectors.

Definition 4.3.3 (Hilbert metric)**.** [2, Remark 4.12] *For two vectors $u, u' \in \mathbb{R}^n_+$ (the set of positive $n-$dimensional vectors), the* **Hilbert metric** *is defined as*

$$d_{\mathcal{H}}(u, u') := \log \max_{i,j} \frac{u_i u'_j}{u_j u'_i}.$$

Equipped with the equivalence relation $u \sim u' \Leftrightarrow \exists r \in \mathbb{R}^+, u = ru'$, it can be shown that the Hilbert metric satisfies all the properties of a distance on the projective cone $\mathbb{R}^n_+/\sim$ [2, Remark 4.12]. We directly employ the following lemma in our proof of the convergence of Sinkhorn's algorithm.

Lemma 4.3.4. [2, Theorem 4.1] *Let $K \in \mathbb{R}^{m\times m}_+$, then for $(v, v') \in \mathbb{R}^m_+ \times \mathbb{R}^m_+$,*

$$d_{\mathcal{H}}(Kv, Kv') \leq \lambda(K) d_{\mathcal{H}}(v, v'), \ \ \textit{where} \ \begin{cases} \lambda(K) := \frac{\sqrt{\eta(K)}-1}{\sqrt{\eta(K)}+1} < 1, \\ \eta(K) := \max_{i,j,k,l} \frac{K_{i,k}K_{j,l}}{K_{j,k}K_{i,l}}. \end{cases}$$

Theorem 4.3.5 (Convergence of Sinkhorn's algorithm)**.** [2, Theorem 4.2, Remark 4.12] *Let* $P^* \in M_+^{m\times m}(\mathbb{R})$ *be the optimal transport plan that minimizes* (4.5)*, decomposed as a matrix* $\mathrm{diag}(u^*)K\mathrm{diag}(v^*)$*, where* $K \in M_+^{m\times m}(\mathbb{R})$ *and* $u^*, v^* \in \mathbb{R}_+^m$*. We initialize Sinkhorn's algorithm with the vector* $v^0 = \mathbb{1}_m$ *and derive* u^ℓ, v^ℓ *iteratively for arbitrary* $\ell \in \mathbb{N}$*. One has* $(u^\ell, v^\ell) \to (u^*, v^*)$ *and*

$$d_{\mathcal{H}}(u^\ell, u^*) = O(\lambda(K)^{2\ell}), d_{\mathcal{H}}(v^\ell, v^*) = O(\lambda(K)^{2\ell}). \tag{4.11}$$

One also has

$$d_{\mathcal{H}}(u^\ell, u^*) \leq \frac{d_{\mathcal{H}}(P^{(\ell)}\mathbb{1}_m, a)}{1-\lambda(K)^2} \text{ and } d_{\mathcal{H}}(v^\ell, v^*) \leq \frac{d_{\mathcal{H}}(P^{(\ell)}\mathbb{1}_m, a)}{1-\lambda(K)^2} \tag{4.12}$$

where $P^\ell := \mathrm{diag}(u^\ell, u^*)K\mathrm{diag}(v^\ell, v^*)$*. Lastly, one has*

$$\|\log(P^\ell) - \log(P^*)\|_\infty \leq d_{\mathcal{H}}(u^\ell, u^*) + d_{\mathcal{H}}(v^\ell, v^*). \tag{4.13}$$

Proof. Notice that for any $(v, v') \in \mathbb{R}_+^m \times \mathbb{R}_+^m$, one has

$$d_{\mathcal{H}}(v, v') = d_{\mathcal{H}}(v/v', \mathbb{1}_m) = d_{\mathcal{H}}(\mathbb{1}/v, \mathbb{1}/v').$$

Therefore,

$$d_{\mathcal{H}}(u^{\ell+1}, u^*) = d_{\mathcal{H}}\left(\frac{a}{Kv^\ell}, \frac{a}{Kv^*}\right) = d_{\mathcal{H}}(Kv^\ell, Kv^*) \leq \lambda(K) d_{\mathcal{H}}(v^\ell, v^*),$$

using Lemma 4.3.4. To prove (4.12), we employ the triangle inequality for the Hilbert metric, and examine the first inequality,

$$\begin{aligned} d_{\mathcal{H}}(u^\ell, u^*) &\leq d_{\mathcal{H}}(u^{\ell+1}, u^\ell) + d_{\mathcal{H}}(u^{\ell+1}, u^*) \\ &\leq d_{\mathcal{H}}(\frac{a}{Kv^\ell}, u^\ell) + \lambda(K)^2 d_{\mathcal{H}}(u^\ell, u^*) \\ &= d_{\mathcal{H}}(a, u^\ell \odot (Kv^\ell)) + \lambda(K)^2 d_{\mathcal{H}}(u^\ell, u^*). \end{aligned}$$

This proves the first inequality, and the second follows similarly. To prove the final inequality (4.13), we use the fact that the Hilbert metric on the projective cone is isometric to the variation seminorm $\|\cdot\|_{\mathrm{var}}$ after a logarithmic change of variables, i.e.,

$$d_{\mathcal{H}}(u, u') = \|\log(u) - \log(u')\|_{\mathrm{var}} \tag{4.14}$$

where $\|f\|_{\text{var}} := (\max_i f_i) - (\min_i f_i)$ for $f \in \mathbb{R}^m_+$. Furthermore, if for some fixed $i \in \{1, \ldots, m\}$, it is true that $f_i = 0$, then $\|f\|_\infty \leq \|f\|_{\text{var}}$. Variables solving (4.9) are defined up to an added constant as dictated by the equivalence relation $\sim$, so one can always find some constant such that $f_i = 0$ for an arbitrary $i \in \{1, \ldots, m\}$. Therefore,

$$\begin{aligned}
\|\log(P^\ell) - \log(P^*)\|_\infty &= \|\log(u^\ell K v^\ell) - \log(u^* K v^*)\|_\infty \\
&= \|\log(u^\ell) - \log(u^*) + \log(v^\ell) - \log(v^*)\|_\infty \\
&\leq \|\log(u^\ell) - \log(u^*)\|_\infty + \|\log(v^\ell) - \log(v^*)\|_\infty \\
&\leq \|\log(u^\ell) - \log(u^*)\|_{\text{var}} + \|\log(v^\ell) - \log(v^*)\|_{\text{var}} \\
&= d_{\mathcal{H}}(u^\ell, u^*) + d_{\mathcal{H}}(v^\ell, v^*). \qquad \square
\end{aligned}$$

The convergence of Sinkhorn's algorithm and its reduced computational complexity relative to network flow solvers have contributed to the emergence of optimal transport in applied mathematics problems. On the hardware side, the increasing accessibility of graphics processing units (GPUs) to the general public has contributed to Sinkhorn's algorithm's applicability in scientific research, as the operations in Sinkhorn's algorithm are particularly suited to modern GPU calculations [2, Remark 4.16].

4.4 Optimal transport and trajectory analysis

One recent application of entropic regularization to optimal transport is in the field of computational biology. Optimal transport allows biologists to understand how cells transform into different cell types and to trace out the lineages of cells as they develop. The genes that a cell expresses determine its cell type, and to assess what cell types are in a given population, we can take a genetic snapshot by counting the genes expressed in each cell. This technique is called single cell RNA-sequencing (scRNA-seq), and it tells us what genes the cells are expressing at a given time [42]. While scRNA-seq can identify the cells present in a sample, it does not have a temporal component, i.e., it does not tell us how the cells transform, or *differentiate*, into the cell types present in our sample. scRNA-seq destroys the cells it studies, which prevents biologists from seeing how gene expression changes over time. To circumvent this limitation, an exciting paper by Schiebinger et al. details how optimal transport has been used to fill in the gaps between scRNA-seq experiments taken at specific developmental time-points [26]. This paper combines displacement interpolation, entropically regularized optimal transport, and unbalanced optimal transport, a variation on optimal transport that does not require the measures μ and ν to have identical mass [43]. Their new method for identifying trajectories, paths that cells traverse in transcriptional space, is called *WaddingtonOT*, named after the developmental biologist Conrad Waddington. The authors think about populations of cells at given times as distinct distributions (but not probabilities, since the number of cells changes time point to time point). At a given time t_i, a set of cells C has both an "ancestor distribution" at time t_{i-1} and a "descendant distribution" at time t_{i+1}, and the trajectory from C is the sequence of descendant distributions at each subsequent time point. The coupling between a set of cells C and a descendant population at t_{i+1} can be calculated using optimal transport algorithms.

What this allows us to do is to look at a particular lineage of cells and see what sort of gene expression changes are necessary for a cell to differentiate into another cell. For instance, in Schiebinger's paper, mouse embryonic fibroblasts (METs) are converted into induced pluripotent stem cells (iPSCs), which hold promise for regenerative medicine [44]. Optimal transport allows biologists to understand the transcriptional trajectory cells traverse to become iPSCs, i.e., what *genes are turned off/on*, and at what *time points*. WaddingtonOT is one example of a tool biologists use to perform "trajectory analysis", the identification of a path through transcriptional space that traverses cellular states associated with cell differentiation [45]. Trajectory analysis is one example of of optimal transport in the world of applied and pure math. We hope that the baseline knowledge in this thesis can empower you to seek out new applications of optimal transport.

Appendix A

Results in Analysis

A.1 Measure Theory

Definition A.1.1 (Measure space)**.** *A* **measure space** (X, Σ, μ) *is comprised of a set X, a σ-algebra Σ, and a measure μ. A σ-algebra is a collection of sets closed under complements, countable unions, and countable intersections. The elements of the σ-algebra are known as* measurable sets*. A measure μ is a function from Σ to $\mathbb{R}$ (or $\mathbb{R} \cup \{\infty\}$ that satisfies the following conditions: for $E \in \Sigma$,*

1. $0 \leq \mu(E) \leq \infty$,

2. $\mu\left(\bigcup_k E_k\right) = \sum_k \mu(E_k)$ *for every countable family $\{E_k\}$ of disjoint sets in Σ.*

A **measurable space** *is a set and a σ-algebra (X, Σ).*

Definition A.1.2 (Borel σ-algebra, sets, measures)**.** [19, pgs. 22, 33] *Let X be a topological space. The σ-algebra generated by the open sets in X is called the* **Borel σ-algebra** *on X, denoted $\mathcal{B}_X$. Members of this σ-algebra are* **Borel sets.** *A* **Borel measure** *is a measure defined on the* **Borel space** $(X, \mathcal{B}_X)$.

Definition A.1.3 (Signed measure)**.** [19, pg. 85] *Let (X, Σ) be a set and a σ-algebra. A* **signed measure** *is a function $\rho : \Sigma \to [-\infty, \infty]$ such that*

- $\rho(\varnothing) = 0$;

- *ρ assumes at most one of the values $\pm\infty$;*

- *if* $\{E_j\}$ *is a sequence of disjoint sets in* Σ*, then* $\rho(\bigcup_{j=1}^{\infty} E_j) = \sum_1^{\infty} \rho(E_j)$*, where the latter sum converges absolutely if* $\rho(\bigcup_{j=1}^{\infty} E_j)$ *is finite.*

The set of signed measures of X *is denoted* $\mathcal{M}(X)$.

Definition A.1.4 (Characteristic function)**.** *The* **characteristic function***, or indicator function, of a set* $A \subseteq X$ *is denoted* $\chi(A)$*. If* $x \in A$*,* $\chi_A(x) = 1$*, and if* $x \notin A$*,* $\chi_A(x) = 0$.

Definition A.1.5 (Measurable function)**.** [19, Corollary 2.2] *Let* (X, Σ) *be a measurable space, and* $f : X \to \mathbb{R}$ *be a function defined on the set* X*. The following are equivalent:*

- f *is a measurable function.*
- $f^{-1}((a, \infty)) \in \Sigma$ *for all* $a \in \mathbb{R}$.
- $f^{-1}([a, \infty)) \in \Sigma$ *for all* $a \in \mathbb{R}$.
- $f^{-1}((-\infty, a)) \in \Sigma$ *for all* $a \in \mathbb{R}$.
- $f^{-1}((-\infty, a]) \in \Sigma$ *for all* $a \in \mathbb{R}$.

Definition A.1.6 (Coupling)**.** [3, pg. 17]

Let (X, μ) *and* (Y, ν) *be two probability spaces. Coupling* μ *and* ν *means constructing two random variables* X *and* Y *on some probability space* $(\Omega, \mathbb{P})$*, such that law*$(X) = \mu$*, law*$(Y) = \nu$*. The couple* (X, Y) *is called a coupling of* (μ, ν)*. By abuse of language, the law of* (X, Y) *is also called a coupling of* (μ, ν).

Coupling two measures μ and ν is equivalent to constructing a measure π on $X \times Y$ such that π admits μ and ν as marginals on X and Y.

Theorem A.1.7 (Characterization of coupling measures)**.** *The following are equivalent:*

- *For all measurable sets* $A \subset X, B \subset Y$*, we have* $\pi[A \times Y] = \mu[A]$ *and* $\pi[X \times B] = \nu[B]$.
- *For all integrable measurable functions* φ, ψ *on* X *and* Y *respectively,*

$$\int_{X \times Y} (\varphi(x) + \psi(y))\, d\pi(x, y) = \int_X \varphi\, d\mu + \int_Y \psi\, d\nu.$$

Proof. (⇐) Let A be a measurable set in X. Consider $\varphi(x) = \chi_A(x)$ and $\psi(y) = 0$.

$$\begin{aligned}
\mu[A] = \int_X \chi_A(x) d\mu &= \int_{X\times Y} \chi_A(x)\, d\pi(x,y) \\
&= \int_{X\times Y} \chi_A(x)\chi_Y(y)\, d\pi(x,y) \\
&= \int_{X\times Y} \chi_{A\times Y}(x,y)\, d\pi(x,y) \\
&= \pi[A\times Y].
\end{aligned}$$

Mutatis mutandis, the same is true for $\nu[B] = \pi[X\times B]$.

(⇒) Without loss of generality, we can prove that if $\pi[A\times Y] = \mu[A]$ when $A\subset X$ and A is measurable, then

$$\int_X \varphi(x) d\mu = \int_{X\times Y} \varphi(x)\, d\pi(x,y).$$

From that result, it immediately follows that

$$\int_{X\times Y} (\varphi(x) + \psi(y))\, d\pi(x,y) = \int_X \varphi\, d\mu + \int_Y \psi\, d\nu.$$

We prove this theorem for indicator functions, simple functions, and finally, general measurable functions.

1) Indicator functions: Let $\varphi(x) = \chi_A(x)$ for some measurable $A\subset X$. Then

$$\int_X \chi_A(x)\, d\mu = \mu[A] = \pi[A\times Y] = \int_{X\times Y} (x,y) d\pi(x,y) = \int_{X\times Y} \chi_A(x) d\pi(x,y).$$

2) Simple functions: Let $\varphi(x) = \sum_{i=1}^n a_i\chi_{A_i}(x)$ for characteristic functions of measurable sets $A_i\subset X$ for all $i\in\{1,\dots,n\}$.

$$\begin{aligned}
\int_X \varphi(x)\, d\mu &= \int_X \left(\sum_{i=1}^n a_i\chi_{A_i}(x)\right) d\mu \\
&= \sum_{i=1}^n a_i \left(\int_X \chi_{A_i}(x)\, d\mu\right) \text{ (Tonelli's theorem)} \\
&= \sum_{i=1}^n a_i \left(\int_{X\times Y} \chi_{A_i}(x)\, d\pi(x,y)\right) \\
&= \int_{X\times Y} \left(\sum_{i=1}^n a_i\chi_{A_i}(x)\right) d\pi(x,y) \\
&= \int_{X\times Y} \varphi(x)\, d\pi(x,y).
\end{aligned}$$

3) Nonnegative measurable functions: Let $\varphi(x)$ be an nonnegative measurable function on X. It can be approximated as a limit of a sequence of nonnegative simple functions $(\varphi_n(x))_{n=1}^{\infty}$ such that these functions converge pointwise to $\varphi(x)$ and $\varphi_n(x) \leq \varphi_{n+1}(x)$ for all $n \in \mathbb{N}$.

$$\begin{aligned}
\int_X \varphi(x)\, d\mu &= \int_X \left(\lim_{n\to\infty} \varphi_n(x)\right) d\mu \\
&= \lim_{n\to\infty} \int_X \varphi_n(x)\, d\mu \text{ (Monotone convergence)} \\
&= \lim_{n\to\infty} \int_{X\times Y} \varphi_n(x)\, d\pi(x,y) \text{ (2)} \\
&= \int_{X\times Y} \left(\lim_{n\to\infty} \varphi_n(x)\right) d\pi(x,y) \\
&= \int_{X\times Y} \varphi(x)\, d\pi(x,y).
\end{aligned}$$

4) General measurable functions: Let $\varphi(x)$ be an arbitrary measurable function on X. It can be expressed as the difference of its positive and negative part, which we call $\varphi^+(x)$ and $\varphi^-(x)$, and are defined as follows:

$$\varphi^+ = \max\{\varphi, 0\}, \varphi^- = -\min\{\varphi, 0\}.$$

By construction, $\varphi(x) = \varphi^+(x) + \varphi^-(x)$. Therefore,

$$\begin{aligned}
\int_X \varphi(x)\, d\mu &= \int_X \varphi^+(x)\, d\mu - \int_X \varphi^-(x)\, d\mu \\
&= \int_{X\times Y} \varphi^+(x)\, d\pi(x,y) + \int_{X\times Y} \varphi^-(x)\, d\pi(x,y) \\
&= \int_{X\times Y} \varphi(x)\, d\pi(x,y).
\end{aligned}$$

□

Definition A.1.8 (L^p space, norm)**.** [12, pg. 252] $L^p(E;\mu) = L^p(E,\Sigma,\mu)$, $0 < p < \infty$ *is the collection of all measurable real or complex-valued* f *such that* $\int_E |f|^p d\mu < \infty$. *The* L^p *norm is defined as*

$$\|f\|_{L^p(E,\mu)} = \left(\int_E |f|^p\, d\mu\right)^{1/p}, 0 < p < \infty.$$

Definition A.1.9 (Minkowski's inequality)**.** [12, Theorem 8.10] *If* $1 \leq p \leq \infty$, *then* $\|f + g\|_{L^p(E,\mu)} \leq \|f\|_{L^p(E,\mu)} + \|g\|_{L^p(E,\mu)}$.

Theorem A.1.10 (Lebesgue's dominated convergence theorem)**.** [12, Theorem 10.31] *Let ϕ, $\{f_k\}$, and f be measurable functions on E such that $|f_k|\leq \phi$ almost everywhere with respect to μ on E and $\phi \in L(E;\mu)$, i.e., $\int_E \phi d\mu$ exists and is finite. Then*

$$f_k \to f \, a.e. \text{ in } E \Rightarrow \int_E f_k \, d\mu \to \int_E f \, d\mu.$$

Definition A.1.11 (Absolute variation and total variation)**.** [59, pg. 176] *The* ***absolute variation*** *of the signed measure μ defined on a measurable space (X, Σ) is the function*

$$|\mu|(E) = \overline{W}(\mu, E) + |\underline{W}(\mu, E)|,$$

and the total variation of a measure is defined as $\|\mu\|_{TV} := |\mu|(X)$.

The total variation norm induces a metric on $\Pi(\mu, \nu)$. For a generic measure space $(\Omega, \mathcal{F})$ with two measures $\mu, \nu \in \mathcal{P}(\Omega)$, the total variation metric is

$$d_{TV}(\mu, \nu) = \sup_{A \in \mathcal{F}} |\mu(A) - \nu(A)|.$$

A.1.1 Disintegration of measure example

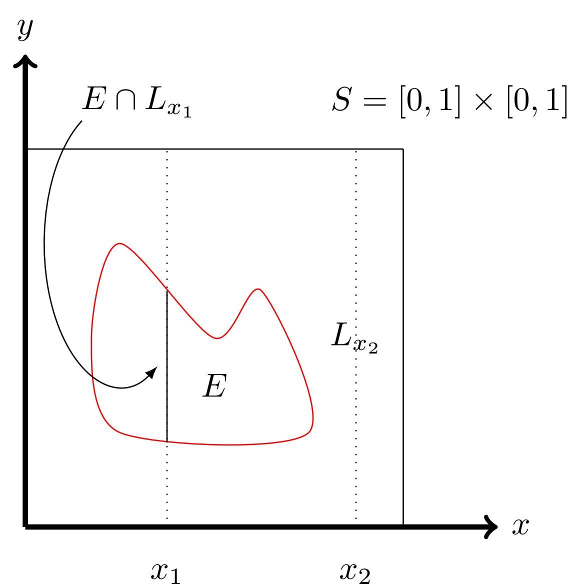


Figure A.1: A depiction of the disintegration of measure of a measurable subset E of $[0, 1] \times [0, 1]$.

Consider the unit square $S = [0, 1] \times [0, 1]$ in $\mathbb{R}^2$, and a measurable subset $E \subset S$. We can determine the area of sets contained in the unit square by using the two-dimensional

Lebesgue measure. Because the two-dimensional Lebesgue measure is defined on a product space ($\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$), we can "decompose" it with respect to a one-dimensional Lebesgue measure. In particular, we can take vertical slices of the set S which we call $L_x = \{x\} \times [0, 1]$, and examine the vertical slices of the set $E = L_x \cap E$. If we add up the uncountable number of vertical slices from E by integration, we should reconstruct the area of E, or the two-dimensional Lebesgue measure of E. We can define a family of measures $\{\mu_x\}_{x \in [0,1]}$ that are the one-dimensional Lebesgue measures on the sets L_x.

$$\mu(E) = \int_{[0,1]} \mu_x(E \cap L_x)\, dx.$$

A.2 Probability theory

Definition A.2.1 (Probability space)**.** *A* **probability space** *is a measure space* $(\Omega, \mathcal{F}, \mathbb{P})$ *such that* $\mu(X) = 1$. *The measure* μ *defines a* **probability distribution**. Ω *is referred to as the* **sample space**.

Definition A.2.2 (Random variable)**.** [29, Definition 3.1.1] *Given a probability space* $(\Omega, \mathcal{F}, \mathbb{P})$, *a* **random variable** *is a function* $X : \Omega \to \mathbb{R}$ *such that*

$$\{\omega \in \Omega : X(\omega) \leq x\} \in \mathcal{F} \text{ for every } x \in \mathbb{R}.$$

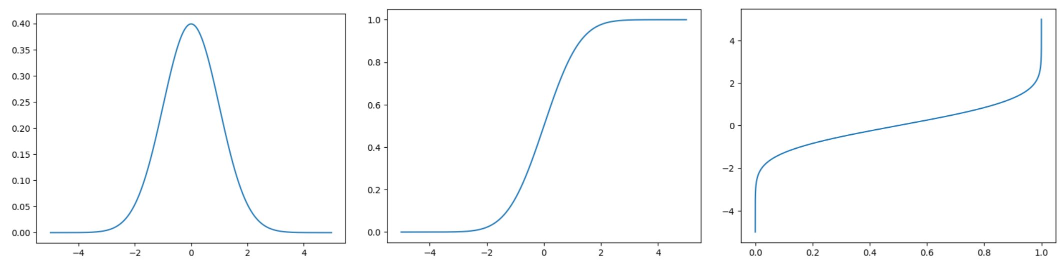
In other words, X *is a* $\mathcal{F}$*-measurable function.*

Definition A.2.3 (Distribution or Law)**.** [29, Definition 6.0.1] *Given a random variable* X *on a probability space* $(\Omega, \mathcal{F}, \mathbb{P})$, *its* **distribution** *or* **law** *is the probability measure defined on* $\mathcal{B}$, *the Borel subsets of* $\mathbb{R}$, *by*

$$\mu(B) = P(X \in B) = P(X^{-1}(B)) \text{ for all } B \in \mathcal{B}.$$

Definition A.2.4 (Probability density function)**.** [29, Remark 12.1.3] *If* $\mu(A) = \int_A f d\lambda$ *for all Borel* A, *we write* $\frac{d\mu}{d\lambda} = f$, *and call* f *the* **density** *or* **Radon-Nikodym derivative** *of* μ *with respect to* λ.

Definition A.2.5 (Kantorovich Problem, probabilistic form)**.** [4, Chapter 1] *Given random variables* $X \sim \alpha$ *and* $Y \sim \beta$, *and* (X, Y) *a coupling of random variables over* $X \times Y$, *the*



(a) Probability density function of the standard normal distribution $\mathcal{N}(0,1)$.

(b) Cumulative distribution function of the standard normal distribution $\mathcal{N}(0,1)$.

(c) The quantile function of the standard normal distribution $\mathcal{N}(0,1)$.

Figure A.2: Let $X \sim \mathcal{N}(0,1)$. Plotted above is its probability density function, cumulative distribution function, and quantile function.

Kantorovich problem is the following:

$$\min_{\pi \in \Pi(\mu,\nu)} \left\{ \int_{X \times Y} c(x,y) d\pi(x,y) : \pi \in \Pi(\mu,\nu) \right\} = \min_{(X,Y)} \left\{ \mathbb{E}_{(X,Y)}(c(X,Y)) : X \sim \alpha, Y \sim \beta \right\}.$$

Definition A.2.6 (Gaussian measure)**.** [53, pg. 1] *A Borel probability measure γ on $\mathbb{R}$ is a* ***Gaussian measure*** *if it is defined by*

$$\gamma(A) := \frac{1}{\sigma\sqrt{2\pi}} \int_A \exp\left(-\frac{(x-\mu)^2}{2\sigma^2} \right) d\lambda(x)$$

where λ is the 1-dimensional Lebesgue measure. The mean of this measure is denoted $\mu \in \mathbb{R}$ and standard deviation is denoted $\sigma \in \mathbb{R}$.

A.3 Functional Analysis

Definition A.3.1 (Duality)**.** *Let X be a Banach space over a field F. The dual space X^* of X is the set of all linear maps/functionals $f : X \to F$.*

Theorem A.3.2 (Riesz representation theorem)**.** [1, Theorem C.18] *Let X be a locally compact Hausdorff space, and $\mu \in M(X)$. Define $F_\mu : C_0(X) \to \mathbb{R}$ as $F_\mu(f) = \int f d\mu$. Then $F_\mu \in C_0(X)^*$ and the map $\mu \mapsto F_\mu$ is an isometric isomorphism from $M(X)$ to $C_0(X)^*$.*

Theorem A.3.3 (Banach-Alaoglu Theorem)**.** *Let X be a normed linear space, and let $B = \{x \in X : \|x\| \leq 1\}$. Then B^* is compact in X^* with respect to the weak-$*$ topology.*

Theorem A.3.4 (Dual of $C_0(X)$). *$C_0(X)$, the set of continuous functions on X vanishing at infinity, is dual to $\mathcal{M}$, the set of regular Borel measures on X.*

A.4 Convex Analysis and Optimization

Definition A.4.1 (Convex set). [13, pg. 1] *A subset K of a real vector space V is convex if and only if for any $x, y \in K$ and $\theta \in [0,1], \theta x + (1-\theta)y \in K$.*

Definition A.4.2 (Convex, concave, affine function). [13, pg. 2] *Let K be a convex subset of a vector space V. A function $f : K \to \mathbb{R}$ is*

1. *convex if for all $x, y \in K$ and $\theta \in [0,1]$,*

$$f(\theta x + (1-\theta)y) \geq \theta f(x) + (1-\theta)f(y).$$

2. *concave if $-f$ is convex,*

3. *affine if f is convex and concave.*

If the inequality is strict, then the function is strictly convex or strictly concave respectively.

Theorem A.4.3. [13, Theorem 8.8] *Let $A \subset \mathbb{R}^\nu$ be a convex set. Then there is a unique affine subspace W of $\mathbb{R}^\nu$ so that $A \subseteq W$, and as a subset of W, A has a nonempty interior.*

Definition A.4.4 (Intrinsic interior, intrinsic boundary). [13, pg. 125] *The interior of A as a subset of W is written A^{iint} and is the intrinsic interior of A. $\partial^i A$ is the intrinsic boundary of A, and is defined as $\overline{A} \backslash A^{iint}$.*

Definition A.4.5 (Face). [13, pg. 121] *A* face *of a convex set is a nonempty subset $F \subset A$ with the property that if $x, y \in A$, $\theta \in (0,1)$ and $\theta x + (1-\theta)y \in F$, then $x, y \in F$. A face F that is strictly smaller than A is a* proper face.

Theorem A.4.6 (The intrinsic interior is the union of all proper faces of a set.). [13, Proposition 8.9] *Let A be a compact convex subset of $\mathbb{R}^\nu$. Then*

1. *$\partial^i A$ is the union of all the proper faces of A.*

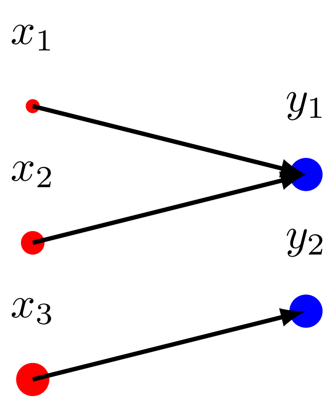
2. *If $x \in \partial^i A$ and $y \in A^{iint}$, $\{\theta | (1-\theta)x + \theta y \in A\} = [0, \alpha]$ for some $\alpha > 1$.*

3. *If $x \in A^{iint}$ and $y \in A$, $\{\theta | (1-\theta)x + \theta y \in A\} \cap (-\infty, 0) \neq \emptyset$.*

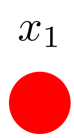
Appendix B

Tools for the thesis

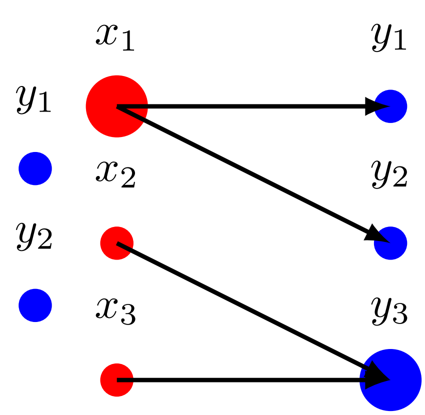
B.1 Discrete Monge



(a) Monge transport map



(b) Monge cannot split mass



(c) Kantorovich transport

Figure B.1: The Monge problem assigns mass from starting points to ending points by creating a surjective map $T : X \to Y$. However, when the source mass is a Dirac mass, it is impossible to map it to multiple points. The Kantorovich problem relaxes this restriction by allowing mass splitting.

Optimal transport problems in the discrete setting involve transporting discrete quantities of mass from a source distribution to a target distribution. These masses are modeled by *Dirac measures.*

Definition B.1.1 (Dirac measure)**.** *Let* (X, Σ) *be a set and* Σ *be a* σ*-algebra on* X*. The* **Dirac measure** *at a point* $x \in X$ *is a probability measure denoted* δ_x*. For all subsets*

$A \subseteq X$,

$$\delta_x(A) := \begin{cases} 1 \text{ if } x \in A \\ 0 \text{ if } x \notin A. \end{cases}$$

Informally speaking, the Dirac measure can be thought of as a probability distribution with all of the mass concentrated on a single point, or a "point mass." A *discrete measure* is a finite linear combination of Dirac measures.

Let us consider what happens when we are transporting discrete masses from one set $X = \{x_1, \ldots, x_n\}$ to another set $Y = \{y_1, \ldots, y_m\}$, (where $n, m \in \mathbb{N}, n \geq m$). We allocate "mass" to each of these points by assigning them Dirac measures and a corresponding weight. The discrete measures on the spaces X and Y, called $\alpha = \sum_{i=1}^{n} a_i \delta_{x_i}$ ($a_i \in \mathbb{R}$, $i \in \{1, \ldots, n\}$) and $\beta = \sum_{j=1}^{n} b_j \delta_{y_j}$ ($b_j \in \mathbb{R}$, $j \in \{1, \ldots, n\}$), respectively, describe the masses assigned to the points in X and Y.

The Monge problem finds a surjective map (when it exists) that associates each point x_i with a point y_j such that mass is conserved. In other words, we seek a "transport map" $T : X \to Y$ such that for all $j \in \{1, \ldots, m\}$, $b_j = \sum_{i:T(x_i)=y_j} a_i$. This restriction guarantees that all masses from positions in x_i that send mass to y_j must sum to the total mass at y_j.

Definition B.1.2 (Discrete Monge problem)**.** [2, Remark 2.4, Definition 2.1]

For a specified transportation cost $c : X \times Y \to [0, \infty)$, find a map $T : X \to Y$ that satisfies

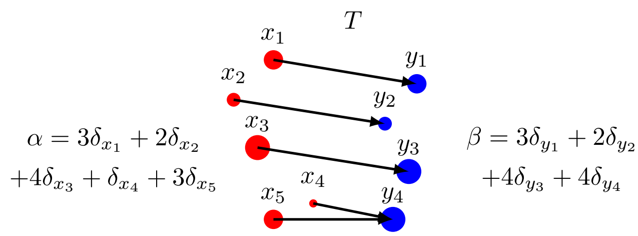
$$\min_T \left\{ \sum_{i=1}^{n} c(x_i, T(x_i)) : T_\# \alpha = \beta \right\},$$

where the **push-forward operator** $T_\# : \mathcal{M}(X) \to \mathcal{M}(Y)$ *satisfies*

$$T_\# \alpha := \sum_{i=1}^{n} a_i \delta_{T(x_i)},$$

whenever $\alpha = \sum_{i=1}^{n} a_i \delta_{x_i} \in \mathcal{M}(X)$.

Figure B.2 depicts a transport map between discrete measures α and β. As seen in the figure, the masses of x_4 and x_5 are transported to the point y_4, i.e., $T(x_4) = y_4, T(x_5) = y_4$, and correspondingly the sum of the weights 1 and 3 for the Dirac measures δ_{x_4} and δ_{x_5} equals the weight at δ_{y_4}.



$$\alpha = \sum_i a_i \delta_{x_i} \qquad \beta = T_\# \alpha = \sum_i a_i \delta_{T(x_i)}$$

$$T(x_1) = y_1, T(x_2) = y_2, T(x_3) = y_3, T(x_4) = y_4, T(x_5) = y_4.$$

Figure B.2: The push-forward operator in the discrete setting "pushes" the mass of the weights of α to the points y_j which receive them. In other words, mass is conserved when transferred from probability measure α to β.

What happens when the number of point masses grows infinitely large? In such a setting, one turns to examining a continuum version of the Monge problem where instead of discrete measures, mass is transported between continuous probability distributions.

Discrete	Continuous
• **Discrete Measures:** $\alpha = \sum_{i=1}^n a_i \delta_{x_i}$ and $\beta = \sum_{j=1}^n b_j \delta_{y_j}$ • **Pushforward:** $T_\# \alpha := \sum_i a_i \delta_{T(x_i)}$ • **Optimization problem:** For a specified transportation cost $c : X \times Y \to [0, \infty)$, find a map $T : \mathcal{X} \to \mathcal{Y}$ that satisfies $\min \{\sum_{i=1}^n c(x_i, T(x_i)) : T_\# \alpha = \beta\}$	• **Probability Measures:** $\mu \in \mathcal{P}(X)$, $\nu \in \mathcal{P}(Y)$ • **Pushforward:** $(T_\# \mu)(A) := \mu(T^{-1}(A))$ • **Optimization problem:** For a specified transportation cost $c : X \times Y \to [0, \infty)$, find a map $T : X \to Y$ that satisfies $\inf \left\{\int_X c(x, T(x)) d\mu(x) : T_\# \mu = \nu\right\}$

B.2 Discrete Kantorovich

In Kantorovich's problem, masses from x_i can be broken up into parts and distributed among the points y_j. Instead of moving mass with a transport plan $T : X \to Y$ which transfers masses from x_i to y_j, we have a coupling matrix $P \in \mathbb{R}_+^{n \times m}$ constructed such that each entry of the matrix P_{ij} contains the mass transferred from x_i to y_j, ensuring that mass is conserved. We present the definition of a compling matrix that satisfies mass conservation, which we call an *admissible coupling.* Let $\mathbf{a} = \{a_1, \ldots, a_n\} \in \mathbb{R}^n$ and $\mathbf{b} = \{b_1, \ldots, b_n\} \in \mathbb{R}^m$.

Definition B.2.1 (Admissible coupling)**.** [2, Section 2.3]

$$U(a, b) := \left\{ P \in \mathbb{R}_+^{n \times m} : \sum_j P_{ij} = \boldsymbol{a}, \sum_i P_{ij} = \boldsymbol{b} \right\}.$$

Now that we have the admissible coupling definition, we can define the Kantorovich problem in the discrete setting.

Definition B.2.2 (Discrete Kantorovich problem)**.** [2, Section 2.3] [21, 2.1] *For a specified transportation cost matrix $C \in \mathbb{R}_{+}^{n \times m}$ where $C_{ij} = c(x_i, y_j)$, find a coupling matrix $P \in U(a, b)$ satisfying*

$$\min_{P \in U(a,b)} \langle C, P \rangle = \sum_{i,j} C_{ij} P_{ij},$$

where $\langle \cdot, \cdot \rangle$ is the Frobenius inner product of two matrices.

Similarly to the Monge problem, we can generalize the Kantorovich problem to the continuous setting.

B.3 Discrete Wasserstein

In analogy with the discrete optimal transport problems we have examined, we first define the discrete Wasserstein distance and move to the continuous setting.

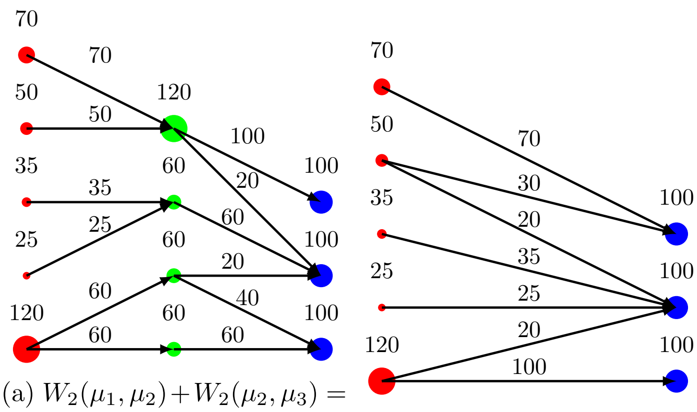
Definition B.3.1 (Discrete Wasserstein distance)**.**

Let $\alpha = \sum_{i=1}^{n} a_i \delta_{x_i}$ and $\beta = \sum_{j=1}^{m} b_j \delta_{y_j}$. Then the Wasserstein distance is defined as

$$W_p(\alpha, \beta) = \min_{P \in U(\alpha,\beta)} \left\{ \sum_{i=1}^{n} \sum_{j=1}^{m} |x_i - y_j| P_{ij} \right\}^{1/p}.$$

The discrete Wasserstein distance satisfies the definition of a metric. Symmetry and nonnegativity of this distance are relatively straightforward to understand in the discrete setting. To informally motivate the triangle inequality for the discrete distance, consider Figure B.3.

If we continue the bakery and cafe analogy for optimal transport, we can think of transport between three discrete measures as a distribution channel. Farmers (μ_1) have to transport resources to an intermediate distribution company (μ_2), who then transport to the cafes (μ_3). The triangle inequality in this context implies that it would be cheaper for the bakeries to forgo the intermediate company and transport directly to the cafes, i.e.,



(a) $W_2(\mu_1, \mu_2) + W_2(\mu_2, \mu_3) = 75.28$

(b) $W_2(\mu_2, \mu_3) = 72.42$

Figure B.3: An example of discrete optimal transport. The measures μ_1, μ_2, and μ_3 satisfy the triangle inequality $W_2(\mu_1, \mu_2) + W_2(\mu_2, \mu_3) \leq W_2(\mu_1, \mu_3)$.

transporting from (μ_1) to (μ_3) costs less than transporting from (μ_1) to (μ_2) and then (μ_2) to (μ_3).

While the discrete Wasserstein distance has wide applications in computer science (COMP OT), we can also define the Wasserstein distance for generic probability measures.

B.4 Existence of optimal transport between discrete measures

For discrete measures $\mu = \frac{1}{n}\sum_{i=1}^{n} \delta_{x_i}$ and $\nu = \frac{1}{n}\sum_{j=1}^{n} \delta_{y_j}$, there exists an optimal transport plan between them.

Theorem B.4.1 (Minkowski-Caratheodory/Choquet's Theorem)**.** [8, Theorem 2.5]

Let $B \subset \mathbb{R}^M$ be a non-empty, convex, compact set. Then for any $\pi^\dagger \in B$ there exists a measure η supported on the exterior points $\mathcal{E}(B)$ such that for any affine function f,

$$f(\pi^\dagger) = \int f(\pi)\, d\eta(\pi).$$

Proof. We first prove that any point $\pi^\dagger$ is a convex combination of at most $\dim(B) + 1$ extreme points by using induction on the dimension of B. In other words, there exist $\pi^{(i)} \in \mathcal{E}(B)$ and a_i with $\sum_{i=0}^{d} a_i = 1$ such that $\pi^\dagger = \sum_{i=0}^{d} a_i \pi^{(i)}$. Let $d = \dim(B)$. When $d = 0$, the space B becomes a point, and the result is vacuously true. For the inductive step, we assume that the result is true for sets of dimension $d - 1$ at most. Pick $\pi^\dagger \in B$

and assume $\pi^\dagger \notin \mathcal{E}(B)$. If $\pi^\dagger \in \mathcal{E}(B)$, then it is itself a convex combination of extreme points. Pick $\pi^{(0)}$ from $\mathcal{E}(B)$ and consider the line crossing through the points $\pi^{(0)}$ and $\pi^\dagger$ extended to the boundary of B. Let θ parametrize this line. θ belongs to the following set: $\{\theta : (1-\theta)\pi^{(0)} + \theta\pi^\dagger\}$, and ranges over $[0, \alpha]$ for some $\alpha \geq 1$, where α corresponds to the point at which the line reaches $\mathcal{E}(B)$. By the convexity and compactness of B, α exists and is finite. Let $\xi = (1-\alpha)\pi^{(0)} + \alpha\pi^\dagger$. This implies that $\pi = (1-\theta_0)\xi + \theta_0\pi^{(0)}$ where $\theta_0 = 1 - 1/\alpha$. By construction, $\xi \in \partial^i B$, the *intrinsic boundary* of B. By Proposition 8.9 in Convexity, An Analytic Viewpoint, the intrinsic boundary $\partial^i B$ is the union of all *proper faces* of B, so ξ must reside on some proper face $F \subset B$. By Proposition 8.10, $\dim(F) \leq d-1$, so by the inductive hypothesis, there exist $\{\pi^{(i)}\}_{i=1}^d$ such that $\xi = \sum_{i=1}^n \theta_i \pi^{(i)}$ with $\sum_{i=1}^d \theta_i = 1$. Therefore, $\pi^\dagger = \sum_{i=1}^d (1-\theta_0)\theta_i\pi^{(i)} + \theta_0\pi^{(0)}$. Because $(1-\theta_0)\sum_{i=1}^d \theta_i + \theta_0 = 1$, we have just written $\pi^\dagger$ as a convex combination of extremal points $\{\pi^{(0)}\}_{i=0}^d$.

An function $f : B \to \mathbb{R}$ is affine if for points $x_i \in B$ and constants $a_i \in \mathbb{R}$ such that $\sum_i a_i = 1$,

$$f\left(\sum_i a_i x_i\right) = \sum_i a_i f(x_i).$$

Therefore, for an affine function f,

$$f(\pi)^\dagger = f\left(\sum_{i=0}^d a_i \pi^{(i)}\right) = \sum_{i=0}^d a_i f(\pi^{(i)}).$$

If we define $\eta = \sum_{i=0}^d a_i \delta_{\pi^{(i)}}$, then

$$f(\pi)^\dagger = f\left(\sum_{i=0}^d a_i \pi^{(i)}\right) = \sum_{i=0}^d a_i f(\pi^{(i)}) = \int f(\pi^{(i)}) d\eta.$$

Thus, we have constructed a measure η that satisfies the necessary conditions. □

Theorem B.4.2 (Birkhoff's theorem)**.** [8, Theorem 2.6]

Let B be the set of $n \times n$ bistochastic matrices, i.e.

$$B = \left\{\pi \in \mathbb{R}^{n\times n} : \forall ij, \pi_{ij} \geq 0; \forall j, \sum_{i=1}^n \pi_{ij} = 1; \forall i, \sum_{j=1}^n \pi_{ij} = 1\right\}.$$

This would be an example of a bistochastic matrix:

$$\begin{bmatrix} 0 & 2/3 & 1/3 \\ 0 & 1/3 & 2/3 \\ 1 & 0 & 0 \end{bmatrix}$$

Proof. We start by showing that every permutation matrix is an extremal point. Let $\pi_{ij} = \delta_{j=\sigma(i)}$ where σ is a permutation. Assume seeking contradiction that $\pi \neq \mathcal{E}(B)$. Then there exists $\pi^{(1)}, \pi^{(2)} \in B$, with $\pi^{(1)} \neq \pi \neq \pi^{(2)}$, and $t \in (0,1)$ such that $\pi = t\pi^{(1)} + (1-t)\pi^{(2)}$. Let ij be such that $0 = \pi_{ij} \neq \pi_{ij}^{(1)}$, then

$$0 = \pi_{ij} = t\pi_{ij}^{(1)} + (1-t)\pi_{ij}^{(2)} \Rightarrow \pi_{ij}^{(2)} = -\frac{\pi_{ij}^{(1)}}{1-t} < 0.$$

But since $\pi_{ij}^{(2)} \geq 0$, this is a contradiction, so $\pi \in \mathcal{E}(B)$. Now we show that every $\pi \in \mathcal{E}(B)$ is a permutation matrix, which are bistochastic matrices comprised of 0s and 1s. To do this, we show that the extremal points of B are i) composed of 0s and 1s and ii) that $\pi = \delta_{j=\sigma(i)}$ for a permutation σ. We suppose seeking contradiction that there exist i_1, j_1 such that $\pi_{i_1 j_1} \in (0,1)$. Since $\sum_{i=1}^n \pi_{ij_1} = 1$, there exists some $i_2 \neq i_1$ such that $\pi_{i_2 j} \in (0,1)$. Continuing this procedure until $i_m = i_1$, we find $j_2 \neq j_1$ such that $\pi_{i_2 j_2} \in (0,1)$. We eventually obtain the sequences

$$I = \{i_k j_k : k \in \{1, \dots, m-1\}\}, I^+ = \{i_{k+1} j_k : k \in \{1, \dots, m-1\}\}.$$

Define $\pi^{(\delta)}$ as the following:

$$\pi_{ij}^{(\delta)} = \begin{cases} \pi_{i_k j_k} + \delta & \text{if } ij = i_k j_k \text{ for some } k \\ \pi_{i_{k+1} j_k} - \delta & \text{if } ij = i_{k+1} j_k \text{ for some } k \\ \pi_{ij} & \text{else} \end{cases}$$

Then it follows that

$$\sum_{i=1}^n \pi_{ij}^{(\delta)} = \sum_{i=1} \pi_{ij} + \delta|\{ij \in I : i \in \{1, \dots, n\}\}| - \delta|\{ij \in I^+ : i \in \{1, \dots, n\}\}|.$$

Note that if $ij \in I$ there exists some i' such that $i'j \in I^+$, and if $ij \in I^+$, there exists some i' such that $i'j \in I$. This implies that the cardinalities of $\{ij \in I : i \in \{1, \dots, \}\}$ and $\{ij \in I^+ : i \in \{1, \dots, n\}\}$ are equal.

Thus, $\sum_{i=1}^n \pi_{ij}^{(\delta)} = \sum_{i=1}^n \pi_{ij} = 1$. Similarly, $\sum_{j=1}^n \pi_{ij}^{(\delta)} = \sum_{j=1}^n \pi_{ij} = 1$. This means that $\pi^{(\delta)}$ (and $\pi^{(-\delta)}$) are bistochastic matrices.

Choose $\delta = \min\{\min\{\pi_{ij}, 1-\pi_{ij}\} : ij \in I \cup I^+\} \in (0,1)$. Define $\pi^{(1)} = \pi^{(-\delta)}, \pi^{(2)} = \pi^{(\delta)}$. We know that $\pi^{(1)}$ and $\pi^{(2)} \in B$ with $\pi^{(1)} \neq \pi^{(2)}$. As $\pi = \frac{1}{2}\pi^{(1)} + \frac{1}{2}\pi^{(2)}$, we find that

$\pi \notin \mathcal{E}(B)$, a contradiction. Therefore, there does not exist a pair $i_1 j_1$ such that $\pi_{i_1 j_1} \in (0,1)$, and that if $\pi \in \mathcal{E}(B)$, then $\pi_{ij} \in \{0,1\}$.

To show that $ii)$: that $\pi \in \mathcal{E}(B)$ is a permutation matrix, we note that for each i there exists some j^* such that $\pi_{ij^*} = 1$. We let $\sigma(i) = j^*$ so by construction we have $\pi_{i\sigma(i)} = 1$. We claim that σ is a permutation. Because σ is a mapping from a set to itself, it is sufficient to show that σ is injective to demonstrate its bijectivity. If $j = \sigma(i_1) = \sigma(i_2)$ where $i_1 \neq i_2$ then

$$1 = \sum_{i=1}^{n} \pi_{ij} \geq \pi_{i_1 j} + \pi_{i_2 j} = 2.$$

This is a contradiction, so $i_1 = i_2$ and σ is injective. This means π is a permutation matrix. □

Theorem B.4.3 (Existence of solution to Discrete Monge)**.** [8, Theorem 2.7] *Let* $\mu = \frac{1}{n}\sum_{i=1}^{n} \delta_{x_i}$ *and* $\nu = \frac{1}{n}\sum_{j=1}^{n} \delta_{y_j}$. *Then there exists a solution to the Monge Problem between* μ *and* ν.

Proof. Let M be the minimum of the Kantorovich optimal transport problem, and fix $\epsilon > 0$. Find a measure $\pi^\epsilon \in B$ such that $M \geq \sum_{ij} c_{ij}\pi^\epsilon - \epsilon$. If we let $f(\pi) = \sum_{ij} c_{ij}\pi_{ij}$ then assuming B is compact and convex, we have by the Minkowski-Caratheodory Theorem that there exists a measure η supported on $\mathcal{E}(B)$ such that

$$f(\pi^\epsilon) = \int f(\pi) d\eta(\pi).$$

Therefore

$$M \geq \int \sum_{ij} c_{ij}\pi_{ij} d\eta(\pi) - \epsilon \geq \inf_{\pi \in \mathcal{E}(B)} \sum_{ij} c_{ij}\pi_{ij} - \epsilon \geq M - \epsilon.$$

Since this is true for all ϵ it holds that $\inf_{\pi \in \mathcal{E}(B)} \sum_{ij} c_{ij}\pi_{ij} = M$. We claim that $\mathcal{E}(B)$ is compact, in which case there exists a minimizer $\pi^\dagger \in \mathcal{E}(B)$. By Birkhoff's theorem $\pi^\dagger$ is a permutation matrix, i.e. there exists some permutation $\sigma^\dagger$ such that $\pi^\dagger_{ij} = \delta_{j=\sigma^\dagger(i)}$. Let $T^\dagger : X \to Y$ be defined by $T^\dagger(x_i) = y_{\sigma(i)}$.

We know that the set of transport maps is non-empty (consider the trivial transport map that moves mass from x_i to y_i for all i). Let T be any transport map and define

$\pi_{ij} = \delta_{y_j = T(x_i)}$. Then

$$\sum_{i=1}^{n} c(x_i, T(x_i)) = \sum_{ij} c_{ij}\pi_{ij} \geq \sum_{ij} c_{ij}\pi^{\dagger}_{ij} = \sum_{i=1}^{n} c(x_i, T^{\dagger}(x_i)).$$

This implies that $T^{\dagger}$ is a solution to the Monge problem. We now show that B is compact and convex, and $\mathcal{E}(B)$ is compact. To show B is compact we consider the ℓ^1 norm without loss of generality: $\|\pi\|_1 = \sum_{ij}|\pi_{ij}|$. For all $\pi \in B$, we have $\|\pi\|_1 \leq n^2$, so B is bounded. To show that B is closed, consider a sequence $\pi^{(m)} \to \pi$. Each element of $\pi^{(m)}$ must converge to the corresponding element in π, i.e. $\pi^{(m)}_{ij} \to \pi_{ij}$ for all ij. Similarly, $\sum_{i=1}^{n} \pi_{ij} = \lim_{m\to\infty} \sum_{i=1}^{n} \pi^{(m)}_{ij} = 1$ and $\sum_{j=1}^{n} \pi_{ij} = 1$. Hence $\pi \in B$ and B is closed, so by the Heine-Borel theorem, B is compact. To check that B is convex, we consider $\pi^{(1)}, \pi^{(2)} \in B$ and $\pi = t\pi^{(1)} + (1-t)\pi^{(2)}$ for $t \in [0, 1]$. Clearly, $\pi_{ij} \geq 0$,

$$\sum_{i=1}^{n} \pi_{ij} = t \sum_{i=1}^{n} \pi^{(1)}_{ij} + (1-t) \sum_{i=1}^{n} \pi^{(2)}_{ij} = t + (1-t) = 1,$$

and similarly $\sum_{j=1}^{n} \pi_{ij} = 1$. This implies $\pi \in B$ and B is convex. To show $\mathcal{E}(B)$ is compact, we just have to show it is closed (because $\mathcal{E}(B)$ is a subset of a bounded set B.) If $\pi^{(m)} \to \pi$ is a sequence contained in $\mathcal{E}(B)$, then we already know that that $\pi \in B$ by the compactness of B. By pointwise convergence of $\pi^{(m)}_{ij} \to \pi_{ij}$ we also have that $\pi_{ij} \in \{0, 1\}$. Hence $\pi \in \mathcal{E}(B)$ and therefore $\mathcal{E}(B)$ is closed. □

Thus, we find that not only does there exist an optimal transport plan in the discrete setting, the plan solves both the Monge and Kantorovich problems.

B.5 Gradient flows and the Fokker-Planck equation

What Jordan, Kinderleher, and Otto proved in their paper was that the Fokker-Planck equation is a gradient flow in the space of probability measures $\mathcal{P}_p(\mathbb{R}^n)$ [58]. To discuss this seminal work in optimal transport, we first define some of the important ideas that pertain to the main result in their paper.

Definition B.5.1 (Gradient flow). [60, pg. 64] *Let $\phi : \mathbb{R}^n \to \mathbb{R}$ be convex and everywhere differentiable. Given $x_0 \in \mathbb{R}^n$, the **gradient flow** of ϕ starting at x_0 is a parametric curve*

$x(t)$ starting at time $t = 0$, which moves in the direction of steepest descent. It is given by the ordinary differential equation (ODE):

$$\begin{cases} x(0) = x_0, \\ \dot{x}(t) = -\nabla\phi(x(t)). \end{cases}$$

However, this expression does not give us an explicit way to understand where the gradient flow is located at $x(t+\tau)$ if we know the gradient flow is at $x(t)$, for some small $\tau > 0$. This motivates the discretization of the gradient flow ODE using the **backward/implicit Euler scheme** as an approximate way to tell where the gradient flows from one position to the next, approximating $\dot{x}(t)$ with the difference quotient $\frac{x(t+\tau)-x(t)}{\tau}$:

$$\frac{x(t+\tau) - x(t)}{\tau} = -\nabla\phi(y). \tag{B.1}$$

There are two different ways to approximate $\dot{x}(t)$, which are called the **forward/explicit Euler scheme** and **backward/implicit Euler scheme** respectively, where in the place of y in (B.1), we set $y = x(t)$ or $y = x(t + \tau)$ respectively. The implicit Euler scheme thus looks like

$$\frac{x(t+\tau) - x(t)}{\tau} = -\nabla\phi(x(t+\tau)). \tag{B.2}$$

We now consider what happens when we take incremental time steps of size τ from starting time $t = 0$. Given x_k, the position along the gradient flow that has taken k steps of size τ away from $x(0)$, we want to find x_{k+1} by solving

$$\frac{x_{k+1} - x_k}{\tau} = -\nabla\phi(x_{k+1}) \Leftrightarrow \nabla_x \left(\frac{|x - x_k|^2}{2\tau} + \nabla\phi(x_k) \right) = 0. \tag{B.3}$$

Since the function ϕ is convex and $\mathbb{R}^n$ is a metric space, solving (B.3) is equivalent to solving

$$x_{k+1} = \text{argmin}_x \left(\frac{|x - x_k|^2}{2\tau} + \phi(x) \right). \tag{B.4}$$

Equation (B.3) is sometimes referred to as the Minimizing Movement Scheme. The scheme that Jordan, Kinderleher, and Otto construct in their paper is defined as the following:

$$\rho_{k+1} = \text{argmin}_\rho \left(\frac{W_2(\rho, \rho_k)^2}{2\tau} + \mathcal{F}(\rho) \right),$$

where W_2 is the 2-Wasserstein distance on the space of probability measures $\mathcal{P}_2(\mathbb{R}^n)$, $\rho(t)$ defines a gradient flow in the Wasserstein space $(W_2, \mathcal{P}_2(\mathbb{R}^n))$, $\rho(t) = \rho_k$ if $t \in ((k-1)\tau, k\tau]$, and the functional $\mathcal{F}(\rho)$ satisfies a variational principle explained in [58, pgs. 4,5]. The main result of the paper is that this iterative process of defining new measures ρ_{k+1} converges to the solution of the Fokker-Planck equation

$$\frac{\partial \rho}{\partial t} = \nabla(\rho \Psi) + \beta^{-1} \Delta \rho, \tag{B.5}$$

where Δ denotes the Laplacian, i.e. $\Delta := \nabla \cdot \nabla$, β is some positive real constant, and Ψ is a positive potential function defined in [58, pgs. 6]. In informal terms, the Fokker-Planck equation has two components to it, where $\beta^{-1}\Delta\rho$ is the "diffusion" component (which resembles the **heat equation**, another famous partial differential equation) and $\nabla \cdot \nabla$, is the "continuity" component, which resembles part of the continuity equation (as defined in section 3.3.3).

B.6 A discussion of single-cell biology and trajectory analysis

Single-cell transcriptomics is much too vast of a topic to do it justice in just one page. Here, we expand on some of the topics described in section 4.4, such as cellular differentiation, which is a fascinating topic in its own right. One of the most fascinating concepts in developmental biology is that complex organisms such as humans all originate from single cells, which divide and turn into different tissues. These cells effectively all contain the same DNA, or have the same **genome**, which is a term used to describe the sum total of genetic information in organisms, which tends to be DNA. You may know that the human body is made of different types of tissue types, such as the epithelium of the skin or the tissue lining your gut. The reason why cells have different functions and morphologies is not because they have different genes, but that they turn different genes on and off. This process is called **differential gene expression**. In fact, one can determine what genes are being expressed by looking at the RNA being produced by cells (as gene expression is defined as the process by which genes are transcribed to produce RNA, which is translated to produced proteins). The sum total of a cell or a cell population's RNA is called its **transcriptome.**

Transcriptomics, the study of transcriptomes, is important because it allows researchers to 1) determine the cell type of the cells being sequenced, and 2) to capture a snapshot of gene expression in time. RNAs degrade quickly, so the RNA that is captured in trancriptomics studies are therefore representative of the current genes being expressed by a cell, or cell population. The way in which we gather transcriptomics data is a procedure called **RNA sequencing**, or **RNA-seq.** A variant of RNA sequencing that has gained popularity in the last decade is called **single-cell RNA sequencing**, or **scRNA-seq** for short. Single cell RNA sequencing isolates single cells, sequences their individual transcriptomes, and collates all of the single cell transcriptomes together in a matrix containg the number of "reads" aligning to specific genes. As the genes expressed by a cell dictates cell type, or can be even used as a criterion to define novel cell types, comparing cells based on the expression levels of different genes is an intuitive way to classify single cells into different cell types. Each gene can be thought of as a dimension along which a single cell can vary. When you consider the fact that each cell has differing expression levels for thousands of genes, the thought of simply visualizing differences in cell expression levels is unrealistic given that one would have to plot cells in high-dimensional space. To capture differences in cell expression patterns with such high-dimensional data, dimensionality reduction algorithms are employed, which are usually **t-SNE** (t-stochastic neighbor embedding) and **UMAP** (uniform manifold approximation and projection). Both of these are nonlinear dimensionality reduction techniques, otherwise known as **manifold learning**, to be differentiated from linear dimensionality reduction techniques such as **principal component analysis**, or **PCA**. Manifold learning techniques rely on what is called the **manifold hypothesis**, which assumes that cell gene expression patterns lie on a high-dimensional manifold with lower dimensionality than the number of total genes. This is because biology inherently constrains the ways in which genes vary with respect to each other, as for example, genes can be co-regulated and increase simultaneously in one specific cell type. The plots used in Scheibinger et al. that describe cell populations are UMAP-based plots [26].

As mentioned in section 4.4, one shortcoming of trajectory analysis is that scRNA-seq is a destructive process. Luckily, there are researchers in the wet-lab biology world working to overcome these limitations, one of which is an Amherst alumnus, Felix Horns! New sequencing procedures aim to sample RNA from cells without killing them, and are promising options for studying cell differentiation in time without using optimal transport for trajectory analysis [62] [63].

B.7 Miscellaneous proofs

Lemma B.7.1 (Convergent implies Cauchy in metric spaces)**.** *In metric spaces, a convergent sequence is also a Cauchy sequence.*

Proof. Let (X, d) be a metric space, and (x_n) be a sequence contained in X converging to $x \in X$. Fix $\epsilon > 0$. There exists some integer N such that $d(x_n, n) < \epsilon/2$. For any $n, m \geq N$, it follows that

$$d(x_m, x_n) \leq d(x_m, x) + d(x_n, x) < \epsilon,$$

which implies that the sequence is Cauchy. □

Lemma B.7.2 (Sequential continuity is equivalent to continuity for metric spaces)**.** *Let $f : X \to Y$ be a function between two metric spaces (X, d_X) and (Y, d_Y), and $x \in X$. f is continuous at x if and only if f is sequentially continuous on X, i.e. for a sequence $x_n \in X$ converging to x, it follows that $f(x_n) \to f(x)$.*

Proof. ($\Rightarrow$) Suppose f is continuous at x. Let (x_n) be a sequence in X converging to x, i.e. for all $\delta > 0$, there exists a $N \in \mathbb{N}$ such that for all $n \geq N$, it follows that $d_X(x_n, x) < \delta$. Fix $\epsilon > 0$. By the definition of continuity, there exists a $\delta > 0$ such that any point x' satisfying $d_X(x', x) < \delta$ satisfies $d_Y(f(x'), f(x)) < \epsilon$. Since $(x_n) \to x$, choose $N \in \mathbb{N}$ such that for all $n \geq N$, $d_X(x_n, x) < \delta$. By continuity, $d_Y(f(x_n), f(x)) < \epsilon$. This shows that it is possible to fix an $\epsilon > 0$, and for such an ϵ, there exists an $N \in \mathbb{N}$ such that for all $n \geq N$, it follows that $d_Y(f(x_n, x)) < \epsilon$.

($\Leftarrow$) We prove the contrapositive statement: f not continuous implies that f is not sequentially continuous. For f to not be continuous, there must exist an $\epsilon_0 > 0$ such that for all $\delta > 0$, there exists a $y \in X$ such that $d_X(x, y) < \delta$ and $d_Y(f(x), f(y)) \geq \epsilon_0$. For $n \geq 1$, we specify a sequence of δ_ns defined as $\delta_n = 1/n$. Choose a sequence of y_ns in X such that $d_X(x, y_n) < \delta_n$, $d_Y(f(x), f(y_n)) \geq \epsilon_0$. By construction, $y_n \to x$, but $f(y_n) \nrightarrow f(x)$.

□

Lemma B.7.3 (Sequential compactness is equivalent to compactness for metric spaces). *Let (X, d) be a metric space and $A \subseteq X$. A is sequentially compact if and only if every open cover of A has a finite subcover.*

Proof. ($\Rightarrow$) Suppose A is sequentially compact and $\{U_i\}_{i \in I}$ is a countable open cover of A. There is a $\delta > 0$ such that any ball $B(x, \delta)$ with $x \in A$ is contained in some U_i. To prove this, suppose towards contradiction that for any $\delta > 0$, there exists a $B(x, \delta)$ with $x \in A$ such that the ball is not contained within any U_i. Then for every integer $n \geq 1$ there is an $x_n \in A$ such that $B(x_n, \frac{1}{n})$ is not contained in any individual U_i. By sequential compactness, there is a subsequence of (x_n) which we call (x_{n_j}) converging to $x \in A$ as $n_j \to \infty$. x is contained in some U_i, and because each set U_i is open, there is an $\epsilon > 0$ such that $B(x, \epsilon) \subseteq U_k$. Picking a n_j such that $d(x_{n_j}, x) < \epsilon/2$, i.e., when $1/n_j < \epsilon/2$, we see that

$$B(x_n j, 1/n_j) \subseteq B(x, \epsilon) \subseteq U_k,$$

but this is a contradiction because we supposed that each $B(x_n, 1/n)$ was not contained in any U_i.

Next, we show that there are finitely many open balls, with radius δ and centers in A, whose union contains A. Suppose towards contradiction that any assortment of finite open balls with radius δ and centers in A will not contain A in their union. As a result, no individual ball will contain A, as we assume there is no finite subcover. Pick $x_1 \in A$. As $B(x_1, \delta)$ does not cover A, we pick $x_2 \in A \backslash B(x_1, \delta)$. We repeat this process iteratively where we pick points

$$x_n \in A \backslash \bigcup_{i=1}^{n-1} B(x_i, \delta),$$

generating a sequence of points x_n contained in A. By sequential compactness, we can take a subsequence of x_n called x_{n_j} converging to a point x as $n_j \to \infty$. By construction, $d(x_{n_j}, x_{n'_j}) \geq \delta$ for all distinct indices of x_{n_j}. This implies that (x_{n_j}) is not Cauchy, and therefore that it is not convergent, which contradicts our assumption of sequential compactness. This implies that there are finitely many balls $B(x_1, \delta), \ldots, B(x_n, \delta)$ with centers in A which cover A each one of which can be contained in an open set U_i, generating a finite subcover.

($\Leftarrow$) Let every open cover of A have a finite subcover, and (x_n) be a sequence contained in A. Suppose towards contradiction there is no convergent subsequence of x_n with a limit contained in A. This implies that for every $y \in A$ there is a $\delta_y > 0$ such that $x_n \in B(y, \delta_y)$ for finitely many n. The open cover $\{B(y, \delta_y) : y \in A\}$ of A has a finite subcover $\{B(y_1, \delta_{y_1}), \ldots, B(y_k, \delta_{y_k})\}$. But there are only finitely many indices n such that $x_n \in B(y_1, \delta_{y_1}) \cup \ldots \cup B(y_k, \delta_{y_k})$, which is a contradiction. $\square$

Lemma B.7.4. $\frac{1}{2}\sqrt{4 + (ax+b)^2} + \frac{1}{2}(ax+b)$ *is strictly convex and satisfies* $0 \leq f(x) \leq 1 + d(x)$.

Here is the proof of this fact:

Proof. We first prove that the function $g(x) = \sqrt{4 + x^2}$ is strictly convex. We can interpret this function as the Euclidean norm in $\mathbb{R}^2$ of the point $(2, x)$, for $x \in \mathbb{R}$. The Euclidean norm satisfies the Cauchy-Schwarz-Buniakowski inequality in $\mathbb{R}^n$.

Therefore,

$$\begin{aligned} g(tx_1 + (1-t)x_2) &= \|(2, tx_1 + (1-t)x_2)\|_2 \\ &= \|t(2, x_1) + (1-t)(2, x_2)\|_2 \\ &\leq |t| \cdot \|(2, x_1)\|_2 + |1-t| \cdot \|(2, x_2)\|_2 \\ &= tg(x_1) + (1-t)g(x_2). \end{aligned}$$

This implies that g is a convex function.

We know that equality in the Cauchy-Schwarz-Buniakowski inequality is attained if and only if for $u, v \in \mathbb{R}^n$, $u = \kappa v$ for some $\kappa \in \mathbb{R}$. This means that $t(2, x_1) = \kappa(1-t)(2, x_2)$ for some $\kappa \in \mathbb{R}$, which is equivalent to $\begin{cases} 2t = \kappa \cdot 2 \cdot (1-t) \\ tx_1 = \kappa(1-t) \cdot x_2. \end{cases}$ The equation $2t = \kappa \cdot 2 \cdot (1-t)$ implies that $\kappa = \frac{t}{1-t}$. Examining the second equation, we see that $x_1 = x_2$. Therefore, because equality is attained only when $x_1 = x_2$, $g(x)$ is strictly convex. Now we prove a lemma that allows us to show that the function $\sqrt{4 + (ax+b)^2}$ is strictly convex.

Lemma B.7.5. *Let g be a strictly convex function and h an affine function. Then $g \circ h$ is strictly convex.*

We know that by definition,

$$h(tx_1 + (1-t)x_2) = th(x_1) + (1-t)h(x_2)$$

Therefore,
$$\begin{aligned}(g \circ h)(tx_1 + (1-t)x_2) &= g(h(tx_1 + (1-t)x_2)) \\ &= g(th(x_1) + (1-t)h(x_2) \\ &< t(g \circ h)(x_1) + (1-t)(g \circ h)(x_2).\end{aligned}$$

Because $\sqrt{4 + (ax+b)^2}$ is the composition of a strictly convex function and an affine function, it is therefore strictly convex.

Furthermore, the sum of a strictly convex function and an affine function is itself strictly convex, so $\sqrt{4 + (ax+b)^2} + (ax+b)$ is strictly convex, and so is $\frac{1}{2}\sqrt{4 + (ax+b)^2} + \frac{1}{2}(ax + b)$. □

Lemma B.7.6. *For sets A, B, C, D, where $C \subseteq A$ and $D \subseteq B$, it follows that*

$$(A \times B) \backslash (C \times D) \subseteq ((A \backslash C) \times B) \cup (A \times (B \backslash D)).$$

Proof. Let $(x, y) \in (A \times B) \backslash (C \times D)$. If this is the case, then $(x, y) \notin (C \times D)$. For this to be true, either $x \notin C$, $y \notin D$, or both. This is simply equivalent to stating that

$$(x, y) \in (A \times (B \backslash D)) \cup ((A \backslash C) \times B) \cup ((A \backslash C) \times (B \backslash D)).$$ □

Lemma B.7.7. [12, Chapter 3, Problem 10]

Let (X, Σ, μ) be a measure space. For measurable sets E_1 and E_2 in Σ,

$$\mu(E_1) + \mu(E_2) = \mu(E_1 \cup E_2) + \mu(E_1 \cap E_2).$$

Proof. Because E_1 and E_2 are measurable, it follows that $E_1 \cup E_2, E_1 \cap E_2, E_2 - E_1$, and $E_1 - E_2$ are all measurable. Furthermore, because $E_1 - E_2, E_2 - E_1$, and $E_1 \cap E_2$ are disjoint, and $E_1 \cup E_2 = (E_1 - E_2) \cup (E_2 - E_1) \cup (E_1 \cap E_2)$, it follows that

$$\begin{aligned}\mu(E_1 \cup E_2) + \mu(E_1 \cap E_2) &= \mu(E_1 - E_2) + \mu(E_1 \cap E_2) + \mu(E_2 - E_1) + \mu(E_1 \cap E_2) \\ &= \mu(E_1) + \mu(E_2).\end{aligned}$$ □

It follows from Lemma B.7.7 that for any measurable sets $A, B \in \Sigma$, the measure of $A \cup B$ is less than or equal to $\mu(A) + \mu(B)$.

Proposition B.7.8 (Entropy is 1-strongly concave)**.** *The function $-H(P)$ is 1-strongly convex.* [51, pg. 281]

Proof. A necessary and sufficient condition for a function to be $\alpha-$strongly convex for some $\alpha \in \mathbb{R}^+$ is that $\nabla^2 f(x) \succeq \alpha I$, where $\nabla^2 f$ denotes the Hessian matrix of f defined term-wise by $(\nabla^2)_{ij} := \frac{\partial^2 f}{\partial x_i \partial x_j}$ and I denotes the identity matrix of dimension nm. The symbol $\succeq$ means that every entry in the matrix $\nabla^2 f$ is greater or equal to the corresponding entry in the matrix αI. We can view $H(P)$ as a function from $\mathbb{R}_+^{nm}$ to $\mathbb{R}$. Let i, j be chosen from $\{1, \ldots, nm\}$. For such a choice of i, j, we have that

$$\frac{\partial^2(-H)}{\partial x_i \partial x_j} = \frac{\partial}{\partial x_j}(\log(x_i) + 1) = \begin{cases} \frac{1}{x_i} & \text{if } i = j \\ 0 & \text{if } i \neq j. \end{cases}$$

Because all values of P_{ij} are less than or equal to 1, it follows that $\frac{1}{x_i} \geq 1$, implying that $\nabla^2 f \succeq I$ and that $-H$ is 1-strongly convex. Equivalently, H is 1-strongly concave. $\square$

Appendix C

Code Listing

Besides Figures 2.3, 3.3, and 4.1, all other figures in the body of the thesis were made using the Tikz package. The author is happy to provide the code for those figures. Figures 2.3, 3.3, and 4.1 were all made in Python, and referenced heavily from the package PythonOT [9]. In particular, Figure 3.3 uses the implementation for displacement interpolation that was used to generate Figure 2.11 in [2].

```
#!/usr/bin/env python
# coding: utf-8

# In[17]:


# sphinx_gallery_thumbnail_number = 5

import numpy as np
import matplotlib.pylab as pl
import matplotlib.pyplot as plt
from scipy.interpolate import CubicSpline
import scipy
import ot
import ot.plot
import matplotlib.gridspec as gridspec
import cv2

# necessary for 3d plot even if not used
from mpl_toolkits.mplot3d import Axes3D
from matplotlib.collections import PolyCollection

from colour import Color


# ## Displacement interpolation between 1-D probability distributions

# In[18]:


```

```
#Gaussian helper function
def gaussian(x, mu, sig):
    return (
        1.0 / (np.sqrt(2.0 * np.pi) * sig) * np.exp(-np.power((x - mu) / sig, 2.0) / 2)
    )


# In[5]:


#Initialize the y-coordinates of the distributions:
# alpha is the starting distribution

n=100
x = np.arange(-20,80,1, dtype=np.float64)

mu1 = 5
sig1 = 10

mu2 = 40
sig2 = 8

func = 0.4*gaussian(x, mu1, sig1)+0.6*gaussian(x, mu2, sig2)

alpha = [0.4*gaussian(i, mu1, sig1)+0.6*gaussian(i, mu2, sig2) for i in x ]
alpha=alpha/sum(alpha)


# In[6]:


#Initialize the y-coordinates of the distributions:
# beta is the ending distribution

beta = gaussian(x, mu=60, sig=10)


# In[7]:


#Visualize the starting and ending distributions (red, blue respectively)
plt.plot(x, alpha, color='red')
plt.plot(x, beta, color='blue')


# In[8]:


#Visualize CDFs of the starting and ending distributions (red, blue respectively)
plt.plot(x, np.cumsum(beta), color='blue')
plt.plot(x, np.cumsum(alpha), color='red')


# In[9]:
```

```


#Calculating the quantile function of alpha and beta
#CubicSpline performs spline interpolation between points
#Reversing np.cumsum and x inverts the function
cs_alpha = CubicSpline(np.cumsum(alpha), x)
cs_beta = CubicSpline(np.cumsum(beta), x)


# In[10]:


#Plotting the quantile functions of alpha and beta
x2 = np.arange(0,1,0.00001)
plt.plot(x2, cs_alpha(x2), color='red')
plt.plot(x2, cs_beta(x2), color='blue')


# In[13]:


#This code block performs displacement interpolation between alpha and beta.
#t is the time parameter in displacement interpolation.
#The quantile of alpha_t (denoted icm)
#is given by (1-t)*quantile of alpha + (t)*quantile of beta
#cm takes the quantile of alpha_t and converts it back to a CDF
#np.diff differentiates the CDF to produce the measure alpha_t.
t=0
icm = (1-t)*cs_alpha(x2) + (t)*cs_beta(x2)
cm = CubicSpline(icm, x2)
m_test = np.diff(cm(np.linspace(-20,80,100000)),n=1)
plt.plot(np.linspace(-20,80,99999),m_test)


# In[15]:


#np.linspace(-20,80,99999)

plt.figure()
cmap = plt.cm.get_cmap('bwr')
verts = []

zs = t_list
for i, z in enumerate(zs):
    ys = B_wass[:, i]
    verts.append(list(zip(np.linspace(-20,80,99999), ys)))

ax = plt.gcf().add_subplot(projection='3d')
print(np.shape(verts))

poly = PolyCollection(verts, facecolors=[cmap(a) for a in reversed(t_list)])
poly.set_alpha(0.7)
ax.add_collection3d(poly, zs=zs, zdir='y')
ax.set_xlabel('x')
ax.set_xlim3d(-20, 80)
```

```python
ax.set_ylabel('$t$')
ax.set_ylim3d(0, 1)
ax.set_zlabel('y')
ax.set_zlim3d(0, B_wass.max() * 1.01)
plt.title('Displacement interpolation')
plt.tight_layout()

plt.show()


# In[16]:


#This code block performs displacement interpolation between alpha and beta
#and plots the interpolants with 11 steps.
times = 11
t_list = np.linspace(0,1,times)
cmaplist = [cmap(a) for a in t_list]

for i in range(times):
    t = t_list[i]
    weights = np.array([1 - t, t])
    icm = (1-t)*cs_alpha(x2) + (t)*cs_beta(x2)
    cm = CubicSpline(icm, x2)
    #plt.axis('off')
    m_test = np.diff(cm(np.linspace(-20,80,100000)),n=1)
    plt.plot(np.linspace(-20,80,99999),m_test, color=cmaplist[10-i])
```

Listing C.1: Code for Figure 2.3 and Figure 3.3

```python
#!/usr/bin/env python
# coding: utf-8

#
# # Regularized OT with generic solver
#
# Illustrates the use of the generic solver for regularized OT with
# user-designed regularization term. It uses Conditional gradient as in [6] and
# generalized Conditional Gradient as proposed in [5,7].
#
#
# [5] N. Courty; R. Flamary; D. Tuia; A. Rakotomamonjy, Optimal Transport for
# Domain Adaptation, in IEEE Transactions on Pattern Analysis and Machine
# Intelligence , vol.PP, no.99, pp.1-1.
#
# [6] Ferradans, S., Papadakis, N., Peyr , G., & Aujol, J. F. (2014).
# Regularized discrete optimal transport. SIAM Journal on Imaging
# Sciences, 7(3), 1853-1882.
#
# [7] Rakotomamonjy, A., Flamary, R., & Courty, N. (2015). Generalized
# conditional gradient: analysis of convergence and applications.
# arXiv preprint arXiv:1510.06567.
#

```

```
# In[114]:


# sphinx_gallery_thumbnail_number = 5

import numpy as np
import matplotlib.pylab as pl
import matplotlib.pyplot as plt
import ot
import ot.plot
import matplotlib.gridspec as gridspec
import cv2


# ## New ot function ##

# In[57]:


#I defined this function because it didn't allow me to play with the colors.
#Previously, we had used red for source and blue for sink.
#By changing source to red and blue to sink, our graphs are visually consistent with the rest of the thesis.
def plot1D_mat(a, b, M, title=''):
    r""" Plot matrix :math:`\mathbf{M}`  with the source and target 1D distribution

    Creates a subplot with the source distribution :math:`\mathbf{a}` on the left and
    target distribution :math:`\mathbf{b}` on the top. The matrix :math:`\mathbf{M}` is shown in between.


    Parameters
    ----------
    a : ndarray, shape (na,)
        Source distribution
    b : ndarray, shape (nb,)
        Target distribution
    M : ndarray, shape (na, nb)
        Matrix to plot
    """
    na, nb = M.shape

    gs = gridspec.GridSpec(3, 3)

    xa = np.arange(na)
    xb = np.arange(nb)

    ax1 = plt.subplot(gs[0, 1:])
    plt.plot(xb, b, 'b', label='Target distribution')
    plt.yticks(())
    plt.title(title)

    ax2 = plt.subplot(gs[1:, 0])
    plt.plot(a, xa, 'r', label='Source distribution')
```

```
    plt.gca().invert_xaxis()
    plt.gca().invert_yaxis()
    plt.xticks(())

    plt.subplot(gs[1:, 1:], sharex=ax1, sharey=ax2)
    plt.imshow(M, interpolation='nearest')
    plt.axis('off')

    plt.xlim((0, nb))
    plt.tight_layout()
    plt.subplots_adjust(wspace=0., hspace=0.2)


# ## Generate data
#
#

# In[58]:


n=100
x = np.arange(n, dtype=np.float64)

# Gaussian distributions
a = ot.datasets.make_1D_gauss(n, m=20, s=5)  # m= mean, s= std
b = ot.datasets.make_1D_gauss(n, m=60, s=10)

# loss matrix
M = ot.dist(x.reshape((n, 1)), x.reshape((n, 1)))
M /= M.max()


# In[59]:


#Gaussian helper function
def gaussian(x, mu, sig):
    return (
        1.0 / (np.sqrt(2.0 * np.pi) * sig) * np.exp(-np.power((x - mu) / sig
    , 2.0) / 2)
    )


# In[60]:


#Simple gaussian distribution
# bin positions (evenly spaced values from 1-100.)
x = np.linspace(-40, 70, 100)

mu = 5
sig = 10

plt.plot(x, gaussian(x, mu, sig))


```

```
# In[61]:


#Bimodal distribution helper function
def func_bimodal(x, mu, sig):
    return (
        1.0 / (np.sqrt(2.0 * np.pi) * sig) * np.exp(-np.power((x - mu) / sig
    , 2.0) / 2)
    )


# In[97]:


#Simple gaussian distribution
# bin positions (evenly spaced values from 1-100.)
x = np.linspace(-40, 70, 100)

mu1 = 5
sig1 = 10

mu2 = 40
sig2 = 8

func = 0.4*gaussian(x, mu1, sig1)+0.6*gaussian(x, mu2, sig2)

gausslist = [0.4*gaussian(i, mu1, sig1)
 +0.6*gaussian(i, mu2, sig2) for i in x ]
gausslist=gausslist/sum(gausslist)


plt.plot(x, gausslist, color='red')


# In[96]:


plt.plot(x, b, color='blue')


# ## Solve EMD
#
#

# In[67]:


G0 = ot.emd(gausslist, b, M)

pl.figure(1, figsize=(5, 5))
plot1D_mat(gausslist, b, G0)


# In[89]:


```

```python
lambd = 0.000000000001

Gs = ot.sinkhorn(gausslist, b, M, lambd, verbose=True)

pl.figure(4, figsize=(5, 5))

#plot1D_mat(gausslist, b, Gs, 'Epsilon =' + str(lambd))

pl.imshow(Gs, interpolation='nearest')

pl.show()


# In[90]:


lambd = 1e-3

Gs = ot.sinkhorn(gausslist, b, M, lambd, verbose=True)

pl.figure(4, figsize=(5, 5))
#plot1D_mat(gausslist, b, Gs, 'OT matrix Sinkhorn')

pl.imshow(Gs, interpolation='nearest')

pl.show()


# In[91]:


lambd = 1e-2

Gs = ot.sinkhorn(gausslist, b, M, lambd, verbose=True)

pl.figure(4, figsize=(5, 5))
#plot1D_mat(gausslist, b, Gs, 'OT matrix Sinkhorn')
pl.imshow(Gs, interpolation='nearest')
pl.show()


# In[92]:


lambd = 1e-1

Gs = ot.sinkhorn(gausslist, b, M, lambd, verbose=True)

pl.figure(4, figsize=(5, 5))
#plot1D_mat(gausslist, b, Gs, 'OT matrix Sinkhorn')
pl.imshow(Gs, interpolation='nearest')
pl.show()


# In[125]:

```

```python

fig = plt.figure(figsize=(10, 7))

rows = 1
columns = 4

Image1 = cv2.imread('erot e-9.png')
Image2 = cv2.imread('erot e-3.png')
Image3 = cv2.imread('erot e-2.png')
Image4 = cv2.imread('erot e-1.png')

fig.add_subplot(rows, columns, 1)
plt.axis('off')
plt.imshow(Image1[:,:,::-1])

fig.add_subplot(rows, columns, 2)
plt.axis('off')
plt.imshow(Image2[:,:,::-1])

fig.add_subplot(rows, columns, 3)
plt.axis('off')
plt.imshow(Image3[:,:,::-1])

fig.add_subplot(rows, columns, 4)
plt.axis('off')
plt.imshow(Image4[:,:,::-1])


# ## Solve EMD with entropic regularization
#
#

# In[5]:


def f(G):
    return np.sum(G * np.log(G))


def df(G):
    return np.log(G) + 1.


reg = 1e-3

Ge = ot.optim.cg(a, b, M, reg, f, df, verbose=True)

pl.figure(3, figsize=(5, 5))
ot.plot.plot1D_mat(a, b, Ge, 'OT matrix Entrop. reg')
```

Listing C.2: Code for Figure 4.1

```python
#!/usr/bin/env python
# coding: utf-8

# In[17]:
```

```


import numpy as np
import math
import matplotlib.pyplot as plt
from scipy.stats import norm
import scipy.stats as ss
import statistics


# PDF

# In[18]:


#normal


# In[19]:


# plt.hist(put_data_here, normed=True, cumulative=True, label='CDF',
#           histtype='step', alpha=0.8, color='k')


# In[20]:


x_axis = np.arange(-5, 5, 0.01)
mean = 0
sd = 1

plt.plot(x_axis, norm.pdf(x_axis, mean, sd))
plt.show()


# CDF

# In[22]:


y_axis = ss.norm.cdf(x_axis)
plt.plot(x_axis, y_axis)


# Quantile

# In[23]:


plt.plot(y_axis, x_axis)
```

Listing C.3: Code for Figure A.2

Bibliography

[1] Conway, J. B. (1985). A Course in Functional Analysis (Vol. 96). Springer New York. https://doi.org/10.1007/978-1-4757-3828-5

[2] Peyré, G., & Cuturi, M. (2020). Computational Optimal Transport (arXiv:1803.00567). arXiv. http://arxiv.org/abs/1803.00567

[3] Villani, C. Optimal transport: Old and new; Springer: Berlin, 2009.

[4] Villani, C. Topics in optimal transportation; American Mathematical Society: Providence, RI, 2021.

[5] Santambrogio, F. (2015). Optimal Transport for Applied Mathematicians – Calculus of Variations, PDEs and Modeling.

[6] van Gaans, O. (2002). Probability measures on metric spaces.

[7] Morrison, S. 2017. Banach spaces. Australian National University, MATH3325.

[8] Thorpe, M. (2018). Introduction to Optimal Transport.

[9] Rémi Flamary, Nicolas Courty, Alexandre Gramfort, Mokhtar Z. Alaya, Aurélie Boisbunon, Stanislas Chambon, Laetitia Chapel, Adrien Corenflos, Kilian Fatras, Nemo Fournier, Léo Gautheron, Nathalie T.H. Gayraud, Hicham Janati, Alain Rakotomamonjy, Ievgen Redko, Antoine Rolet, Antony Schutz, Vivien Seguy, Danica J. Sutherland, Romain Tavenard, Alexander Tong, Titouan Vayer, POT Python Optimal Transport library, Journal of Machine Learning Research, 22(78):1-8, 2021. Website: https://pythonot.github.io/

[10] Graf. S. and Maudlin, R. D. A classification of disintegrations of measures. In Measure and measurable dynamics (Rochester, NY, 1987), no. 94 in Contemp Math. Amer. Math. Soc., Providence, RI, 1989, pp. 147-158

[11] Cover, T. M., & Thomas, J. A. (2006). Elements of Information Theory.

[12] Wheeden, R. L.; Zygmund, A. Measure and integral: An introduction to real analysis; CRC Press: Boca Raton etc., 2015.

[13] Simon, B. (2011). Convexity: An Analytic Viewpoint (Cambridge Tracts in Mathematics).

[14] Ambrosio, L.; Gigli, N.; Savaré, G. Gradient flows in metric spaces and in the space of probability measures; Birkhäuser: Basel, 2005.

[15] Wasserman, L. Optimal Transport and Wasserstein Distance. Carnegie Mellon University, 36-708.

[16] Ambrosio, L., Brué, E., & Semola, D. (2021). Lectures on Optimal Transport (Vol. 130). Springer International Publishing. https://doi.org/10.1007/978-3-030-72162-6

[17] Polyanskiy, Y. 2018. Random Variables. Massachusetts Institute of Technology, 6.436J / 15.085J.

[18] Kolouri, S., Park, S. R., Thorpe, M., Slepcev, D., & Rohde, G. K. (2017). Optimal Mass Transport: Signal processing and machine-learning applications. IEEE Signal Processing Magazine, 34(4), 43–59. https://doi.org/10.1109/MSP.2017.2695801

[19] Folland, G. B. Real analysis: Modern techniques and their applications; Wiley: New York, 2021.

[20] Aude Genevay. Entropy-Regularized Optimal Transport for Machine Learning. Artificial Intelligence [cs.AI]. PSL University, 2019.

[21] Cuturi, M. (2013). Sinkhorn Distances: Lightspeed Computation of Optimal Transportation Distances (arXiv:1306.0895). arXiv. http://arxiv.org/abs/1306.0895

[22] Burkard, R. E., Dell'Amico, M., & Martello, S. (2009). Assignment problems. SIAM, Society for Industrial and Applied Mathematics.

[23] Cormen, T. H., Leiserson, C. E., Rivest, R. L., & Stein, C. (2022). Introduction to algorithms. The MIT Press.

[24] Weed, J., & Bach, F. (2017). Sharp asymptotic and finite-sample rates of convergence of empirical measures in Wasserstein distance (arXiv:1707.00087). arXiv. http://arxiv.org/abs/1707.00087

[25] Varadarajan, V. S. (1958). On the Convergence of Sample Probability Distributions. Sankhyā: The Indian Journal of Statistics (1933-1960), 19(1/2), 23–26. http://www.jstor.org/stable/25048365

[26] Schiebinger, G., Shu, J., Tabaka, M., Cleary, B., Subramanian, V., Solomon, A., Gould, J., Liu, S., Lin, S., Berube, P., Lee, L., Chen, J., Brumbaugh, J., Rigollet, P., Hochedlinger, K., Jaenisch, R., Regev, A., & Lander, E. S. (2019). Optimal-Transport Analysis of Single-Cell Gene Expression Identifies Developmental Trajectories in Reprogramming. Cell, 176(4), 928-943.e22. https://doi.org/10.1016/j.cell.2019.01.006

[27] Agarwal, Pankaj K., et al. "Near-Linear Time Approximation Algorithms for Curve Simplification." Algorithmica, vol. 42, no. 3–4, July 2005, pp. 203–19. DOI.org (Crossref), https://doi.org/10.1007/s00453-005-1165-y.

[28] Kosorok, M. R. (2008). Introduction to Empirical Processes and Semiparametric Inference (1st ed.). Springer.

[29] Rosenthal, J. S. (2006). A first look at rigorous probability theory (2nd ed). World Scientific.

[30] Boyd, S. P., & Vandenberghe, L. (2004). Convex optimization. Cambridge University Press.

[31] Bishop, C. M. (2006). Pattern recognition and machine learning. Springer.

[32] Christian Léonard. A survey of the Schrödinger problem and some of its connections with optimal transport. Discrete and Continuous Dynamical Systems, 2014, 34(4): 1533-1574.

[33] Csiszar, I. (1975). *I*-Divergence Geometry of Probability Distributions and Minimization Problems. The Annals of Probability, 3(1), 146–158.

[34] Amari, S. (2016). Information Geometry and Its Applications (Vol. 194). Springer Japan.

[35] G. Monge, Mémoire sur la théorie de déblais et de remblais. Histoire de l'Académie Royale des Sciences de Paris, 1781, 666-704.

[36] Kantorovich, L. (1958). On the Translocation of Masses. Management Science, 5(1), 1–4.

[37] Arjovsky, M., Chintala, S., & Bottou, L. (2017). Wasserstein GAN (arXiv:1701.07875). arXiv. https://arxiv.org/abs/1701.07875

[38] Stein, E. M., & Shakarchi, R. (2011). Functional analysis: Introduction to further topics in analysis. Princeton University Press.

[39] Dvurechensky, Pavel, et al. Computational Optimal Transport: Complexity by Accelerated Gradient Descent Is Better Than by Sinkhorn's Algorithm. arXiv:1802.04367, arXiv, 7 June 2018. arXiv.org, http://arxiv.org/abs/1802.04367.

[40] Franklin, J., & Lorenz, J. (1989). On the scaling of multidimensional matrices. Linear Algebra and Its Applications, 114–115, 717–735. https://doi.org/10.1016/0024-3795(89)90490-4

[41] Jaynes, E. T. (2003). Probability Theory: The Logic of Science. Cambridge University Press.

[42] Kolodziejczyk, A. A., Kim, J. K., Svensson, V., Marioni, J. C., & Teichmann, S. A. (2015). The Technology and Biology of Single-Cell RNA Sequencing. Molecular Cell, 58(4), 610–620. https://doi.org/10.1016/j.molcel.2015.04.005

[43] Séjourné, T., Peyré, G., & Vialard, F.X. (2023). Unbalanced Optimal Transport, from theory to numerics. In Handbook of Numerical Analysis (Vol. 24, pp. 407–471). Elsevier. https://doi.org/10.1016/bs.hna.2022.11.003

[44] Takahashi, K., & Yamanaka, S. (2006). Induction of Pluripotent Stem Cells from Mouse Embryonic and Adult Fibroblast Cultures by Defined Factors. Cell, 126(4), 663–676. https://doi.org/10.1016/j.cell.2006.07.024

[45] Amezquita, R.A., Lun, A.T.L., Becht, E. et al. Orchestrating single-cell analysis with Bioconductor. Nat Methods 17, 137–145 (2020). https://doi-org.amherst.idm.oclc.org/10.1038/s41592-019-0654-x

[46] Bendall, S. C., Davis, K. L., Amir, E. D., Tadmor, M. D., Simonds, E. F., Chen, T. J., Shenfeld, D. K., Nolan, G. P., & Pe'er, D. (2014). Single-Cell Trajectory Detection Uncovers Progression and Regulatory Coordination in Human B Cell Development. Cell, 157(3), 714–725. https://doi.org/10.1016/j.cell.2014.04.005

[47] Durruthy-Durruthy, R., & Heller, S. (2015). Applications for single cell trajectory analysis in inner ear development and regeneration. Cell and Tissue Research, 361(1), 49–57. https://doi.org/10.1007/s00441-014-2079-2

[48] De Philippis, G., & Figalli, A. (2015). Partial regularity for optimal transport maps. Publications Mathématiques de l'IHÉS, 121(1), 81–112. https://doi.org/10.1007/s10240-014-0064-7

[49] Frogner, Charlie, et al. Learning with a Wasserstein Loss. arXiv:1506.05439, arXiv, 29 Dec. 2015. arXiv.org, https://doi.org/10.48550/arXiv.1506.05439.

[50] Bourne, David P., et al. "Semi-Discrete Optimal Transport Methods for the Semi-Geostrophic Equations." Calculus of Variations and Partial Differential Equations, vol. 61, no. 1, Feb. 2022, p. 39. arXiv.org, https://doi.org/10.1007/s00526-021-02133-z.

[51] Bubeck, Sébastien. Convex Optimization: Algorithms and Complexity. arXiv:1405.4980, arXiv, 16 Nov. 2015. arXiv.org, https://doi.org/10.48550/arXiv.1405.4980.

[52] Dowson, D. C., and B. V. Landau. “The Fréchet Distance between Multivariate Normal Distributions.” Journal of Multivariate Analysis, vol. 12, no. 3, Sept. 1982, pp. 450–55. DOI.org (Crossref), https://doi.org/10.1016/0047-259X(82)90077-X.

[53] Bogachev, V. I. (2015). Gaussian measures. Amer Mathematical Soc.

[54] Brasco, L. “A Survey on Dynamical Transport Distances.” Journal of Mathematical Sciences, vol. 181, no. 6, Mar. 2012, pp. 755–81. DOI.org (Crossref), https://doi.org/10.1007/s10958-012-0713-7.

[55] Stroock, D. W. (2011). Probability theory: An analytic view. Cambridge University Press.

[56] McCann, Robert J. “A Convexity Principle for Interacting Gases.” Advances in Mathematics, vol. 128, no. 1, June 1997, pp. 153–79. DOI.org (Crossref), https://doi.org/10.1006/aima.1997.1634.

[57] Otto, Felix. “THE GEOMETRY OF DISSIPATIVE EVOLUTION EQUATIONS: THE POROUS MEDIUM EQUATION.” Communications in Partial Differential Equations, vol. 26, no. 1–2, Jan. 2001, pp. 101–74. DOI.org (Crossref), https://doi.org/10.1081/PDE-100002243.

[58] Jordan, Richard, et al. “The Variational Formulation of the Fokker–Planck Equation.” SIAM Journal on Mathematical Analysis, vol. 29, no. 1, Jan. 1998, pp. 1–17. DOI.org (Crossref), https://doi.org/10.1137/S0036141096303359.

[59] Bogachev, V. I. (2007b). Measure theory (Vol. 1). Springer.

[60] Figalli, A., & Glaudo, F. (2021). An Invitation to Optimal Transport, Wasserstein Distances, and Gradient Flows. EMS Press.

[61] Bunne, Charlotte, et al. “Learning Single-Cell Perturbation Responses Using Neural Optimal Transport.” Nature Methods, vol. 20, no. 11, Nov. 2023, pp. 1759–68. DOI.org (Crossref), https://doi.org/10.1038/s41592-023-01969-x.

[62] Horns, Felix, et al. “Engineering RNA Export for Measurement and Manipulation of Living Cells.” Cell, vol. 186, no. 17, Aug. 2023, pp. 3642-3658.e32. DOI.org (Crossref), https://doi.org/10.1016/j.cell.2023.06.013.

[63] Chen, Wanze, et al. “Live-Seq Enables Temporal Transcriptomic Recording of Single Cells.” Nature, vol. 608, no. 7924, Aug. 2022, pp. 733–40. DOI.org (Crossref), https://doi.org/10.1038/s41586-022-05046-9.

Corrections

When originally submitted, this honors thesis contained some errors which have been corrected in the current version. Here is a list of the errors that were corrected.

Various places in the thesis. 4 grammatical errors were corrected, including replacing commas with periods.

p. 1, l. 1. The phrase "seeks to establish results" was changed to "establishes results".

p. 1, l. 8. The phrase "optimal transport plan" was changed to to "transport scheme".

p. 4, l. 10. A sentence was added to direct the reader to measure theory references: "We assume in Chapter 2 and 3 that the reader is familiar with basic definitions and results from Measure Theory, many of which we provide in Appendix A.1, and which can also be found in [12]."

p. 5, l. 1. The phrase "Borel measures" was changed to "Borel probability measures".

p. 8, l. 13. $\pi_{y\#}\gamma = \mu$ was replaced with $\pi_{y\#}\gamma = \nu$.

p. 8, l. 14. π_x and π_y were replaced with μ and ν.

p. 10, l. 10. Replaced the previous sentence with the following: "By compactness, there exists a subsequence (x_{n_k}) of this minimizing sequence which converges to some $\overline{x} \in X$."

p. 14, l. 2. Removed broken reference.

p. 15, l. 21. Changed $\sup_{\varphi,\psi\in\Phi_c}$ to $\sup_{(\varphi,\psi)}$.

p. 15, l. 22. Added the phrase "where the supremum is taken over $(\varphi, \psi) \in C_b(X) \times C_b(Y)$."

p. 16, l. 18-20. Added the expression $\inf_{\pi \in \Pi(\mu,\nu)} I[\pi]$ and reformatted the equations to contain them within the margins.

p. 18, l. 27, p. 19, l. 1. Changed $(F^{-1} \times G^{-1})_{\#} L\lfloor_{[0,1]}$ to $(F^{[-1]} \times G^{[-1]})_{\#} L\lfloor_{[0,1]}$.

p. 19, l. 7. Added a right curly bracket to $\min\{F(x), G(y))$.

p. 21, l. 16. Changed $F^{-1}(t)$ and $F^{-1}(t)$ to $F^{[-1]}(t)$ and $G^{[-1]}(t)$.

p. 21, l. 17. Changed $(F^{-1} \times G^{-1})_{\#} L\lfloor_{[0,1]}$ to $(F^{[-1]} \times G^{[-1]})_{\#} L\lfloor_{[0,1]}$.

p. 21, l. 19. Removed the integral

$$\int_{\mathbb{R}^2} d(x-y) d\pi^{\dagger}(x,y)$$

and replaced L with $L\lfloor_{[0,1]}$.

p. 22, l. 6. Replaced t_1 with x_1.

p. 22, l. 8. Replaced t with t_1.

p. 23, l. 12,14. The phrase "singleton sets" was changed to "atoms".

p. 26, l. 10. Changed the definition of the p-Wasserstein distance by defining $\mu, \nu \in \mathcal{P}(X)$ instead of havingg $\nu \in \mathcal{P}(Y)$, and rewrote the definition as

$$W_p(\mu,\nu) := \min_{\gamma \in \Pi(\mu,\nu)} \left\{ \int_{X \times X} |x-y|^p \, d\gamma \right\}^{1/p}$$

instead of having the integral evaluated over $X \times Y$.

p. 27, l. 5. $\pi = \int_X (\delta_x \otimes \pi_x) \, d\mu_x$ was replaced with $\pi = \int_X (\delta_x \otimes \pi_x) \, d\mu$.

p. 27, l. 19. Specified that $X_1 = X_2 = X_3 = X$.

p. 30, l. 10. Changed $D_{KL}(P \| Q)$ to $D_{KL}(p \| q)$.

p. 30, l. 14. Specified that if μ is not absolutely continuous with respect to ν, then $D_{KL}(\mu \| \nu) = \infty$.

p. 34, l. 8. Replaced $\mu_t = ((1-t)\pi_\mu + t\pi_\nu)_\#\gamma$ with $\mu_t = ((1-t)\mu + t\nu)_\#\gamma$ and specified that $t \in [0, 1]$.

p. 35, l. 7. Replaced mu with μ.

p. 36, l. 15, 17. Replaced $\frac{\partial_t \rho t}{\partial t}$ with $\frac{\partial \rho_t}{\partial t}$.

p. 36, l. 15, 21. Replaced $\min_{\rho_t, v} A[\rho, v_t]$ with $\min_{(\rho_t, v_t)} A[\rho_t, v_t]$.

p. 44, l. 8. Rephrased "Definition 4.3" as "(4.3)".

p. 44, l. 10. Renamed Definition 4.3.1 as Theorem 4.3.1.

p. 44, l. 19-21. Replaced λ_i^* and ν_i^* with λ_i and ν_i.

p. 46, l. 4. Replaced L_c^ε with L_C^ε.